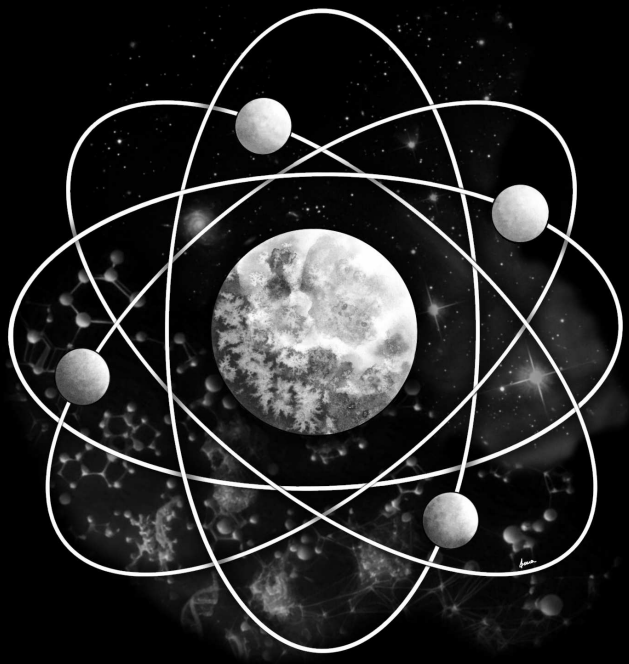
ALUMINIUM MODULATION OF THE DYNAMICS OF SHORT DISORDERED PEPTIDES

DOCTORAL THESIS



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ALUMINIUM MODULATION OF THE DYNAMICS OF SHORT DISORDERED PEPTIDES

A Computational Study

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Abstract

Proteins and metals are two biological elements with many essential functions in organisms. While they often work together for shared purposes, their aberrant interaction typically leads to the development of disorders. In the case of intrinsically disordered proteins (IDPs) and intrinsically disordered regions (IDRs), a type of proteins that move away from the traditional structure-function paradigm, the abnormal metal coordination alters the natural accessible conformation distributions of the protein. The resultant metal-IDP/IDR complex has been linked to medical conditions multiple times, including Alzheimer's Disease (AD), Parkinson's Disease (PD) or Huntington's Disease (HD). Understanding their interactions can pave the way for improved treatments and prevention strategies, ultimately enhancing public health and quality of life. In this thesis, we have focused on studying two key processes involving the coordination of peptides displaying intrinsically disordered behaviour to an aluminium cation (Al^{3+}): (i) the potential of *de novo* mimosine-containing peptides as aluminium chelator agents for metal overload, and (ii) the role of aluminium in protein aggregation mechanisms of AD, including his integration in the Metal Ion Hypothesis (MIH), a hypothesis proposed to explain the formation of pathological aggregates in AD due to the presence of metal ions. These two processes present opposing natures as the problem and the solution to their related medical conditions. Therefore, an adequate understanding of the metal-peptide interaction is fundamental to our capacity to take advantage of this duality. We have used a combination of quantum mechanical

and molecular dynamics techniques to address our systems' multi-scaling nature, allowing us to simultaneously evaluate both the statistical ensembles of conformations and the specific molecular interactions governing them. Our analysis of the metal-peptide interactions across the evaluated systems shows that aluminium cation presents remarkable affinity with the proteins, especially towards residues with oxygen atoms —such as phosphoserines or mimosines— forming strongly coordinated complexes. Factors such as the peptide length, the coordinated residue types, the intra-chain protein interactions, or the system charge distribution can modulate the interactions between the metal and the peptide, having a non-negligible impact on the energetics of the systems, particularly through the entropic contribution. Our results suggest that these structural factors can offset the entropic cost associated with the metal binding to the peptide. In the case of mimosine-containing peptides, the entropic penalty is a contribution that could be modified by design to achieve suitable chelation affinities. However, in the aggregation mechanisms of AD, the character of the interactions would naturally reduce this entropic cost, resulting in unfortunate favourable metal coordination and subsequent aggregate growth. Leveraging the duality of metal-protein interaction systems, these results could be utilized to design efficient chelating agents and to understand the molecular mechanisms of protein aggregation, potentially leading to the development of therapeutic strategies.

Declaration of Originality

I hereby declare that the work presented in this thesis is my own unless otherwise stated. To the best of my knowledge, the work is original, and ideas developed in collaboration with others have been appropriately referenced.

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List of Acronyms

AD Alzheimer's Disease

BO Born-Oppenheimer

BAT Bonds, Angles and Torsions

CA α-Carbon

CAM Coulomb Attenuated Method

CCZ Charge Condensation Zone

CD Circular Dichroism

CG Coarse-Grained

DFT Density Functional Theory

DOF Degree Of Freedom

FCS First Coordination Shell

GGA Generalized Gradient Approximation

GPU Graphics Processing Unit

GTO Gaussian Type Orbitals

HD Huntington's Disease

HDBSCAN Hierarchical Density-Based Spatial Clustering of Applications with Noise

HF Hartree-Fock

HK Hohenberg-Kohn

IDP Intrinsically Disordered Proteins

IDR Intrinsically Disordered Regions

KS Kohn-Sham

LFT Ligand Field Theory

LJ Lennard-Jones

LINCS Linear Constraint Solver

MD Molecular Dynamics

MIE Mutual Information Expansion

MIH Metal Ion Hypothesis

MIST Maximum Information Spanning Tree

ML Machine Learning

MM Molecular Mechanics

MRI Medical Resonance Imaging

NMR Nuclear Magnetic Resonance

NFT Neurofilament Tangles

NN Neural Network

PCA Principal Component Analysis

PCM Polarizable Continuum Model

PD Parkinson's Disease

PDF Probability Distribution Function

PME Particle Mesh Ewald

QM Quantum Mechanics

QTAIM Quantum Theory of Atoms in Molecules

RE Replica Exchange

REST2 Replica Exchange with Solute Tempering

RESP Restrained Electrostatic Potential

SCF Self - Consistent Field

SCS Second Coordination Shell

STO Slater Type Orbitals

TF Thomas-Fermi

1 Introduction

«The first principle is that you must not fool yourself and you are the easiest person to fool.» — Richard Feynman.

This thesis investigates the influence of aluminium on the conformational landscape of selected short peptides. The metal binding process has varying effects on the dynamics and energetics of proteins, and due to the complexity of these molecules, the implications of metal binding are often not fully understood, even for well-characterized proteins. Furthermore, the vast sequence and structural diversity of proteins and the range of metals —some of which are bioavailable but not biologically active— make the study of protein-metal interactions highly valuable from scientific, medical, and technological perspectives. In this Introduction, I review key aspects of proteins and metals and their binding process. The general and specific objectives of each chapter are also outlined.

1.1 Protein structure and Folding

1

Proteins are linear biopolymers formed by α–amino acids [1]. Each amino acid contains an amino (-NH2) and a carboxyl (-COOH) functional group and differs from the rest in its side chain (-R). Except for glycine, the rest of the amino acids present chirality as a result of the four different ligands carried by the α–carbon, resulting in (L) and (D) enantiomers. In nature, only the α-L-amino acids are found in proteins. Figure 1.1 shows the 20 α-L-amino acids organisms use to build proteins, called *proteinogenic* amino acids.

The amino acids form the protein chain by binding through the peptide bond. This bond is formed by the condensation reaction of the amino group and the carboxyl group of consecutive amino acids. The resultant C–N bond is shorter than conventional due to the mesomeric effect with the C=O double bond, resulting in a partially planar behaviour [1]. This particular structure does not allow the peptide bond rotation, which is fixed at a ω dihedral angle of 180° —*trans*— due to the repulsion of amino acid's atoms.

Moreover, two other bonds are present in the so-called *backbone* of the protein. The N–C^{α} and C^{α}–C‘ bonds —with ϕ and ψ rotation dihedral angles respectively— are normal single bonds that can rotate. However, the type of side chain and the fact that, at certain ϕ and ψ rotation angles, the atoms of two consecutive amino acids would collide results in the presence of sterically forbidden regions.

The Ramachandran plots [2] are representations of the forbidden, unfavourable - allowed and favourable - allowed conformational regions of the space defined by these $\{\phi,\psi\}$ dihedral angles. All these regions set up the conformational space of a protein. Given the variety of side chains and the

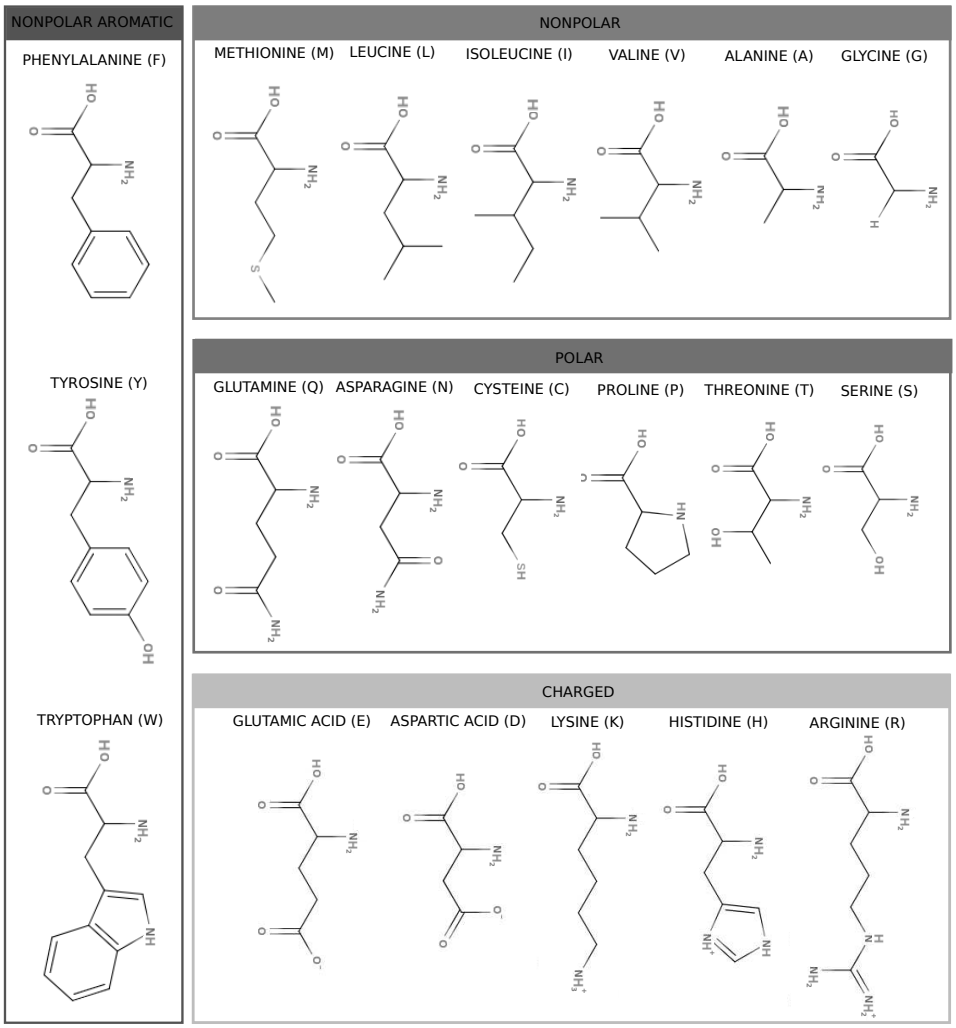


Figure 1.1: The 20 proteinogenic amino acids, grouped by side chain nature.

high number of residue combinations, the particular conformational space Degree Of Freedom (DOF) of a N amino acid protein are the N–*1* sets of $\{\phi,\psi,\omega\}$ dihedral angles —although ω could be considered negligible—. The energy profile of the resultant ensemble of structural conformations accessible for a particular protein is called its *conformational free energy landscape or surface*.

Although a protein can adopt many conformations, its biological functions are carried out in a specific conformation called *native conformation*. The progressive structural change of a protein into its native conformation is called *folding*. Anfinsen conjectured that the native conformation was completely defined by the specific interaction of the amino acid sequence and that this conformation corresponded to the system's free energy minimum [3]. This is known as the Anfinsen's Thermodynamic Hypothesis. But how does a protein go from a given conformation in the conformational energy landscape to the free energy minimum proposed by Anfinsen? As pointed out by Levinthal [4], if the folding process were a completely random procedure, the time taken would be enormously long given the great number of degrees of freedom of a protein. Since the folding time of proteins is of the order of milliseconds to seconds [5], the conclusion was that there must exist factors that assist the folding. This is called the Levinthal's Paradox. This seminal idea was used to propose a folding pathway concept, where once the protein enters the pathway it leads to the native conformation.

In the decade of 1990, combining insights from lattice simulation models and spin glasses, Wolynes, Onuchic, Thirumalai and their co-workers contributed to the development of the energy landscape theory of protein folding [6–10]. In this model, the protein folds by trickling down the funnel-shaped energy landscape through several possible pathways that converge into the

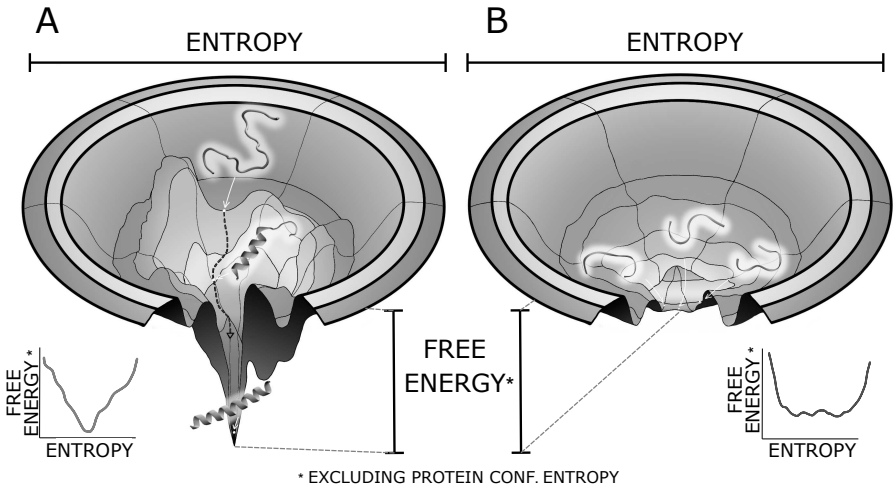


Figure 1.2: Representation of the folding funnel model where the conformational energy landscape is a surface of points each corresponding to a specific conformation of the protein. The Y axis represents the free energy, while the X axis represents the entropy. The process of folding —red dashed line— represents a pathway over the surface from an unfolded conformation to the native conformation.

conformation of minimum free energy indicated by Anfinsen. The width of the funnel corresponds to the entropy while the depth of the funnel represents the free energy. During the descent of the funnel, the protein could get trapped temporarily in local energy minima. Figure 1.2A represents the folding funnel model where the red dashed line corresponds to a folding pathway over the conformational energy landscape.

Surprisingly, decades of research on protein folding resulted in the discovery of some proteins that do not fold into a native conformation and still perform their biological function. Proteins that follow this conduct are called Intrinsically Disordered Proteins (IDP). Since their discovery, they have been found to comprise between 30% to 50% of the eukaryotic genomes [11]. Figure 1.2B shows the folding funnel of these types of proteins.

1.2 Intrinsically Disordered Proteins (IDPs)

1

The 20th Century marked significant advances in understanding the relationship between protein structure and function. This relationship, often summarized as Amino Acid Sequence → 3D Structure → Protein Function, indicated that a well-defined 3D structure was essential for protein function. This idea was rooted in Emil Fischer's 1894 "lock and key" model [12], which suggested that proteins require specific structures to bind correctly to other molecules. Later, Mirsky, Pauling [13], and Hsien Wu [14] reinforced this view by demonstrating that protein function depends on having a stable prior structure.

However, in 1950, this paradigm faced a challenge when Karush observed that serum albumin could bind to a wide range of small molecules with high affinity [15]. He proposed that its binding site could adopt multiple conformations, each of which was in equilibrium, and the best fit would be selected based on the interaction with its partner. He called this phenomenon *Configurational Adaptability*. Shortly after, Koshland independently proposed a similar idea [16], named *Induced Fit*, to describe the conformational changes required during protein synthesis to accommodate different conformations from a similar amino acid sequence. These two proposals hinted at the possibility that significant conformational changes, rather than fixed structures, could influence protein function in some cases. Yet, the exact mechanism was unclear, leaving open questions about whether the binding induced the best-fit conformations or whether these were selected from an ensemble of pre-existing ones.

By the 1980s, discoveries further complicated the structure - function paradigm. Researchers found that certain protein segments essential for

function showed no discernible electron density during crystallography, suggesting they lacked stable structure [17–22]. The reason for the absence of electron density is typically attributed to the inability of structural regions to scatter X-rays coherently due to their flexibility and disorder. This observation, combined with later findings from Nuclear Magnetic Resonance (NMR) studies [23–25], revealed proteins that appeared functional yet lacked a stable 3D structure. These findings led to the identification of IDPs, proteins or peptides that are functional despite their lack of a fixed structure.

The discovery of IDPs prompted a rethinking of the traditional protein structure-function paradigm in the early 21st Century. The nowadays accepted model expands the functionality of the proteins beyond a single native structure. Moreover, the intrinsic absence of folded structure is considered to confer the protein an ability to bind several partners or a precise control of the thermodynamics of the binding processes.

To be strictly correct, the term IDP refers specifically to those proteins that display intrinsic disorder in their whole structure. In contrast, the term Intrinsically Disordered Regions (IDR) refers to specific regions of proteins that present intrinsic disorder, with other regions being structured. Although their phenomenology is equivalent [26], it is important to clarify the terminology for accurate interpretation.

The inherent properties of IDPs/IDRs make them involved in an astonishing amount of crucial biological processes. For example, in cell signalling and regulation [27], gene regulation [28] or cellular response [29]. The capability of IDPs to interact with collections of partners is vastly exploited by organisms in organizing complex protein-protein interaction networks [30–35]. However, the functional versatility of IDPs/IDRs also makes them more prone to be involved in a broad range of human diseases. The misfolding,

misidentification, missignalling, or unnatural folding results in diseases including cancer, cardiovascular disease, amyloidosis, neurodegenerative diseases, and diabetes. The association of disorder proteins and diseases is so strong that in 2008, Uversky, Oldfield, and Dunker —three authorities in disorder proteins— introduced the concept of D^2, disorder in disorders, to summarize the work in this area of study [36].

Metallic cations play a crucial role in various biological processes due to their beneficial chemical properties. Like IDPs/IDRs, they are essential for numerous molecular functions, but their dysfunction can contribute to a wide range of diseases. In addition, many diseases arise from unexpected interactions between IDPs/IDRs and metal ions.

1.3 Metals in Biological Processes

By mass, the elements that constitute more than 1% of the Earth's composition include: iron —Fe— at 32%, oxygen —O— at 30%, silicon —Si— at 15%, magnesium —Mg— at 14%, sulphur —S— at 3%, nickel —Ni— at 2%, calcium —Ca— at 1.5%, and aluminium —Al— at 1.4%. Collectively, these elements account for 98.9% of the Earth's composition, with the remaining 1.1% made up of trace elements [37]. Notably, when considering only these most abundant elements, metals comprise more than half of the Earth's composition. These elements are dispersed across various environments, including soil, water, air, and living organisms, and they participate in specific biogeochemical cycles. Figure 1.3 illustrates the intricate interrelationship between metals and the biogeochemical environment. The uptake of metals by organisms is influenced not only by their presence and relative abundance but also by their bioavailability. For instance, although aluminium is

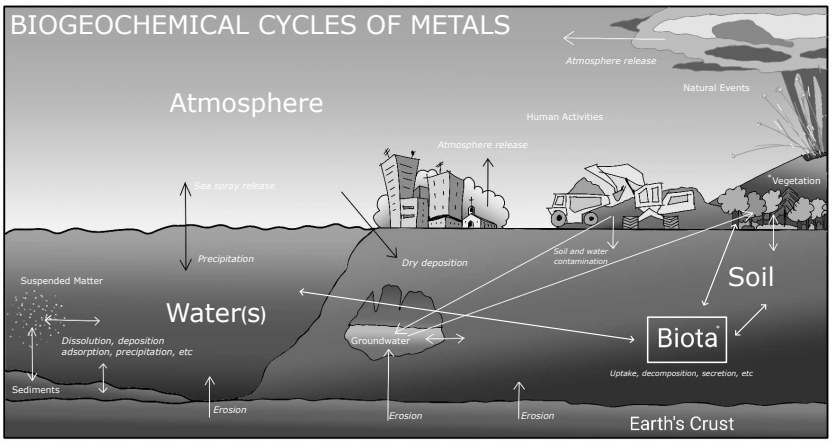


Figure 1.3: Metals are involved in a multitude of cyclic processes that move them along the biosphere.

the third most abundant element in the Earth's crust, its retention in the lithosphere is remarkably efficient, with 99.999% of Al remaining unavailable for biological uptake [38].

Metal ions are an essential part of all life forms [39–42]. Approximately half of all proteins contain metal ions [43], and most ribozymes cannot function without them, as they constitute an integral part of many enzymes and are indispensable in a multitude of catalytic reactions [39–41, 44–47]. For example, the binding of an iron centre in the haemoglobin-myoglobin metalloprotein is essential for our respiration process. Metal ions such as Zn(II), Mg(II), and Ca(II) help stabilize folded protein structures or regulate specific physiological conformations. Additionally, transition metals like Fe, Cu, and Mn are vital for electron transfer processes involved in redox reactions, while alkali and alkaline ions, such as Na(I) and K(I), are vital for cellular responses.

The coordination of metals with proteins is a complexation process whose

interactions are described by the Ligand Field Theory (LFT) [48]. According to the LFT, the interactions between the metallic centre and its ligands —the protein in this case— have a mixed ionic and covalent character. The orbitals of the metal and the orbitals of the protein atoms involved in the coordination are combined into a set of molecular orbitals classified into bonding, anti-bonding, and non-bonding molecular orbitals. The energetic differences between these molecular orbitals and their electronic occupation influence the preferences of metals for particular ligands and vice versa.

A given metal ion with a set of ligands has a well-defined coordination shell. Consequently, in metal-protein complexation, the protein chain must adapt its structure to accommodate the coordination shell required by the metal, often leading to significant structural or conformational changes. In the case of metalloproteins —proteins with metal ion cofactors— the binding sites are usually well-defined and unique. However, multiple binding sites are present when the binding is spurious [49].

As discussed previously, the energy landscape of proteins can be defined by all the accessible conformations of the protein. Thus, incorporating a metal ion modifies the energy landscape of the protein. If the coordination involves only a minor or localized rearrangement, the impact on the conformational energy landscape is likely to be small. However, there are instances where the presence of a metal ion dramatically alters the energy landscape. For many proteins, these modulations in the energy landscape are beneficial, supporting coherent biological functions. Nevertheless, unexpected or incorrect coordination can disrupt functionality and is often implicated in disease development. For instance, the miss-coordination of metals with proteins can lead to metal overload, causing multiple diseases with the different metals and producing significant damage to organs, particularly the liver and

kidneys [50, 51].

Since IDPs/IDRs and metals are implicated in a multitude of biological processes, there is increasing evidence that both are involved simultaneously in many diseases. For example, in Alzheimer's Disease (AD), Parkinson's Disease (PD) and Huntington's Disease (HD), there is substantial evidence that suggests that heavy metals contribute to the aggregation of the IDPs/IDRs that constitute their pathological hallmarks [49, 52–55]. In general terms, metal binding promotes folding states and stabilizes folded or partially folded states by bringing together charged residues present on the surface of proteins, and by neutralizing the charge-charge repulsions, especially for multivalent metal ions [56–58]. This is particularly notable for IDPs/IDRs due to their flexibility and dearth of hydrophobic residues.

The nature of each metal ion defines its specific thermodynamics and kinetics, governing its interaction reaction rates in different conditions —such as association-dissociation or intra-/inter-peptidic metal transfer—. Labile metals tend to display higher rates of these reactions, while inert metals form more persistent complexes [59, 60]. However, effects in aggregation among different metals also present common features: for instance (i) some studies suggest that metal-binding sites in aggregated protein forms can retain structural similarities or binding motifs to those in the monomeric form [61]; (ii) the more labile metal ions show wide partner heterogeneity [62, 63]; and (iii) the second coordination shell has a very strong influence in the resultant binding of the metal ion [64–66].

Given the complexity involved in characterizing the molecular mechanisms of the formation of metal-protein complexes, computational advances play a crucial role in understanding these interactions. Computational methods provide electronic, atomic, and molecular-level insights that are often

difficult or impossible to obtain experimentally, making them highly valuable tools in metal–protein research. However, modeling metal-containing systems remains challenging due to several factors [67]. Transition metals possess high angular momentum atomic orbitals —d- and f-orbitals—, which require advanced computational methods and extensive basis sets to be accurately described. Modeling metals with multiple accessible oxidation states can be non-trivial, as small changes in the environment may significantly alter the electronic structure. Predicting ground and excited states becomes particularly difficult in the presence of degenerate or near-degenerate electronic states, which require multireference methods for accurate treatment. In addition, complex chemical bonding phenomena —such as backdonation and the mixed covalent/ionic character of metal–ligand interactions— are often inadequately captured by standard computational approaches. Finally, a lack of high-quality experimental data, or inconsistencies between experimental and computational results due to methodological limitations or differing conditions, can hinder the validation and calibration of theoretical models.

In the last years, the computational methods applied to protein systems have greatly benefited from recent technological advancements. The most prominent advance in the broader research area of proteins is definitely the integration of machine learning (ML), which has significantly expanded the capabilities of traditional modeling techniques, including those used in the study of metal–protein interactions. ML-based models are being applied to predict protein structures and potential energy surfaces with high accuracy, as seen in tools like AlphaFold [68], SchNet [69], and PhysNet [70]. Additionally, ML is increasingly incorporated into molecular dynamics simulations, enabling enhanced sampling methods such as ML-accelerated metadynamics

[71] and ML-driven coarse-grained modeling [72].

If we narrow our focus to specific methodologies directly applied to metal-protein interactions, tools such as GaudiMM modeling suite [73] —a predictive algorithm to identify and characterize the metal-protein binding by explicitly treating the metal ions— or MetalPDB [74] —a database resource to explore structural metal-binding motifs and patterns — are widely used for studying binding site features. Additionally, methods for computing single-molecule entropies —such as Rigid-Rotor Harmonic-Oscillator approximation and Mutual Information-based approximations [75]— to evaluate the thermodynamic impact of the metal binding are cutting-edge approaches currently [76, 77].

Understanding the interactions between metal and protein is crucial for elucidating their contributions to health and disease. In this thesis, the phenomenon of the modulation of protein dynamics by metal coordination is explored using an aluminium cation bound to different short intrinsically disordered peptides. This work highlights the significance of metal coordination in human diseases and the potential therapeutic approaches for addressing related disorders.

1.4 Objectives

The primary aim of this thesis is to characterize the alterations that occur in short peptides from IDRs and in *de novo* IDP-like peptides upon binding with the aluminium (III) cation. This characterization involves analyzing changes in the structure of the peptides as well as modifications in their conformational ensembles. Additionally, it includes assessing the energetic contributions of the peptide, focusing on both enthalpic and entropic factors.

The specific objectives of each project are outlined below.

- Mimosine Project (Chapter 3)
  - Characterize the key elements of the *de novo* IDP-like mimosine-containing peptide systems for modulating enthalpic and entropic contributions in the chelator design.
    - This is done by using molecular dynamics to simulate the systems with modifications of elements, such as the addition of new residues or side chain modifications.
    - Use previous simulations to characterize and determine the contribution of each modification in the enthalpic/entropic interplay with quantum mechanics calculations and configurational entropy determination techniques.
- Neurofilament Monomer Project (Chapter 4)
  - Evaluate the aluminium coordination with the short peptides from NF-M protein IDRs and the impact of the first and second coordination shells in the binding.
    - This is achieved by performing quantum mechanical calculations of all possible aluminium-neurofilament coordination systems, including first and second coordination shells.
  - Assess the relevance of the phosphorylation groups in the conformational landscape of the neurofilament chains.
    - Using molecular dynamics simulations of free chains with different levels of phosphorylation states and evaluating the results of a series of magnitudes: radius of gyration, contacts between chains, secondary structure formation, and entropy.

  - Analyze the impact of the presence of aluminium centres in the conformational landscape of the neurofilament chains.
    * This is done by using molecular dynamics simulations of aluminium containing neurofilament systems and evaluating the results of several magnitudes: radius of gyration, contacts between chains, secondary structure formation, and entropy.
  - Evaluate the integration of the results in the Metal Ion Hypothesis (MIH) and elaborate a possible list with requirements for the mentioned integration.
    * To do this, we compare the resultant conformational ensembles upon metal binding and their energetic/entropic contributions and evaluate their role and compatibility in the already hypothesized aggregation mechanism.

- Neurofilament Dimers Project (Chapter 5)
  - Assess the relevance of the phosphorylation groups in the interaction between pairs of chains in the absence of aluminium and evaluate the impact of hyperphosphorylation in the chains.
    * This is achieved by analyzing the replica exchange with solute tempering (REST2) dynamics simulations of systems in the absence of aluminium, with different phosphorylation proportions, including all chains fully phosphorylated, known as the hyperphosphorylation state.
  - Characterize the conformational states of the systems by varying the metal coordination and the phosphorylation proportion. Judge these conformations in a possible Metal Ion Hypothesis mechanism following the previously defined requirements.

    * By studying the replica exchange with solute tempering dynamics simulations of three different aluminium coordinations selected based on the previous Chapter 4 energy calculations. Characterizing local and global magnitudes such as radius of gyration and contacts between residues at both inter-chain and intra-chain levels.
    * Including at least three different states of phosphorylation for each aluminium coordination. Comparing the same magnitudes given the change in phosphorylation proportion.

– Obtain simulation time for training a machine-learning coarse-grained (ML-CG) model to elaborate Coarse-Grained simulations of the systems —further work—.

Resumen Introducción

1

Las proteínas son moléculas formadas por una secuencia de α–aminoácidos unidos mediante enlaces peptídicos. Cada aminoácido es único debido a su particular cadena lateral. Exceptuando a la glicina, el resto de aminoácidos presentan quiralidad en su carbono–α debido a sus cuatro ligandos diferentes. Esto hace que los aminoácidos naturales que forman las proteínas sean siempre α–L–aminoácidos. Las diferentes combinaciones posibles de aminoácidos en una secuencia da lugar a que cada proteína pueda adoptar una serie de conformaciones diferentes. Es decir, a tener un "paisaje energético conformacional" particular.

Las proteínas presentan en su paisaje una conformación de energía mínima correspondiente a su conformación nativa, es decir, la estructura específica en la que cumplen su función biológica. Sin embargo, algunas proteínas que no se pliegan completamente o no lo hacen en absoluto pueden seguir cumpliendo sus funciones biológicas. Estas proteínas son conocidas como Proteínas Intrínsecamente Desordenadas (IDPs) y Regiones Intrínsecamente Desordenadas (IDRs) y representan entre el 30% y el 50% de los genomas eucariotas [11]. El descubrimiento de las IDPs/IDRs rompió con el paradigma previamente establecido, en el que la función de una proteína era intrínseca a su forma estructural. A día de hoy, los autores defienden una interpretación estructural continua, donde existen ejemplos de proteínas que cumplen su función con diferentes grados de desorden intrínseco [78, 79].

Las IDPs/IDRs están involucradas en múltiples procesos biológicos cru-

ciales como consecuencia de sus particulares características [27–29]. Sin embargo, a menudo este hecho también tiene como resultado su presencia en diversas enfermedades incluyendo cáncer, diabetes o enfermedades neurodegenerativas como el Alzheimer.

Los metales son otra pieza fundamental en muchos procesos biológicos. Estos elementos constituyen más de la mitad de la composición por masa de la Tierra. Como consecuencia, el desarrollo y la evolución exitosa de los organismos ha estado ligada a la asimilación y uso de metales [39–42]. Los metales son indispensables para los organismos y aproximadamente la mitad de las proteínas requieren iones metálicos para su correcto funcionamiento [43]. De forma similar a las proteínas, los metales están tan integrados en la maquinaria molecular de los organismos que cualquier defecto puede tener como resultado el desarrollo de múltiples enfermedades [80].

Dada la importancia de estos dos elementos en los organismos, la coordinación de metales con proteínas es un proceso enormemente relevante desde el punto de vista científico, médico y tecnológico. Entender sus características, mecanismos moleculares y consecuencias supone poder prevenir o tratar multitud de enfermedades, con un enorme impacto en la calidad de vida de nuestra sociedad.

En los tres capítulos principales de esta tesis hemos investigado la influencia que tiene un ion metálico, en particular el catión de aluminio Al^{3+}, en el espacio energético conformacional de varios péptidos que exhiben un comportamiento intrínsecamente desordenado.

En el Capítulo 3 evaluamos la interacción metal-péptido desde el punto de vista del diseño de péptidos quelantes con mimosina para el tratamiento del exceso de aluminio en el cuerpo. Los agentes quelantes actualmente utilizados, como la deferiprona, presentan múltiples efectos secundarios además

de ser caros [81–86]. Es por ello que existe un creciente interés en desarrollar nuevos agentes quelantes con menores efectos adversos. Desde hace unos años se ha propuesto utilizar como agentes quelantes péptidos con mimosina, un aminoácido no esencial que ha demostrado tener multitud de efectos beneficiosos ligados a su capacidad para coordinar metales [87–92]. Estudiamos por tanto como es la coordinación entre ambos elementos y que modificaciones estructurales del péptido permiten promover la coordinación, factor clave del proceso de quelamiento.

En los Capítulos 4 y 5, evaluamos la interacción metal-péptido entre iones de aluminio y péptidos cortos pertenecientes a la región intrínsecamente desordenada (IDR) de la proteína Neurofilamento Humano de tamaño Medio (NF-M). El estudio se desarrolla en el marco de un mecanismo hipotético de agregación de proteínas denominado Hipótesis del Ion Metálico, con el fin de explicar la formación de agregados proteicos en la enfermedad de Alzheimer. A pesar de conocerse las proteínas involucradas en el proceso de agregación y las formaciones patológicas finales de los agregados en la enfermedad del Alzheimer [49, 52, 53, 93, 94], el mecanismo molecular especifico de este proceso es aún hoy desconocido, aunque se han establecido diversas hipótesis [95]. Una de ellas es la Hipótesis del Ion Metálico, que establece que la agregación se debe a la presencia de iones metálicos, aunque sin aclarar el papel particular que juega el metal [96–99]. Estudiamos como influye la presencia del metal en la dinámica de uno o varios péptidos, su impacto energético y el posible papel que juega el metal en el mecanismo de agregación.

1

2

Theory & Methods

This chapter is dedicated to introducing and describing the theoretical foundations and details of the methods employed during this thesis. Studying proteins and their interactions is a complex task that usually requires a multi-scaling approach. In Figure 2.1 the multi-scale diagram is represented, showing how the different modelling techniques expand over the time and length scales. A typical system studied by biochemists that involves proteins can develop from very rapid processes of small molecules, corresponding to the bottom left corner of the diagram, to roughly the middle of it. Although overlapped, each approach is limited to a specific range of scales, therefore the techniques are often used in combination. This means that a bottom-up combination of quantum mechanics with molecular mechanics or even continuum models such as coarse-graining establishes a symbiosis that allows a fine complementing when it comes to characterizing and understanding the protein dynamics. In practice, however, the differences in length and time scales between each level of the multi-scaling approach mean that the combination of their results must be done carefully.

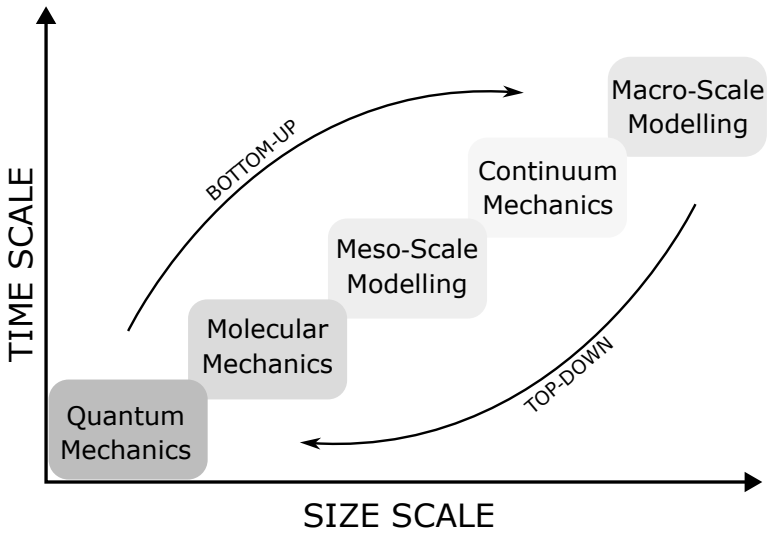


Figure 2.1: The multi-scale diagram is a representation of the different approaches available for studying physical processes. Each process overlaps with the adjacent and often they are used in combination.

2.1 Quantum Mechanics

In this section, I will limit myself to creating a simple and intuitive narrative line to visit the most relevant points of the foundation process and techniques of Quantum Mechanics (QM). This methodological chapter was developed based on these references [100–111].

2.1.1 Foundations of Quantum Mechanics

The 20th century arrived amid an astonishing geopolitical, scientific, and artistic upheaval throughout the planet, and it will be a historical period full of significant events for humanity. In this context, the physicist Max Planck introduced the concept of *quantization* while trying to solve the problem of

black body radiation. According to Planck, energy is quantified in discrete units [112]. This is generally assumed to be the birth of the quantum theory. Some years later, this idea was extended by Albert Einstein to explain the photoelectric effect [113], introducing the concept of *quanta* of energy. These quantization and wave-particle duality of light concepts were later years further applied by other scientists. Niels Bohr proposed the discrete energy levels of electrons [114] and Louis de Broglie hypothesized that wave-particle duality is displayed also by other particles, such as electrons [115]. In 1925, Wolfgang Pauli introduced the Exclusion Principle [116], indicating that fermions cannot simultaneously occupy the same quantum state —an early hint of the existence of the spin—.

Motivated by the hypothesis of de Broglie, Erwin Schrödinger developed his famous equation that describes the changes over time in the quantum state of a system [117]. The relevance of this mathematical framework was immense and entailed subsequent conclusions. The solution of the Schrödinger equation evidenced that quantum systems can exist in multiple states simultaneously — the *superposition principle*—. Max Born suggested the probabilistic interpretation of the wave function [118]. The concept of trajectory was abandoned and a particle's movement was defined by probabilities at a given time. This interpretation of quantum mechanics is known as the *Copenhagen interpretation*.

The maturation of quantum theory came with the establishment of some extraordinary postulates. Werner Heisenberg formulated his *Uncertainty Principle* [119], which states that complementary variables cannot be measured simultaneously with arbitrary precision; Niels Bohr stated that wave and particle aspects cannot be observed simultaneously in his *Complementary Principle*; [120] and John von Neumann set the concept of *wave-function*

collapse [121], indicating that the act of measurement forces quantum system into one of its states. Finally, the *quantum entanglement* was introduced and developed by Albert Einstein, Boris Podolsky, Nathan Rosen, and Erwin Schrödinger, indicating that particles can be correlated regardless of the distance [122–125].

In the span of 40 years, the foundations of QM were settled. Since then, the impact and consequences of this new theory have been extensive. The technology of the modern world, so-called the Digital Revolution, relies on the application of the principles of quantum mechanics. From semiconductors to satellites, through Medical Resonance Imaging (MRI) or the still maturing field of quantum computing.

2.1.2 Born-Oppenheimer Approximation

Although the Schrödinger equation describes the evolution of any particle subject to quantum mechanical behaviour, in reality, it remains analytically tractable only for a very limited number of situations. The presence of multiple interacting particles, as it occurs in a molecule, forces the use of approximations to simplify its complexity. One of the most relevant approximations is the so-called Born-Oppenheimer (BO) Approximation [126], which entails that, as the mass of the atomic nuclei is much heavier than the mass of electrons, their respective movement can be uncoupled. Similar to an adiabatic process where changes occur slowly enough that the system remains in equilibrium, the changes in the nuclei are considered to be slow enough —due to mass difference— so that the electronic part can adjust to it instantaneously. This uncoupling allows independent treatment of the energetic terms of electrons and nuclei, resulting in the separation of the wave function into the electronic and nuclear parts. Hence, the time-independent

non-relativistic Schrödinger equation, appropriate to describe the quantum mechanical behaviour of a molecular system consisting of both electrons and nuclei, can be expressed as,

$$\hat{\mathrm{H}}|\psi(\mathbf{r},\mathbf{R})\rangle = E|\psi(\mathbf{r},\mathbf{R})\rangle, \tag{2.1}$$

where $\psi(\mathbf{r},\mathbf{R})$ is the full wave-function; $\mathbf{r}$ are the electronic coordinates and $\mathbf{R}$ the nuclei coordinates. The uncoupling of the wave function is expressed then as,

$$|\psi(\mathbf{r},\mathbf{R})\rangle \simeq |\psi_{\mathrm{e}}(\mathbf{r};\mathbf{R})\rangle \otimes |\chi_{\mathrm{N}}(\mathbf{R})\rangle, \tag{2.2}$$

where $\psi_{\mathrm{e}}(\mathbf{r};\mathbf{R})$ is the electronic wave-function, which depends explicitly on the electron positions and parametrically on the nuclear positions and $\chi_{\mathrm{N}}(\mathbf{R})$ is the nuclear wave-function, which depends explicitly on the nuclear positions. This results in the separation of the Hamiltonian operator into electronic and nuclear parts. As the energy of the electronic part depends parametrically on the positions of the nuclei $\mathbf{R}$, these are varied and the electronic part is solved at each step repeatedly,

$$\hat{\mathrm{H}}_{\mathrm{e}}|\psi_{\mathrm{e}}(\mathbf{r};\mathbf{R})\rangle = E_{\mathrm{e}}|\psi_{\mathrm{e}}(\mathbf{r};\mathbf{R})\rangle, \tag{2.3}$$

then, the total energy is determined by the following expression,

$$\hat{\mathrm{H}}_{\mathrm{N}}|\chi_{\mathrm{N}}(\mathbf{R})\rangle = E|\chi_{\mathrm{N}}(\mathbf{R})\rangle. \tag{2.4}$$

2.1.3 Density Functional Theory

The complexity of conventional wave-function methods and their computationally expensive cost for accurate results became rapidly a limitation in

the field of electronic structure. In this context, the previously developed —by Llewellyn Thomas [127] and Enrico Fermi [128] independently in the 1920s— description of the electronic structure using a functional of the electronic density was further expanded. The initial Thomas-Fermi (TF) model approximation can be described as,

$$E_{\mathrm{TF}}[\rho] = T_{\mathrm{TF}}[\rho] + E_{\mathrm{ext}}[\rho] + E_{\mathrm{Coulomb}}[\rho], \tag{2.5}$$

where $E_{\mathrm{TF}}[\rho]$ is the total energy functional; $T_{\mathrm{TF}}[\rho]$ is the kinetic energy term, assuming homogeneous electron gas; $E_{\mathrm{ext}}[\rho]$ is energy the interactions of the electrons with the external potential $v(\mathbf{r})$ due to the nuclei; and $E_{\mathrm{Coulomb}}[\rho]$ is the Coulomb repulsion between electrons.

Pierre Hohenberg and Walter Kohn expanded this previous description by formulating their theorems, for non-degenerate ground-states, that gave birth to Density Functional Theory (DFT) [129].

- **First Theorem:** The ground state electron density $\rho(\mathbf{r})$ of a system of interacting electrons uniquely determines the external potential $v(\mathbf{r})$ in which the electrons move and thus the Hamiltonian and all physical properties of the system.
- **Second Theorem:** The ground-state energy E_0 and the ground-state density $\rho(\mathbf{r})$ of a system characterized by the potential $v(\mathbf{r})$ can be obtained from a *variational principle* which involves only the density. There exists a total energy functional $E[\rho]$ for the ground state energy, and the correct ground state density minimizes this functional.

The total energy functional $E_{\mathrm{HK}}[\rho]$ in Hohenberg-Kohn (HK) DFT is described then as,

$$E_{\mathrm{HK}}[\rho] = F_{\mathrm{HK}}[\rho] + E_{\mathrm{ext}}[\rho], \tag{2.6}$$

where $F_{\mathrm{HK}}[\rho]$ term includes kinetic energy $T[\rho]$ and electron-electron interactions; and the second term $E_{\mathrm{ext}}[\rho]$ is the interaction energy of the electron density with the external potential. The HK theorems assure the existence of a functional that describes exactly the ground state of the system.

This theoretical approach became a widely used computational method by the crucial development of the Kohn-Sham equations by Walter Kohn and Lu Jeu Sham —Kohn-Sham (KS)— [130]. In early DFT —before Kohn-Sham—, expressing the kinetic energy $T[\rho]$ solely in terms of the electron density was highly problematic, and the approximations used were often inaccurate. Kohn and Sham addressed this by (i) introducing an auxiliary system of non-interacting electrons that reproduced the same electron density as the interacting system and (ii) introducing the Kohn-Sham orbitals $\psi_{\mathrm{i}}(\mathbf{r})$ for treating these electrons. This allowed the kinetic energy to be treated more accurately while leaving the exchange-correlation energy to be approximated, significantly improving the overall precision of DFT.

The explicit mathematical description of the KS total energy is,

$$E_0[\rho] = -\frac{1}{2}\sum_{i=1}^{N}\langle\psi_i[\rho]|\nabla_i^2|\psi_i[\rho]\rangle + \int v_{\mathrm{ext}}(\mathbf{r})\rho(\mathbf{r})\mathrm{d}\mathbf{r} + \frac{1}{2}\int\int\frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}' + E_{\mathrm{xc}}[\rho], \quad (2.7)$$

where the first term is the non-interacting kinetic energy; the second term is the external potential energy; the third term is the classical Coulomb energy term; and finally $E_{\mathrm{xc}}[\rho]$ is the exchange-correlation energy functional.

The electronic density $\rho[\mathbf{r}]$ is determined in a Self - Consistent Field (SCF) method by solving iteratively the Kohn-Sham orbitals until convergence.

The Kohn-Sham orbitals are defined as,

$$\hat{\mathrm{F}}_{\mathrm{KS}}|\psi_i(\mathbf{r})\rangle = \epsilon_i|\psi_i(\mathbf{r})\rangle, \tag{2.8}$$

where $\hat{\mathrm{F}}_{KS}$ is the Kohn-Sham operator. The electronic density is expressed in Kohn-Sham as,

$$\rho(\mathbf{r}) = \sum_{i=1}^{N} |\psi_i(\mathbf{r})|^2. \tag{2.9}$$

Nevertheless, the exact form of the exchange-correlation functional $E_{\mathrm{xc}}[\rho]$ is unknown. Fortunately, a multitude of accurate approaches have been developed to deal with this term. Usually, the exchange and the correlation terms of the functional also require corrections to obtain accurate chemical descriptions. There exist functionals only for the exchange or the correlation part, and there exist functionals that do not separate the terms.

2.1.4 Functionals used in this work

In this work, we have used the state-of-the-art CAM-B3LYP [131], BP86(B-P) [132, 133] and ωB97X-D [134] functionals. The CAM-B3LYP functional combines two characteristics of the most popular functionals in these times: the benefits of the hybrid functionals —B3LYP [135]— and the long-range corrections —CAM [136]—. The hybrid part integrates a 20% exact exchange from Hartree-Fock theory, and an 80% of approximate exchange —B88 [137]— and correlation —LYP [138]— from DFT, both Generalized Gradient Approximation (GGA) functionals. The mathematical form of the B3LYP functional is,

$$E_{\mathrm{xc}}^{\mathrm{B3LYP}} = (1-a_0)E_{\mathrm{x}}^{\mathrm{LSDA}} + a_0 E_{\mathrm{x}}^{\mathrm{HF}} + a_{\mathrm{x}}\Delta E_{\mathrm{x}}^{\mathrm{B88}} + a_{\mathrm{c}}E_{\mathrm{c}}^{\mathrm{LYP}} + (1-a_{\mathrm{c}})E_{\mathrm{c}}^{\mathrm{VWN}} \tag{2.10}$$

In this expression, the Becke suggested coefficients are a_0=0.20, a_{x}=0.72 and a_{c}=0.81. Also, $E_{\mathrm{x}}^{\mathrm{LSDA}}$ is the exchange from Local Spin Density Approximation, $E_{\mathrm{x}}^{\mathrm{HF}}$ is the exact Hartree-Fock exchange, $\Delta E_{\mathrm{x}}^{\mathrm{B88}}$ is the Becke's gradient correction to the exchange, $E_{\mathrm{c}}^{\mathrm{LYP}}$ is the correlation from LYP —which also includes density gradient terms— and finally $E_{\mathrm{c}}^{\mathrm{VWN}}$ is the local correlation used to provide the different coefficients of local and gradient corrected correlation —Vosko, Wilk, and Nusair functional [139]—.

The Coulomb Attenuated Method (CAM) corrections allow us to overcome situations in which several functionals fail, such as charge transfer excitations [140–142]. This is done by separating both the long-range and the short-range contributions via an Ewald split of the initial r_{12}^{-1},

$$\frac{1}{r_{12}} = \frac{1-\mathrm{erf}(\mu r_{12})}{r_{12}} + \frac{\mathrm{erf}(\mu r_{12})}{r_{12}}. \tag{2.11}$$

In CAM-B3LYP, the CAM corrections are generalized and incorporated into B3LYP. The generalization includes two extra parameters —α and β— for the long/short-range regions. The form of the generalized CAM equation is,

$$\frac{1}{r_{12}} = \frac{1-[\alpha+\beta\cdot\mathrm{erf}(\mu r_{12})]}{r_{12}} + \frac{\alpha+\beta\cdot\mathrm{erf}(\mu r_{12})}{r_{12}}, \tag{2.12}$$

where the first term accounts for the short-range region and the second term accounts for the long-range region. The reported best functional uses $\alpha = 0.19$ and $\alpha+\beta = 0.65$, with $\mu = 0.33$. The resultant CAM-B3LYP with α,β parameters allows extra flexibility for mixing Becke-1988 (DFT) exchange and Hartree-Fock (HF) exchange contributions from r_{12}^{-1}.

2.1.5 Basis Sets

In addition to an appropriate functional, computational calculations require a set of known functions called basis sets to describe the molecular orbitals of the system. A basis set consists of a collection of basis functions used to construct the atomic or molecular orbitals in multi-electron systems. The orbitals can be represented as,

$$\phi_i = \sum_{j=1}^{N} c_{ji}\chi_j, \tag{2.13}$$

where N is the number of basis functions; c_{ji} are the linear combination coefficients; and χ_j is the basis function. The size of the basis set plays a crucial role in determining the precision and accuracy of the calculation, with larger basis sets generally offering a more accurate description. However, increasing the basis set size also leads to higher computational costs. Therefore, the acceptability of a calculation depends on finding an optimal balance between the accuracy gained and the computational resources required.

Given that the accuracy and effectiveness of any Ab Initio calculations profoundly depend on the nature and quality of the basis functions used, this topic has been extensively studied and has caused rivers of ink to flow. However, as a comprehensive review of all the relevant concepts falls outside the scope of this thesis, I will focus specifically on introducing the Pople-style basis functions in a general manner. These are the used references [143, 144].

Some general basis set concepts however are necessary for an adequate exposition. A set of functions containing the minimal functions to represent the orbitals of an atom or molecule is called *minimal basis set or single zeta*. When more than a single function is used for representing each orbital,

this is indicated as *double zeta* for two functions, *triple zeta* for three, and so on. Often, to alleviate the computational burden, only the valence orbitals involved in the bonding are expanded, called *split-valence*. Although the most accurate orbital types are the Slater Type Orbitals (STO), they are restricted to very small molecular systems, and the standard type can be considered the Gaussian Type Orbitals (GTO), which allows an easy evaluation of the multi-centre integrals due to the properties of Gaussian functions. The functional form of the Gaussian primitives orbitals is,

$$\phi(x, y, z; \epsilon, i, j, k) = N x^i y^j z^k e^{-\epsilon r^2}, \tag{2.14}$$

where N is the normalization constant; ϵ is the width exponent; *i, j, k* are non-negative indices that indicate the nature of the orbital; and $r^2 = x^2 + y^2 + z^2$ defines the radial spatial symmetry. The shape of the orbitals determined as $\mathrm{L} = i + j + k$ corresponds to s-type (L = 0), p-type (L = 1), d-type (L = 2), and so on. L = 0 are the spherical GTOs, L = 1 is the axial, etc.

An adequate basis set requires enough flexibility to describe the orbital distortion that occurs during molecule formation accurately. It is common for this condition to require supplementary functions with a higher angular momentum to capture these effects correctly. These supplementary functions are called *polarization functions* and are usually denoted by adding an asterisk (*). Similarly, when in an excited state or when negatively charged atoms are present, one should take care of a proper description of the most external orbitals —effectively describing the extended electron cloud— with additional functions. These additional functions are called *diffuse functions* and are usually denoted by adding a plus (+).

Now, the Pople-style basis functions were introduced by Pople and cowork-

ers [145] in 1972. The naming of the basis set, encoded as n-ijkG**++ is: n is the number of Gaussian primitives in the core orbitals; i, j, k are the number of Gaussian primitives in the valence orbitals, with i, j denoting double split and i, j, k triple split —notice therefore that this is valence-split—; ** and ++ are for augmented basis set, the polarization and diffusion functions added, a single * or + to only heavy atoms and two ** or ++ for also H atoms.

We have used several basis sets in this thesis, depending on the requirements of the system. I will briefly explain one of these, the 6-31G* Pople-style basis set: 6 the Gaussian primitives for core orbitals; 31 the valence double split, indicating the use of 3 Gaussian primitives for s-type orbitals and 1 single Gaussian primitive for p-type orbitals and * indicates that polarization functions are added to only the heavy atoms. The absence of + indicates that no diffuse functions are present in this basis set.

2.2 Molecular Mechanics

Molecular Mechanics (MM) is a general term to refer to the use of Classical Mechanics —Newton's Equation of Motion— to model molecular systems. The two main considerations are i) the Born-Oppenheimer Approximation is assumed valid, meaning that movement of electrons and nuclei is uncoupled, and ii) the interactions between the particles that make up the system, i.e the *potential energy*, is calculated as a function of the coordinates of nuclei using force-descriptor models called *force fields*. The main use of MM is the propagation in time of molecular systems in a computational technique called Molecular Dynamics (MD), creating —predicting— *trajectories*. These trajectories, if certain criteria are met, allow for the calculation of ensemble

averages and the estimation of various dynamical properties.

2.2.1 Foundations of Molecular Dynamics

Given that MD is a method that uses a wide variety of mathematical and physical aspects, I will briefly introduce some of the developments that crystallized into the MD simulations as we understand them today. One could very well say that molecular dynamics starts with the publication of *Philosophiæ Naturalis Principia Mathematica* by Isaac Newton in 1687 [146], given that the equations of motion published there are still used nowadays. Likewise, the development of Statistical Mechanics by Ludwig Boltzmann [147] and Josiah Willard Gibbs [148] in the late 19th Century is also an aspect of paramount importance in the birth of molecular dynamics. Walter Heitler and Fritz London [149] and John Lennard-Jones [150] regularized, with their works during the first half of the 20th Century, the use of potential energy functions to describe interactions in molecules. Yet, the emergence of MD understood as today —or at least as a primitive equivalent—, had to wait for the proper development of computers, which occurred after the middle of the 20th Century. In 1957, Berni Alder and Thomas Wainwright performed the first molecular simulation with hard spheres [151].

The 60's, 70's, and 80's were fruitful years, introducing the famous CHARMM, AMBER, or GROMOS forcefields, and algorithms like Verlet or Parrinello-Rahman barostat. In 1991, the GROMACS MD simulations package was developed becoming one of the most popular molecular simulation software. In 2013, Martin Karplus, Michael Levitt, and Arieh Warshel were awarded the Nobel Prize for their integration of quantum mechanics and molecular mechanics in the QM/MM methodology [152].

In recent years, the development of Graphics Processing Unit (GPU)

and their parallelization via CUDA acceleration —NVIDIA— has enhanced the computing power, allowing an increase in the size and complexity of simulated systems and the incorporation of new techniques such as machine learning to molecular simulations.

2.2.2 Statistical Mechanics

We can use MD simulations to estimate ensemble averages due to their connection with time average properties. This connection derives from the *Ergodic Hypothesis*, which can be stated as follows:

Hypothesis 1 *In an ergodic system, over a sufficiently long time, a system explores all accessible states in its phase space with equal probability.*

This is, any ergodic system visits all its possible states. This foundational assumption is key, allowing us to relate the macroscopic properties of experiments —ensemble average— with the microscopic behaviour of individual particles in simulations —time average—. Consider as an illustrative example the Figure 2.2, where the ensemble average and the time average are compared in an experiment.

It is important to indicate that ergodicity is not straightforward to check and for many real-world problems, only its violation can be evaluated directly. Given the temporal limitations in experiments and simulations, it is often not possible to evaluate if all the possible states of a system are explored or not, and therefore not possible to discern whether it is ergodic or not. For protein dynamics research, ergodicity is often assumed, and the crucial point is to have sufficient sampling so that the system has visited all its states. This is known as the convergence criteria [153].

The observables are comparable between experiments and simulations

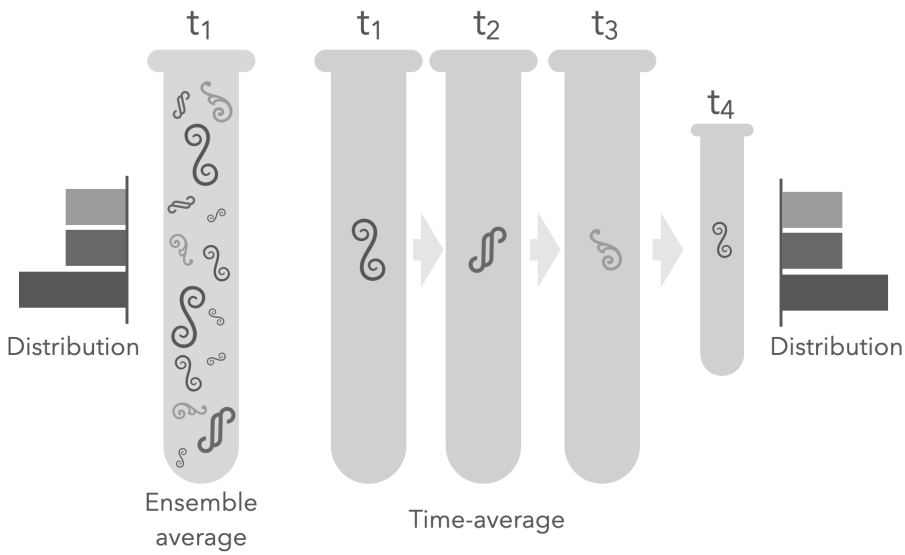


Figure 2.2: In this experiment, from a test tube full of protein copies we pick one and isolate it in another tube. At the time t_1, the tube with multiple protein copies shows that all three spatial conformations are already explored, while in the tube with the isolated protein, only one. However, as time goes on, the isolated protein explores the rest of the spatial conformations. At a sufficient time —convergence—, the statistical distribution of the spatial conformations of the ensemble average is also correctly captured by the time average. This simple experiment shows how the time average is comparable to the ensemble average as assumed by the Ergodic hypothesis.

as long as the type of statistical ensemble used in simulations aligns with the constraints and conditions present in the real experimental system. The mathematical expression for a given observable is,

$$\langle O(\mathbf{p}(t), \mathbf{r}(t)) \rangle = \lim_{\tau \to \infty} \frac{1}{\tau} \int_{t_0}^{t_{0+\tau}} O(\mathbf{p}(t), \mathbf{r}(t)) \mathrm{d}t, \tag{2.15}$$

where $\langle O(\mathbf{p}(t), \mathbf{r}(t)) \rangle$ is the time average observable; τ is the observation time; t_0 and t are the time intervals; $O(\mathbf{p}(t), \mathbf{r}(t))$ is a given observable; and $\mathbf{p}(t), \mathbf{r}(t)$ are the phase space coordinates of the system.

Simulating a molecular system involves the combination of many mathematical operations and tremendously ingenious models. In the following Figure 2.3 a general scheme of an MD simulation is shown. The molecular system is represented as a set of coordinates —positions— of the interaction centres. Moreover, each of these is assigned an initial velocity that makes it easier to reach the thermal equilibrium and equilibrate the system. These velocities are related to the velocity distribution of the simulation's ensemble —usually the Maxwell-Boltzmann distribution—. These two parameters are called the *initial conditions* of the structure. The propagation from the initial point onwards implies the integration of the equations of motion —typically derived from Newton's second law— based on the forces acting on each interaction centre, indicated by the force fields. Integration is done by computational algorithms —integrator— that discretize *time* and iteratively solve —approximate— the differential equations of motion, resulting in a new set of positions and velocities. This process ends when the simulation reaches the number of steps required —or when your High Performance Computing cluster kills your job due to the wall time limit...—.

The discretization of time is primarily driven by a balance of accuracy and computational efficiency. While large time steps would allow reaching

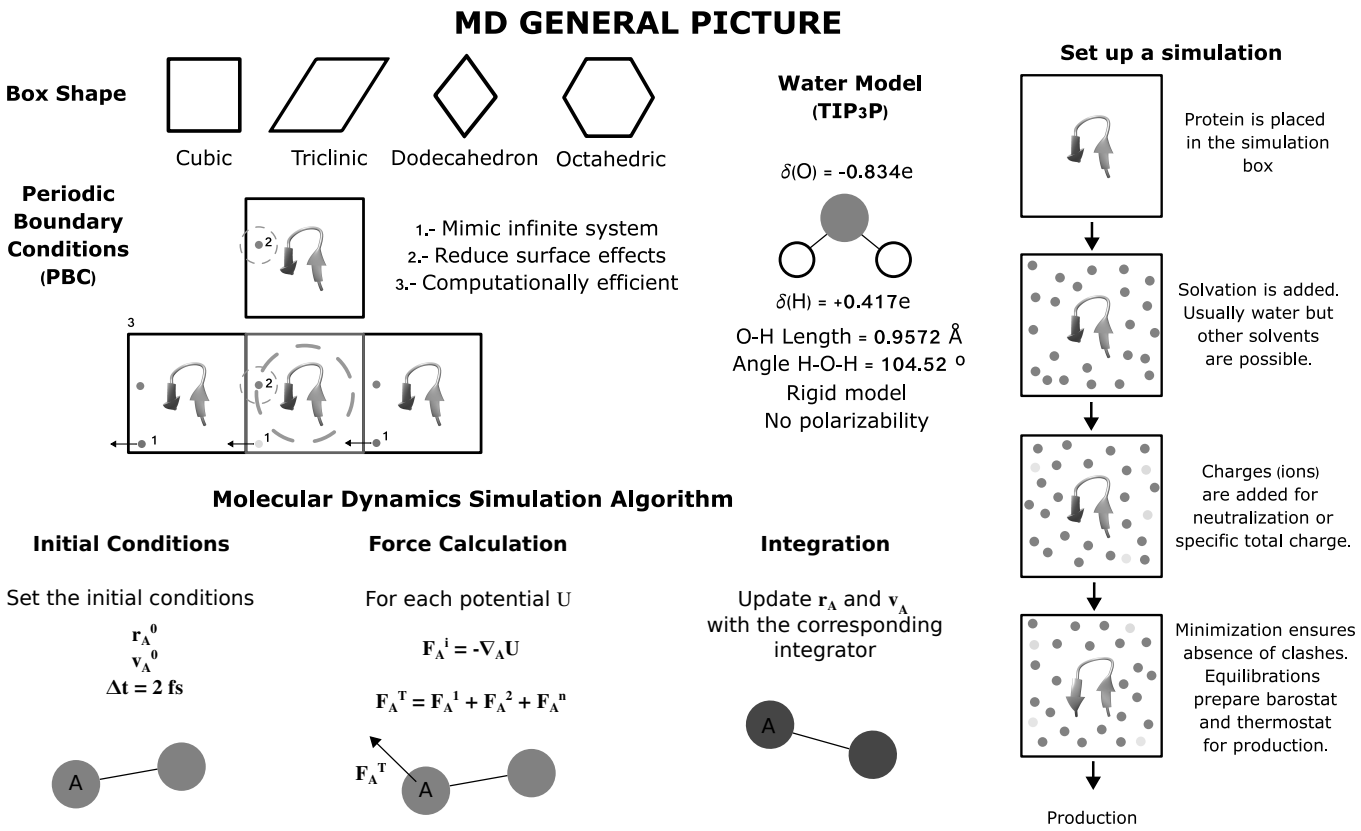


Figure 2.3: Simulating a molecular system requires a multitude of mathematical operations and models. In this general picture, the main aspects of any MD simulation are shown: the size and shape of the simulation box; the solvent model; the periodic boundary conditions; the simulation set-up; and the internal algorithms. Some aspects have been simplified or even omitted for the sake of clarity.

larger time scales, they would not correctly capture the fastest motions of the system, usually the hydrogen vibrations, which is important to ensure the stability and accuracy of the simulation. Given that the hydrogen oscillation period is around 10-13 fs and that the integrators often require at least five integration steps per period of oscillation for accuracy, the time step is typically set to 2 fs [154]. When the fastest motions of a simulation are removed, e.g. in coarse-grained models, the time step can be increased up to the corresponding equivalence with the next fastest oscillation period.

In the following subsections, I will develop further into some of the most relevant elements of a conventional MD simulation.

2.2.3 Integration Algorithms

As mentioned, positions and velocities are obtained by numerically solving differential equations of motion. This essentially means that positions are connected in time, or at least that time plays a meaningful role. However, there exist simulation methods —e.g. Metropolis Monte Carlo— where positions are not temporally related and are dependent only on the previous step. The method used for the calculation of a molecular *trajectory* defines how stochastic or deterministic a simulation is. In this case, we will remain tied to MD simulations, and therefore, the simulations will be completely deterministic.

For a simulation of this type, the integration algorithms are approximate numerical methods that determine —update— the positions and velocities of interaction centres given some initial conditions and a time step. The use of a given force field to the set of initial conditions results in the potential energy U for the system. The application of the gradient operator to each interaction centre results in the forces acting on each of these, a process

described by the following expression,

$$\mathbf{F}_i = -\nabla_i U_i, \tag{2.16}$$

where $\mathbf{F}_i$ is the total force acting on a interaction centre i. The actual form of the potential term U will depend on the specific force field that is used. The resultant $\mathbf{F}_i$ defines the velocity and position of each interaction centre by Newton's second law,

$$\mathbf{F}_i = m_i \mathbf{a}_i = m_i \frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = m_i \frac{\mathrm{d}^2\mathbf{r}_i}{\mathrm{d}t^2}. \tag{2.17}$$

At this point, positions and velocities are fed into an integrator to determine the next position and velocity at the next step in a given time-step size Δt. A commonly used integrator, simple and robust, is the Leapfrog algorithm [155], which updates velocities as,

$$\mathbf{v}_i(t + \frac{1}{2}\Delta t) = \mathbf{v}_i(t - \frac{1}{2}\Delta t) + \frac{\Delta t}{m_i}\mathbf{F}_i(t), \tag{2.18}$$

and positions as,

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i t + \Delta t \mathbf{v}_i(t + \frac{1}{2}\Delta t). \tag{2.19}$$

Notice that positions and velocities are updated asymmetrically —positions at Δt and velocities at $\frac{1}{2}\Delta t$—. This staggered manner helps maintain numerical stability and improves accuracy —avoiding direct coupling reduces error accumulation—. When the simulation ends, the set of conformations ordered in time —the interaction centre's positions at each iteration— is known as *trajectory*.

2.2.4 Force Fields

As mentioned above, the Equation 2.16 defines a general expression for the potential U determined by the force field. The terms inside U vary from one force field to another, depending on the mathematical expressions used to represent the required interactions. We can write a general expression as,

$$U = U_{\text{bonded}} + U_{\text{non-bonded}}, \tag{2.20}$$

where the first term U_{bonded} comprises the bonded interactions, and the second term $U_{\text{non-bonded}}$ comprises the non-bonded interactions.

The bonded term is typically assembled by combining the potentials used to describe bond stretching, angle bending, and dihedral rotations,

$$U_{\text{bonded}} = \sum_{\text{bonds}} V_{\text{bond}}(r_{ij}) + \sum_{\text{angles}} V_{\text{angle}}(\theta_{ijk}) + \sum_{\text{dihedrals}} V_{\text{dihedral}}(\phi_{ijkl}), \tag{2.21}$$

where $V_{\text{bonds}}(r_{ij})$ is the potential function from each bond; $V_{\text{angles}}(\theta_{ijk})$ is the potential function from each angle; and $V_{\text{dihedrals}}(\phi_{ijkl})$ is the potential function from each dihedral. Each of these terms is defined as: Bonds,

$$V_{\text{bond}}(r_{ij}) = \frac{1}{2} k_{\text{bond}}(r_{ij} - r_{0,ij})^2, \tag{2.22}$$

where k_{bond} is the force constant of the bond; $\mathbf{r}_{ij}$ is the bond length; $\mathbf{r}_{0,ij}$ is the equilibrium bond length. Then the angles,

$$V_{\text{angle}}(\theta_{ijk}) = \frac{1}{2} k_{\text{angle}}(\theta_{ijk} - \theta_{0,ijk})^2, \tag{2.23}$$

where k_{angle} is the force constant of the angle; θ_{ijk} is the angle; $\theta_{0,\text{ijk}}$ is the

equilibrium angle. Finally the dihedrals,

$$V_{\text{dihedral}}(\phi_{ijkl}) = \sum_n \frac{k_{\text{dihedral}}}{2}[1 + \cos(n\phi_{ijkl} - \phi_{0,ijkl})]. \tag{2.24}$$

where k_{dihedral} is the force constant of the dihedral; n is the number of minima of the potential; ϕ_{ijkl} is the dihedral angle; and $\phi 0, ijkl)$ is the phase angle parameter of the potential. Bonded interactions are calculated only for interaction centres within a molecule and up to three bonds apart.

Similarly, the non-bonded term $U_{\text{non-bonded}}$ is commonly built by combining the well-known Coulomb and Lennard-Jones (LJ) potentials, which account for the electrostatic interactions and the Van der Waals interactions. It is of the form,

$$U_{\text{non-bonded}} = \sum_{ij} 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \sum_{ij}\frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}, \tag{2.25}$$

where ϵ_{ij} is the depth of the potential well —or dispersion energy—; σ_{ij} is the distance at which the potential is zero; r_{ij} is the distance between the two interacting centres; q is the charge of the corresponding interaction centre; and ϵ_0 is the vacuum permittivity. The calculation of the non-bonded interaction is done for all pairs of interaction centres in different molecules, or in the same molecule but separated by more than three bonds. It is consequently the most computationally intensive part of a simulation. To lighten up the computational burden, the Van der Waals interactions are usually truncated to a cut-off taking advantage of the rapid decay of these, with little impact on the accuracy. However, this does not apply to long-range electrostatic. Thus, techniques such as Ewald Summation or Particle Mesh Ewald (PME), that squeeze the spatial representation of charge distribution and exploit Fourier transforms are employed to handle the electrostatic

interactions efficiently.

2.2.5 Enhanced Sampling

The time required for some observables to be correctly captured in the time average commonly reaches prohibitive computational costs for a conventional MD simulation. While electron transfer motions can occur on the femtosecond to nanosecond time scales, some protein conformational changes can take microseconds or even milliseconds. As of today, the Folding@home project, which aims to distribute the computational burden across thousands of volunteer computers, has been able to run —in a non-stopped non-combined simulation— 1.52 milliseconds for a slow-folding 32-residue NTL9 protein (2010). Although a little out of date, it gives us an idea of the limitations of conventional MD simulations. Furthermore, the other problem associated with simulations is the size of the system. The larger a system is, the more expensive it will be to simulate it.

To overcome these two problems, several approaches have been developed, aimed to facilitate or assist the phase space exploration and the size treatment: *Enhanced Sampling* methods [156] and *Multi-scale* techniques [157]. Some of the most well-known methods of the former are Metadynamics [158, 159], Umbrella Sampling [160, 161], and the different variations of Replica Exchange (RE) [162]. In this thesis, one of the variations of the latter has been used: the Replica Exchange with Solute Tempering (REST2) [163, 164]. One of the most well-known Multi-scale techniques in chemistry is probably Coarse-Grained Modelling [165], also present in the last Further Work chapter of this thesis.

REST2 Methodology

The idea behind the Temperature Replica Exchange method [162] is to use multiple simulations simultaneously —called replicas— that differ from each other slightly in their temperature. At a given time, the replicas try to exchange with the others, swapping their positions and therefore changing the setup to which they are subjected. Replicas at higher temperatures have an increased chance of getting away from equilibrium, thus visiting large areas of the conformational space. Meanwhile, replicas at low temperatures explore specific regions more exhaustively. This combination allows an efficient exploration of the conformational space of a system. The swapping of replicas depends on the energy overlap between replicas, so maintaining a balance between the temperature range and number of replicates is crucial.

However, even with this methodology, for relatively small systems —more than 4000 atoms— a large number of replicas —64— are needed to ensure the energy overlapping, which depends on the Hamiltonian, that grows with the system size. To overcome this issue, REST2 heats only the solute molecule, leaving the solvent "cold" in all the replicas [163, 164]. This translates into easier energy overlapping because the Hamiltonian is not only less complex, but its changes are more localized in the system, making the transitions between replicas smoother, which positively impacts the acceptance ratio of the replica swapping. As a result, fewer replicas are needed.

The total interaction energy of the system for REST2 is decomposed into three components: the solute intra-molecular energy, E_{ss}; the interaction energy between the solute and solvent, E_{pw}; and the self-interaction energy between solvent molecules, E_{ww}. The heating of the solute is effectively accomplished by scaling the interaction terms as shown in the following

expression,

$$E_{\mathrm{m}}^{\mathrm{REST2}}(X) = \frac{\beta_{\mathrm{m}}}{\beta_0} E_{\mathrm{ss}}(X) + \sqrt{\frac{\beta_{\mathrm{m}}}{\beta_0}} E_{\mathrm{sw}}(X) + E_{\mathrm{ww}}(X), \tag{2.26}$$

where $\frac{\beta_{\mathrm{m}}}{\beta_0}$ and $\sqrt{\frac{\beta_{\mathrm{m}}}{\beta_0}}$ are the scaling factors of each term. X is the set of parameters that defines the system's configuration.

2.2.6 Coarse-Grained Modelling

Coarse-Grained (CG) modelling is a widely used multi-scale technique where the original structure of a system is simplified into a more basic representation. This simplification offers two advantages: (i) to treat very large systems by reducing the number of particles and (ii) to accelerate the protein landscape exploration compared with conventional MD simulation. Although these tasks —to treat large systems and explore the protein conformational landscape— are theoretically possible with conventional simulations, the time and energy consumption necessary make them computationally prohibitive in practice.

What may appear as a mere simplification can significantly impact the foundational accuracy or physical correctness of CG simulations. This has led some researchers to evaluate the statistical mechanical foundations of the CG modelling method. Izvekov and Voth [166] introduced the rigorous procedure of coarse-graining a system and determining the CG interaction potential from the atomistic forces using a variational minimization procedure. In landmark works, Noid *et al.* [167, 168] introduced the concept of *thermodynamic consistency*, which evaluates whether or not the assumption that the results from the CG low-resolution model are consistent with those obtained with a more detailed all-atom model is valid. This thermodynamic

consistency is achieved if (i) *"the CG coordinates and momentum are assigned a well-defined meaning as linear combinations of the coordinates and momenta of a subset of atoms of the atomistic simulation"*; and (ii) *"the equilibrium distribution of coordinates and momenta of the CG is equal to the equilibrium distribution of the atomistic model"*. Moreover, they demonstrated that the variational principle forms the fundamental basis of CG and that a thermodynamically consistent model can be developed by the method introduced by Izvekov and Voth.

The key point of the simplification of the structure is that the original structure atoms are *merged* into beads, and these beads represent the atoms that they contain. The process of deciding which and how to merge the original atoms is called *mapping*. Some of the most popular mapping schemes for proteins, shown in Figure 2.4, are the 5-beads mapping, where each residue is merged into 5 beads corresponding to atoms C, N, O, C^{α}, C^{β}; and the C^{α}-bead mapping, where each residue in merged into its C^{α} atom.

The mapping from the all-atom to the CG beads can be expressed mathematically as,

$$\mathbf{R}^N = \mathbf{\Xi}\mathbf{r}^n, \tag{2.27}$$

where $\mathbf{R}$ are the CG coordinates; N is the number of CG beads; $\mathbf{r}$ is the all-atom coordinates; and n is the number of real atoms. $\mathbf{\Xi}$ is the transformation matrix or mapping operator. This mapping operator is different depending on the mapping scheme. As the mapping scheme defines the type and nature of the CG beads, the interactions between beads are completely dependent on the mapping. This means that interactions are parametrized for each mapping scheme and each ensemble of structures considered.

In recent years, some groups have introduced Machine Learning (ML)

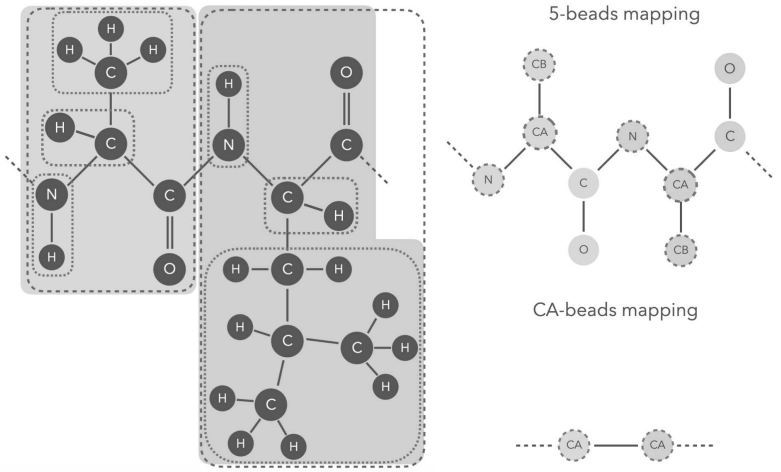


Figure 2.4: Most common mapping schemes: 5-beads and C^{α}-beads. The left structure corresponds to the all-atom structure of two residues. The right panel shows the resultant mapping after the corresponding merge process, a red dotted line for 5-beads and a blue dotted line for C^{α}-beads.

algorithms into the CG modelling method. ML can learn fast by training on large-scale databases, capturing the many-body atomic interactions crucial for accurate descriptions [169–172]. Another recent key focus has been the transferability of CG force fields —the ability to use a force field beyond the structures used in its parametrization—. Although still under development, some groups are advancing toward the goal of broader transferability [173]. One of these groups working in transferable machine-learning coarse-grain models is the group of Prof. Dr. Cecilia Clementi in Berlin, where I completed my international stay. In the following, I develop how ML is integrated into the parametrization process of the CG force field.

As mentioned, the interactions between the beads defined during the initial mapping must be parametrized via ML. The parametrization process is done using a Neural Network (NN) architecture. However, not all the interactions are parametrized. Usually, some of the most basic well-known

interactions, such as the covalent interaction between particles, are described manually by statistical fitting. This is done for several reasons: i) efficiency and complexity reduction: reducing the number of parameters and operations in the training is more efficient and allows lower training times; ii) over-fitting: at a given time, the machine learning model "quits" learning from the training data and starts to just describe all the data in the training set, resulting in a negative impact on its performance outside the training set. By excluding some parameters, the over-fitting problem is slightly prevented; iii) stability and convergence: convergence and stability are achieved more simply by statistical fitting rather than training, especially for already well-known parameters —with enough data—.

Therefore, the global interactions of the CG model can be expressed as follows,

$$V(\mathbf{R}^N;\{\Theta\}) = V_\Delta(\mathbf{R}^N;\{\Theta\}) + V_{\text{prior}}(\mathbf{R}^N), \tag{2.28}$$

where $V(\mathbf{R}^N;\{\Theta\})$ is the global set of interactions; $V_\Delta(\mathbf{R}^N;\{\Theta\})$ is the term describing the interactions parametrized via the NN, where $\{\Theta\}$ is the set of parameters learned by the NN; and $V_{\text{prior}}(\mathbf{R}^N)$ is the term describing the interactions statistically fitted, also known as *priors interactions*.

The prior term $V_{\text{prior}}(\mathbf{R}^N)$ is a set of different interactions that are represented by different potentials. In this case, the interactions represented are the covalent interaction between particles —defined by bonds, angles, and dihedrals—, a repulsive non-bonded term, and in our case an electrostatic term. The potentials for each case are:

- For the bonds and angles, a harmonic potential of the form,

$$V(r) = k(r-r_0)^2 + V_0. \tag{2.29}$$

- For dihedrals, a Fourier truncated expansion of the form,

$$V(\theta) = \sum_n k_{1,n}\sin(n\theta) + k_{2,n}\cos(n\theta) + V_0. \tag{2.30}$$

- For the non-bonded repulsive term, a repulsive potential of the form,

$$V(r) = \left(\frac{\sigma_{ij}}{r_{ij}}\right)^6. \tag{2.31}$$

- Finally, for the electrostatics, a Coulombic potential of the form,

$$V(r_{ij}, q_i, q_j) = \frac{k}{\epsilon_{\mathrm{r}}}\frac{q_i q_j}{r_{ij}}. \tag{2.32}$$

The other term $V_\Delta(\mathbf{R}^N; \{\Theta\})$ is the term optimized by the NN. The optimization process is done by the force-matching approach [174] applied to multi-scale coarse-graining, where the so-called *loss function* $L[\mathbf{F}_{\mathrm{CG}}(\mathbf{R}; \{\Theta\})]$ is minimized. The loss function is of the form,

$$L[\mathbf{F}_{\mathrm{CG}}(\mathbf{R}; \{\Theta\})] = \frac{1}{3N}\sum_{i=1}^{N} ||\boldsymbol{\Xi}\mathbf{F}(\mathbf{r}^n) - \mathbf{F}_{\mathrm{CG},\Delta}(\mathbf{R}^N; \{\Theta\})||^2, \tag{2.33}$$

where $\frac{1}{3N}$ is a normalization factor; $\boldsymbol{\Xi}\mathbf{F}(\mathbf{r}^n)$ is the reference atomistic forces mapped to coarse-grained space; and $\mathbf{F}_{\mathrm{CG},\Delta}(\mathbf{R}^N; \{\Theta\}) = -\nabla V_\Delta(\mathbf{R}^N; \{\Theta\})$ are the forces predicted from the coarse-grained model. The NN optimizes the CG model by correcting the $\{\Theta\}$ parameters at each *epoch* —each training iteration— so that the differences between the reference forces and the model forces decrease until a convergence.

There is an interesting detail to mention at this point. At the first epoch of the training, there exists an already pre-computed forces value

$\nabla V_{\Delta}(\mathbf{R}^N; \{\Theta\})$ sometimes called *Delta Forces*. One can imagine this step as that old procedure of starting the engine of an aeroplane with the hand pushing the propeller, and it is of paramount importance because it influences the convergence speed, the stability of the training process, and overall improves its efficiency.

2.3 Entropy calculations

The presence of the metal cation that we focus on modulates the free energy landscape of the proteins, and one of the most influential contributions is entropy [175, 176]. This energetic term was initially introduced in the field of thermodynamics and was later connected to statistical mechanics and information theory. The significance of this magnitude in this thesis necessitates providing an appropriate description of it. However, entropy is probably one of the most difficult terms to describe due to its multiple interpretations and non-equivalent definitions [177–179].

Intending to clarify the use of entropy in this thesis, I start by introducing the well-known Gibbs-Shannon entropy formula,

$$S = -k_{\mathrm{B}} \int \int \rho(\mathbf{p}, \mathbf{q}) \ln(h^{\mathrm{M}} \rho(\mathbf{p}, \mathbf{q})) \mathrm{d}\mathbf{p} \mathrm{d}\mathbf{q}, \tag{2.34}$$

where k_{B} is the Boltzmann constant; $\rho(\mathbf{p}, \mathbf{q})$ is the Boltzmann-Gibbs probability density function in the phase of M momenta $\mathbf{p}$ and coordinate $\mathbf{q}$; and h^{M} is a factor that ensures a dimensionless quantity. This Equation 2.34 can be factorized into the momentum term S_{p} and spatial term S_{q}, arbitrarily assigning the h^{M} factor to the momentum part. The terms are expressed as,

$$S_{\mathrm{p}} = -k_{\mathrm{B}} \int \rho(\mathbf{p}) \ln \left(\frac{h^{\mathrm{M}} \rho(\mathbf{p})}{q^0} \right) \mathrm{d}\mathbf{p}, \tag{2.35}$$

$$S_{\mathrm{q}} = -k_{\mathrm{B}} \int \rho(\mathbf{q}) \ln(\rho(\mathbf{q}) q^0) \mathrm{d}\mathbf{q}, \tag{2.36}$$

where q^0 ensures dimensionless. We care about the entropy differences —ΔS— between states of the systems, and these often occur at constant temperatures. Consequently, the S_{p} term is fixed at a constant value —since it is assumed to be in thermal equilibrium— and cancels out when comparing two states. Therefore, the calculation of entropy is usually just a matter of determining the spatial entropy term S_{q}. This spatial term of the entropy S_{q} is commonly referred to as the *configurational entropy*.

At this point, it is important to clarify our definitions. A protein can explore different conformers, which are the different spatial arrangements of the atoms —conformations—. Each of these conformers presents a translational S_{trans}, rotational S_{rot} and vibrational S_{vib} entropy contribution. If we ignore the roto-translational contributions, each conformation presents a different vibrational contribution of the entropy. Moreover, from the total variety of conformations, an entropy contribution also arises, a term labelled as *conformational entropy* S_{conform} [180]. This term is combined with the previous vibrational entropy contribution S_{vib} resulting in the following expression,

$$S_{\mathrm{configurational}} = S_{\mathrm{vib}} + S_{\mathrm{conform}}. \tag{2.37}$$

This resultant term is the configurational entropy previously defined,

$$S_{\mathrm{configurational}} = S_{\mathrm{q}}. \tag{2.38}$$

At this point, the configurational entropy S_{q} is typically transformed into non-Cartesian Q coordinates, further separated into external and internal coordinates. The resultant S_{q} can be redefined with this transformation as

S_{Q},

$$S_{\mathrm{Q}} = S_{\mathrm{internal}} + S_{\mathrm{external}} \tag{2.39}$$

Under the assumption of a vanishing external field, the external term remains constant and, for entropy differences, the interest is shifted towards determining the contribution from the internal coordinates $S_{\mathrm{Q,int}}$. These internal coordinates are the Bonds, Angles and Torsions (BAT) degrees of freedom. Notice that the previously introduced S_{vib} contribution is inherently integrated in these internal coordinates. For a single molecule, sampling the internal spatial Probability Distribution Function (PDF) is sufficient for determining the total entropy contribution to the Gibbs free energy change —from the solute—. Moreover, some authors indicate that the change in configurational entropy $S_{\mathrm{configurational}}$ —Equation 2.37— is dominated by the conformational entropy S_{conform} [181]. For this reason, examples in the literature can be found where the terms configurational and conformational entropies are used indiscriminately. Similarly, authors often use the configurational entropy term to define the entropy associated with all possible conformations of the system.

In this thesis, we define the determined entropy as configurational entropy $S_{\mathrm{configurational}}$ because it inherently integrates the vibrational contribution. However, we shall reduce the terminology to S_{conf} from now on. We also carry out some specific configurational entropy determination strategies by not considering bonds and angles and only dihedrals —torsions—. These strategies are indicated in each case and further explained in the corresponding sections.

Entropy calculations from the mutual information expansion

In this work, the configurational entropy S_{conf} has been determined using the Maximum Information Spanning Tree (MIST) scheme, by performing the calculations with its parallel implementation in PARENT Software [182]. [1]

To explain the MIST scheme, I first introduce the Mutual Information Expansion (MIE) approach, a non-parametric many-body expansion method [75]. In the MIE expansion, the PDFs are collected for individual degrees of freedom —$\rho(Q_i)$—, pairs of degrees of freedom —$\rho(Q_i, Q_j)$—, triads of degrees of freedom —$\rho(Q_i, Q_j, Q_k)$—, and so on. Then, the Gibbs-Shannon entropies are determined for each of these elements: $S(X_i)$ for individual degrees of freedom, $S(X_i, X_j)$ for pairs, $S(X_i, X_j, X_k)$ for triads, etc. In Figure 2.5, a scheme of the entropy terms introduced below is shown. Then, the full-dimensional configurational entropy S_{conf} is determined as,

$$\begin{aligned} S_{\text{conf}}^{\text{MIE}} = &\sum_i S(X_i) + \sum_{i<j} [S(X_i, X_j) - S(X_i) - S(X_j)] + \\ &+ \sum_{i<j<k} [S(X_i, X_j, X_k) - S(X_i, X_j) - S(X_i, X_k) - \\ &- S(X_j, X_k) + S(X_i) + S(X_j) + S(X_k)] + ... \end{aligned} \tag{2.40}$$

The correlation effects are often rewritten as *mutual information* terms. For correlations between pairs of degrees of freedom, we define,

$$I^2(X_i; X_j) = S(X_i, X_j) - S(X_i) - S(X_j), \tag{2.41}$$

[1] The code can be downloaded from GitHub at `https://github.com/markusfleck/PARENT`

and for correlations among groups of three degrees of freedom,

$$I^3(X_i;X_j;X_k) = S(X_i,X_j,X_k) - S(X_i,X_j) - S(X_i,X_k) - S(X_j,X_k) + S(X_i) + S(X_j) + S(X_k). \tag{2.42}$$

Then, the expansion is written as

$$S_{\text{conf}}^{\text{MIE}} = \sum_i S(X_i) - \sum_{i<j} I^2(X_i;X_j) + \sum_{i<j<k} I^3(X_i;X_j;X_k) - .... \tag{2.43}$$

The truncation of the expansion at a given order introduces some truncation error. Moreover, the convergence of the expansion is not monotonic beyond the pairwise correlations, meaning that the rest of the correlation terms either raise or lower the entropy.

The MIST expansion [183, 184] captures more efficiently and improves the convergence of the previous method by rewriting the S_{conf} term by the use of conditional entropy terms, $S(X_i|X_j)$ instead of the joint entropy $S(X_i,X_j)$.

$$S(X_i|X_j) = S(X_i,X_j) - S(X_j) \tag{2.44}$$

Furthermore, the MIST method selects only the most relevant couplings, maximizing these mutual information terms. The resultant general expression for total configurational entropy is then,

$$S_{\text{conf}}^{\text{MIST}} = \sum_i S(X_i|X_{1,...,i-1}) = \sum_i [S(X_i) - \max(I^2(X_i;X_{1,...,i-1}))] \tag{2.45}$$

where for every index i there is choice from a conjugate $j \in 1,...,i-1$ concerning the mutual information term.

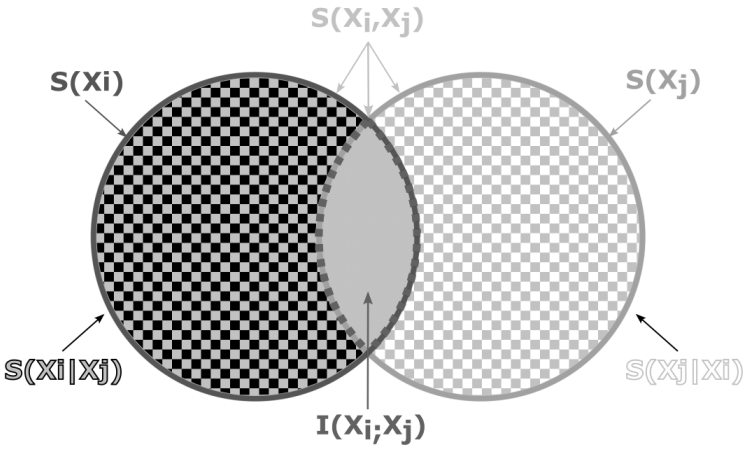


Figure 2.5: Schematic of the entropy terms for two correlated variables X_{i} and X_{j}.

3

Al(III) & Mimosine

3.1 Mimosine Peptides as Metal Chelators: Insights

Under normal circumstances, the concentration of metal ions within cells is tightly regulated through various transporter proteins and storage mechanisms, maintaining a dynamic equilibrium. However, disruptions in the physiological processes that govern cellular homeostasis often lead to disorders associated with metal overload [80].

Metal overload is a common feature across various essential biological ions in organisms. For instance, Wilson's disease is characterized by the abnormal accumulation of copper due to the liver's inability to produce bile that removes excess copper [185, 186]. This dysfunction can lead to severe complications, including cirrhosis, liver failure, and blood disorders, which in some cases may be fatal. Similarly, hypercalcemia presents with

elevated calcium levels in the blood, often resulting from conditions like cancer or abnormal parathyroid hormone production by the parathyroid glands [187]. If left untreated, hypercalcemia can lead to osteoporosis, kidney stones, and even kidney failure, which can ultimately be fatal. In secondary hemochromatosis, iron overload occurs due to repeated blood transfusions, often used to treat other medical conditions [188]. As with other forms of metal overload, excessive iron accumulation can result in cirrhosis, diabetes, and heart disease. Moreover, accumulating non-essential metals, such as aluminium, can lead to similar organ dysfunction or failure.

The technological advances of the past century have significantly increased the presence of aluminium in daily life. Growing evidence of aluminium accumulation in the body [189–191], coupled with its proven neurotoxicity [192, 193], has highlighted the need for effective treatments to address metal overload, such as metal chelation therapies.

Initial treatment strategies for aluminium overload employed iron chelators, based on the similarities between Fe(III) and Al(III). These ions share the same oxidation state, similar ionic radii —Al(III) = 54 pm, Fe(III) = 65 pm—, and a preference for octahedral coordination with six ligands. Both metals are classified as "hard" according to the Hard and Soft Acids and Bases theory, due to their small size and high charge density, and they share common binding trends with ligands, although iron typically exhibits stronger affinities.

Chelating agents like deferiprone —DFP—, originally developed for Fe(III), have also shown promise in chelating Al(III) [81–86]. However, the high cost and side effects associated with these agents have prompted the search for new compounds with a better benefit-to-risk ratio. In this context, the non-proteinogenic mimosine amino acid has emerged as a potential candidate

for chelating high-valent metals, due to its structural similarity to DFP. Mimosine, also known as β-[N-(3-hydroxy-4-oxypyridyl)]-α-amino propionic acid, is found in plants of the Mimosoidae clade (see Figure 3.1). It acts as a reversible inhibitor of DNA replication by tightly binding to metal ions within the active sites of various enzymes, leading to its investigation for a wide range of activities, including antiviral [87], antimicrobial [88], herbicidal [89], anticancer [90, 91], and anti-inflammatory [92] effects.

Despite its potential, there is a lack of experimental studies on the coordination of aluminium with mimosine residues, and insufficient theoretical data to fully understand its impact on polypeptide chains and metal coordination, including its effects on the conformational energy landscape. Recently, biochemical studies have explored the coordination of mimosine-containing peptides to metal cations both experimentally and computationally [194]. Lachowicz and colleagues synthesized mimosine-containing peptides and assessed their coordination with Fe(III). Mujika et al. [195] designed aluminium chelators using mimosine-containing peptides *in silico*, and Urriolabeitia et al. [196] parametrized mimosine residues for Amber force fields, facilitating the study of mimosine-containing peptides with molecular dynamics simulations. In this Chapter, we have developed parameters for a mimosine residue coordinated with the aluminium cation, allowing us to perform both bound and unbound simulations of mimosine-containing peptides. This enables us to evaluate the chelation effects on the conformational energy landscape and gain deeper insights into the potential of mimosine-containing peptides as a chelating agent for aluminium.

3.2 Methodology

3.2.1 Molecular models

In earlier studies, it was demonstrated that *de novo* mimosine-containing peptides consisting of eight residues effectively chelate metal ions such as Al(III) and Fe(III) [194, 195]. These peptides exhibit high conformational flexibility, are capable of forming particular intra-chain interactions without a rigid structure and lack a fixed secondary structure. In this sense, we consider these peptides to present a disordered behaviour similar to IDPs/IDRs. Building on these findings, we selected peptides of this length as our starting point and explored variations in sequence length and composition. Using Chimera [197], we constructed peptides featuring a backbone made of four glycine residues (G), one proline (P), and three additional residues: mimosine (X), phenylalanine (F), or tyrosine (Y) to compare the interactions between these ring-containing residues (see Figure 3.1A). In our designs, the proline residue is positioned before the second X/F/Y residue. To assess the influence of backbone flexibility, we also considered replacing the proline with another glycine. Furthermore, to investigate the effects of chain length, we included a decapeptide with two additional glycine residues on each side of the central mimosine. Regarding the metal-bound systems, we coordinated an aluminium cation to each of the three peptides containing mimosine residues by manually adjusting these mimosine residues into the known optimal octahedral geometry (see Figure 3.1B). All systems were protected by acetylation —ACE— and amidation —NME— in the N-terminus and C-terminus, respectively. Details of all studied systems can be found in Table 3.1 and Figure A.1 of Appendix A.

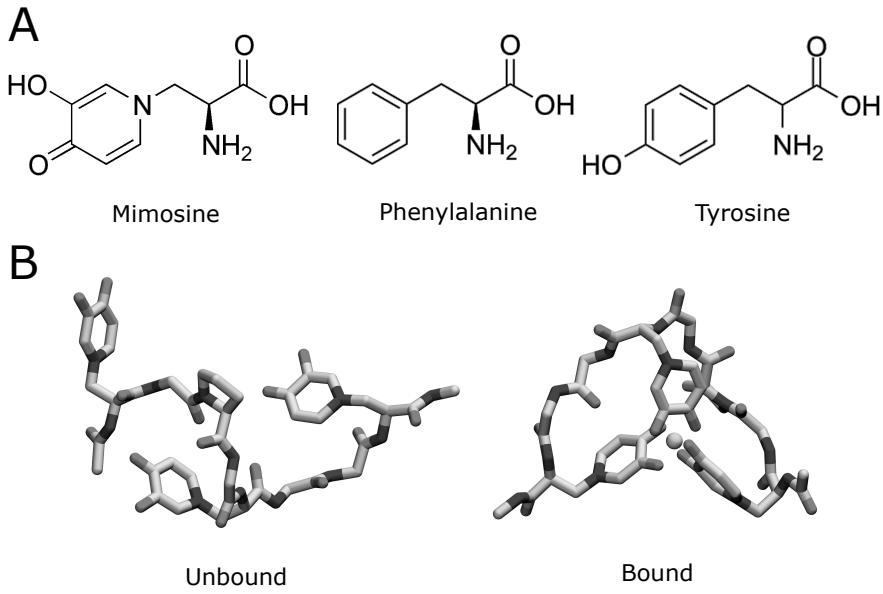


Figure 3.1: A) Ring-containing residues used in this work: Mimosine, Phenylalanine and Tyrosine; B) Comparison between the unbound and bound models with mimosine residues. The aluminium ion is bound in an octahedral geometry to the bidentate mimosines.

Table 3.1: Aluminium-free peptides studied in this Chapter 3. X corresponds to the mimosine residues. The aluminium-bound systems are constructed by coordination of the metal cation to the mimosine residues of the corresponding peptides.

System	**Sequence**
pep^{8}_{mim}	XGPGXGGX
pep^{8}_{phe}	FGPGFGGF
pep^{8}_{tyr}	YGPGYGGY
pep^{8G}_{mim}	XGGGXGGX
pep^{10}_{mim}	XGGGGXGGGX

3.2.2 Force fields

For the aluminium-free peptide simulations, we utilized the Amber03 force field parameters established by Urriolabeitia and colleagues [196]. In the aluminium-bound peptides, mimosine loses the proton from the hydroxyl group in the ring, facilitating bi-dentate binding. To create a bonded model for the aluminium-bound mimosine residue, we employed the MCPB.py tool developed by Li and coworkers [198], which is included with AmberTools [199]. The parametrization of the bonded mimosine residue involved several steps: first (i) we conducted geometry optimization; next (ii) we estimated force constants for bonds, angles, and dihedrals using the Seminario method [200]; and finally (iii) we calculated point charges through the Restrained Electrostatic Potential (RESP) model [201]. The DFT calculations were carried out using the B3LYP/6-31G* method within the Gaussian16 package [202]. All files used for model parametrization, along with the resulting parameters, have been deposited in the OSFHOME online repository[1].

3.2.3 MD simulations

Atomistic MD simulations for all eight systems were conducted using the GROMACS package [154] (version 2020.4) with the Amber03 force field [203], modified to include both bound and unbound mimosine residues. The peptides were placed in a cubic box with a 1 nm buffer from the box edges and were solvated with TIP3P model water molecules [204]. Sodium and chloride ions were added achieving a concentration of 0.1 M.

The equilibrium protocol for all systems began with energy minimization via the steepest descent method, followed by two equilibration runs of

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500 ps each: the first one in the NVT ensemble and the second one in the NPT ensemble. The v-rescale thermostat [205] and the Berendsen barostat [206] were employed during equilibration. Both runs utilized the leap-frog algorithm with a 2 fs time step and maintained a temperature of 300 K, consistent with the production phase. Production runs were performed in the NPT ensemble, applying constraints to all hydrogen bonds using the Linear Constraint Solver (LINCS) algorithm [207]. Electrostatic interactions were calculated using the PME method [208] with a cut-off of 0.8 nm for short-range van der Waals interactions. The Bussi-Donadio-Parrinello thermostat [205] and Parrinello-Rahman barostat [209] —isotropic coupling— were employed for production runs. Frames were saved every 10 ps for data analysis. The production simulations were conducted for 1 μs, which was deemed sufficient for achieving convergence —see Appendix A—.

3.2.4 Characterization of interactions

The peptides analyzed in this study consist solely of glycine, proline, and other ring-containing residues —mimosine, phenylalanine, and tyrosine–. Several geometries exist for the interactions of these rings, with the most relevant ones being: the sandwich geometry, where the rings are stacked face-to-face, nearly overlapping; the parallel displacement geometry, where the rings are also face-to-face but with a horizontal offset; and the T-shaped or Y-shaped geometry, characterized by edge-to-face stacking at approximately a 90º angle between the planes [210–212]. Previous research indicated that the sandwich geometry predominates in mimosine-containing peptides [196].

In this study, we examined ring-ring (R-R) interactions for the unbound systems —$\text{pep}^{8}_{\text{mim}}$, $\text{pep}^{8}_{\text{phe}}$, $\text{pep}^{8}_{\text{tyr}}$, $\text{pep}^{8\text{G}}_{\text{mim}}$, $\text{pep}^{10}_{\text{mim}}$—, focusing on inter-ring distances and orientations. We calculated ring distances by averaging the

distances between the five atoms that form the rings. Ring angles were determined by measuring the angle between the normal vectors of the ring planes. A schematic representation of these interactions is provided in Figure A.2 of Appendix A. Typically, interactions are considered significant at a maximum distance of 0.52 nm and a minimum of 0.36 nm between ring centres [213]. The angle between the planes of the rings can vary widely, from fully planar (0°) to perpendicular (90°). We represented the distribution of snapshots in each type of R-R interaction using pie charts, with a detailed explanation provided in Appendix A.

3.2.5 Entropy calculations

We determined the configurational entropy from the MD trajectories for each of the molecular systems using the MIST method in its parallel implementation included with the PARENT software [182]. We also used a bootstrapping method to compute a distribution of configurational entropies for each system. The configurational entropy was determined using only the dihedral's degrees of freedom. For bound systems, we used only the *backbone dihedrals*. For the aluminium-bound systems, we subtracted the entropy of each system from its corresponding unbound reference, allowing us to compute both the *backbone dihedrals* only and the *backbone and side-chain dihedrals* entropies. This strategy and the bootstrapping details are explained in Appendix A.

3.2.6 Electronic structure calculations

To characterize the coordination shell of the metal centre, we conducted quantum mechanical calculations. On one hand, we optimized the com-

plete aluminium-mimosine structures of pep^{8}_{mim}, pep^{8G}_{mim}, and pep^{10}_{mim} using TURBOMOLE [214], applying the B-P/Def-TZVP method [215] along with the COSMO solvation model [216]. Additional calculations utilized the Quantum Theory of Atoms in Molecules (QTAIM) [217, 218], employing the AIMALL software package [219]. To further understand the interactions within the coordination spheres, we performed an energy decomposition analysis. The total interaction energy was divided into two components.

First, we examined the interaction energy between Al(III) and mimosine residues $\Delta E^{\mathrm{Al}}_{\mathrm{Int}}$, defined as follows,

$$\Delta E^{\mathrm{Al}}_{\mathrm{Int}} = \sum_{i}^{n} E^{\mathrm{Al-Mim}_i}_{\mathrm{Int}}. \tag{3.1}$$

The second interaction energy examined takes into consideration the repulsion between the different mimosine ligands, $\Delta E^{\mathrm{CS}}_{\mathrm{f}}$,

$$\Delta E^{\mathrm{CS}}_{\mathrm{f}} = \sum_{i,j\neq i}^{n} E^{\mathrm{Mim}_i-\mathrm{Mim}_j}_{\mathrm{Int}}. \tag{3.2}$$

The sum of both energy contributions then gives the total interaction energy $\Delta E^{\mathrm{T}}_{\mathrm{Int}}$,

$$\Delta E^{\mathrm{T}}_{\mathrm{Int}} = \Delta E^{\mathrm{Al}}_{\mathrm{Int}} + \Delta E^{\mathrm{CS}}_{\mathrm{f}}. \tag{3.3}$$

All interaction energies are obtained as the sum of all the inter-group atom pair contributions,

$$E^{\mathrm{X-Y}}_{\mathrm{Int}} = \sum_{i}^{n}\sum_{j}^{m} E^{\mathrm{X}_i-\mathrm{Y}_j}_{\mathrm{Int}}, \tag{3.4}$$

where n is the number of atoms of group X and m is the number of atoms of group Y. Similarly, X_i stands for the ith atom of group X and Y_j stands

for the jth number of group Y.

On the other hand, to determine $\Delta G_{\text{aq}}^{\text{Comp}}$ and $\Delta H_{\text{aq}}^{\text{Comp}}$, we conducted DFT optimization calculations of the metal-centre coordination shell — comprising the metal centre and the ligands directly bound to it, truncated at C^{β}— using Gaussian16 software [202]. We employed the ωB97XD functional [134] along with the Def2TZVP basis set [215] and the Polarizable Continuum Model (PCM) of solvation [220]. To verify that the optimized structures represented true minima on the potential energy surfaces, we performed frequency calculations at the same level of theory.

The formation reaction used to define the formation enthalpy and Gibbs free energy is as follows,

$$[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{L}^- \longleftrightarrow \text{AlL}_3 + 6\text{H}_2\text{O}, \tag{3.5}$$

where L is the mimosine ligand.

For this reaction, the equations for the Gibbs free energy and enthalpy of formation are,

$$\Delta H_{\text{aq}}^{\text{Comp}} = H_{\text{aq}}(\text{AlL}_3) + 6H_{\text{aq}}(\text{H}_2\text{O}) - H_{\text{aq}}([\text{Al}(\text{H}_2\text{O})_6]^{3+}) - 3H_{\text{aq}}(\text{L}^-) + \Delta nRT\ln(24.46), \tag{3.6}$$

and,

$$\Delta G_{\text{aq}}^{\text{Comp}} = G_{\text{aq}}(\text{AlL}_3) + 6G_{\text{aq}}(\text{H}_2\text{O}) - G_{\text{aq}}([\text{Al}(\text{H}_2\text{O})_6]^{3+}) - 3G_{\text{aq}}(\text{L}^-) + 6RT\ln(55.34) + \Delta nRT\ln(24.46). \tag{3.7}$$

Since the enthalpies are determined using an ideal gas at 1 atm as the

standard state, the last term in Equation 3.6 corresponds to the volume change due to the transformation from 1 atm to 1 M in solution, where Δn refers to the change in the number of species in the reaction. Equivalently, the term $6RT\ln(55.34)$ in Equation 3.7 is the entropic factor that accounts for the concentration of 55.34 M of six water molecules in liquid water.

The penalty associated with the deprotonation of the hydroxyl group of the mimosine ring ΔG_{deprot} at physiological pH(7.4) was also taken into account following the expression,

$$\Delta G_{\text{aq}}^{\text{Phys}} = \Delta G_{\text{aq}}^{\text{Comp}} + \Delta G_{\text{deprot}} \tag{3.8}$$

where the last term corresponds to the free energy of deprotonation calculated according to the prediction value $\text{p}K_{\text{a}} = 9.05$ —by ChemAxon— and with the following form,

$$\Delta G_{\text{deprot}} = 2.303RT(\text{p}K_{\text{a}} - \text{pH}). \tag{3.9}$$

Moreover, we also considered another equilibrium to examine the competition between the aluminium ion and protons for the binding site of the ligand L. The formation reaction in this case is,

$$[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{LH} \longleftrightarrow \text{AlL}_3 + 3\text{H}_2\text{O} + 3[\text{H}_3\text{O}]^{+}, \tag{3.10}$$

where the proton of the mimosine residue L is given to three water molecules. The enthalpy and Gibbs free energy for this reaction are determined equivalently to Equations 3.6 and 3.7. As highlighted in the work of Jensen *et. al* [221], continuum solvation models like PCM often struggle to accurately predict the solvation energies of small charged ions. To address this issue, we

used an alternative approach to determine the hydronium Gibbs free energy term $G_{\mathrm{aq}}(H_3O^+)$, converting it from the equivalent term determined in the gas phase using the following expression,

$$G_{\mathrm{aq}}(H_3O^+) = G_{\mathrm{gas}}(H_3O^+) + \Delta G_{\mathrm{solv}}(H_3O^+), \tag{3.11}$$

where $G_{\mathrm{gas}}(H_3O^+)$ was determined at the same DFT level as before. The $\Delta G_{\mathrm{solv}}(H_3O^+)$ term is an experimental solvation energy of hydronium (-103.45 kcal/mol) [222]. This $G_{\mathrm{aq}}(H_3O^+)$ term was further corrected [221] to account for the pH following the expression,

$$G_{\mathrm{pH}} = -(pH)mRT\ln(10), \tag{3.12}$$

where physiological pH=7.4 and m is the number of hydronium molecules, in this case, three.

3.3 Results

For the metal-free peptides, our primary focus was on how the introduction of a non-proteinogenic amino acid like mimosine influences the structure compared to similar residues. Our reference system was the $\mathrm{pep}^{8}_{\mathrm{mim}}$ peptide, which included three mimosine residues and was recently synthesized [194]. We compared this peptide to $\mathrm{pep}^{8}_{\mathrm{phe}}$ and $\mathrm{pep}^{8}_{\mathrm{tyr}}$, where mimosine residues were substituted by phenylalanine and tyrosine residues, respectively. Additionally, we examined how variations in the sequence around the mimosine residues impact the conformational landscape of the peptides. This was demonstrated with the $\mathrm{pep}^{8\mathrm{G}}_{\mathrm{mim}}$ system, where the original proline residue was replaced by glycine; and with the $\mathrm{pep}^{10}_{\mathrm{mim}}$ system, which features an

additional glycine on either side of the central mimosine to enhance peptide flexibility. Finally, we analyzed the structural changes upon metal chelation by exploring the systems containing mimosine residues in their aluminium-bound form. We anticipated that aluminium would constrain the peptides and limit their movement, resulting in a significant entropic penalty.

3.3.1 Mimosine residues influence the conformational ensembles of peptides

The influence of mimosine residues on the conformational ensembles of the peptides pep^8_{mim}, pep^8_{phe}, and pep^8_{tyr} was examined by comparing their radius of gyration, contact maps, and Ramachandran plots, as illustrated in Figure 3.2. The distribution of the radius of gyration —shown in Figure 3.2A— was similar across all three systems. However, the mean value for pep^8_{mim}, $\langle R_g \rangle = 0.55 \pm 0.02$ nm, was slightly lower than that of pep^8_{phe}, $\langle R_g \rangle = 0.59 \pm 0.01$ nm, and pep^8_{tyr}, $\langle R_g \rangle = 0.58 \pm 0.01$ nm. Additionally, the maximum R_g^{max} for the mimosine-containing peptides was also observed at slightly shorter values: $R_g^{max} = 0.51$ nm for pep^8_{mim} compared to $R_g^{max} = 0.56$ nm and $R_g^{max} = 0.57$ nm for pep^8_{phe} and pep^8_{tyr} respectively.

This trend suggested stronger intramolecular interactions due to the presence of mimosine, which was further supported by the analysis of the contact maps, shown in Figure 3.2B. These maps revealed a clear increase in ring contacts when mimosine residues were present, in contrast to the contacts observed with tyrosine or phenylalanine. This behaviour indicated more specific ring interactions for the mimosine residues. Conversely, we observed no significant differences in the Ramachandran plots between mimosine residues and either phenylalanine or tyrosine, reported in Figure 3.2C. In all cases, the peptides explored a wide range of Ramachandran space, showing no clear

preference for a specific type of secondary structure, which aligns with the intrinsically disordered nature of these *de novo* peptides.

We also conducted a detailed analysis of the specific interactions between rings in the mimosine peptides and their phenylalanine/tyrosine variants. We report the fraction of simulation frames exhibiting ring stacking interactions in a sandwich geometry for each pair of rings. We denote interactions as Ri-Rj for interactions between ring i and ring j, and Ri-Rj-Rk when all three rings are stacked together. The results of this ring stacking analysis are presented in Figure 3.2D.

For all three peptides, the predominant fraction corresponded to the non-interacting states of the rings. However, notable differences emerged between the systems. In $\text{pep}^8_{\text{mim}}$, approximately 30% of all explored conformations exhibited a sandwich geometry with at least two stacked rings, values significantly higher compared to $\text{pep}^8_{\text{phe}}$ and $\text{pep}^8_{\text{tyr}}$. The latter two displayed much lower occurrences of sandwich geometry ring interactions, with less than 2% for $\text{pep}^8_{\text{phe}}$ and about 6% for $\text{pep}^8_{\text{tyr}}$. Among the sandwich geometry interactions in $\text{pep}^8_{\text{mim}}$, the most common type was R1-R2 (15.5%), followed by R1-R3 (10.6%) and R2-R3 (4.3%). This suggested that the proline residue located between rings R1 and R2 may facilitate the interaction between these two mimosine residues.

3.3.2 Sequence modification can modulate the conformational landscape of mimosine-containing peptides

Next, we examined how subtle changes in the peptide sequence may affect its conformational preferences. We considered two scenarios: (i) substitut-

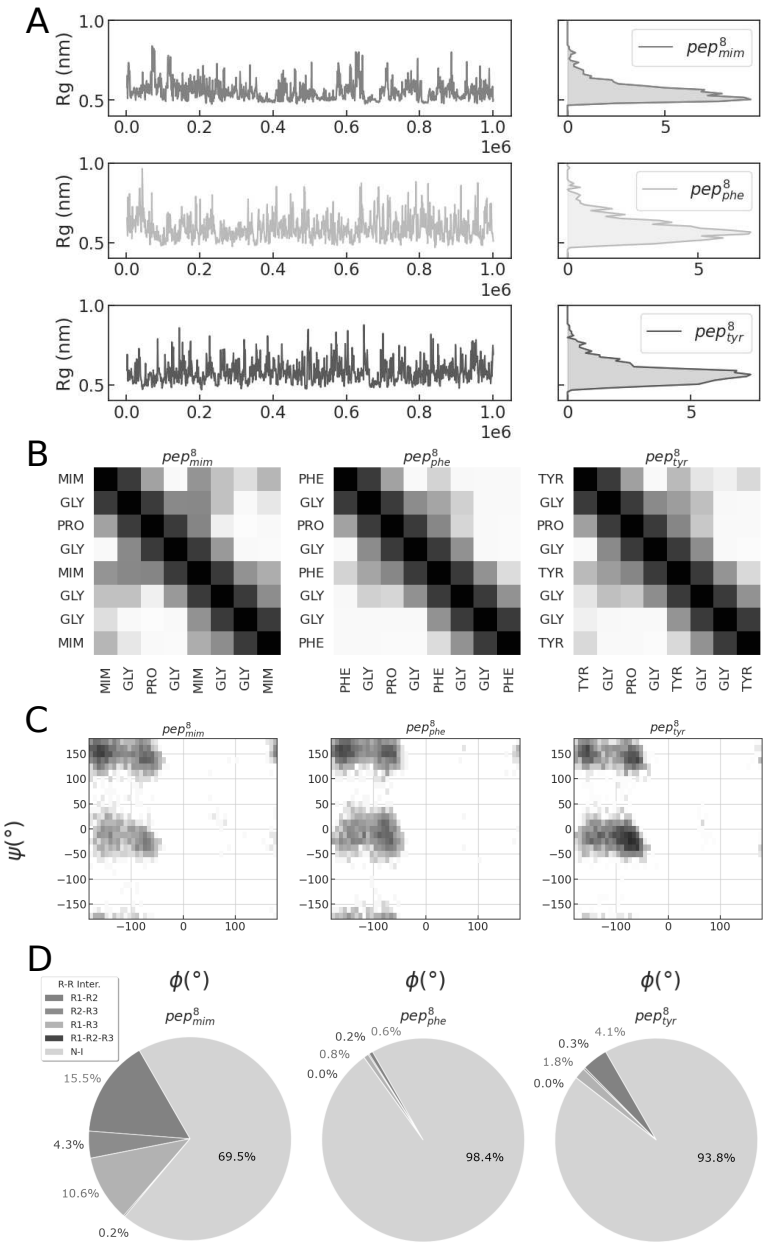


Figure 3.2: Analysis of simulation runs for the mimosine peptide and the phenylalanine and tyrosine variants. (A) Time series data for R_g —left— and distributions from the simulation trajectories —right—. (B) Contact-maps. (C) Density plots for the Ramachandran angles —ϕ and ψ— for the corresponding ring-containing residues of each peptide. (D) Pie charts for the fraction of different ring stacking states.

ing the proline residue with glycine while maintaining the peptide length —pep^{8G}_{mim}—, and (ii) extending the peptide length by adding two more glycine residues —pep^{10}_{mim}—.

For pep^{8G}_{mim}, we found an average radius of gyration $\langle R_g \rangle$ of 0.57 ± 0.01 nm, which was slightly longer than that of the reference peptide, displayed in Figure 3.3A. The addition of two glycine residues —system pep^{10}_{mim}— significantly affected the structure, leading to a larger radius of gyration of $\langle R_g \rangle$ = 0.61 ± 0.01 nm. The maximum of the R_g distribution for these peptides was observed at 0.51 nm and 0.56 nm, respectively.

The contact maps also reflected the effects of sequence variation —see Figure 3.3B—. In pep^{8G}_{mim}, the contacts between mimosine residues were reduced compared to the reference peptide, indicating that the presence of a proline between two mimosine residues was indeed promoting their interaction.

Finally, we determined the influence on ring-stacking interactions, presented in Figure 3.3D. Both pep^{8G}_{mim} and pep^{10}_{mim} showed lower propensities for sandwich geometry ring interactions compared to the original design, with approximately 14% and 22% fractions, respectively. Notably, in pep^{8G}_{mim}, all types of ring interactions were reduced following the substitution of proline with glycine, in the order: R1-R3 (6.7%) > R1-R2 (5.3%) > R2-R3 (1.9%). This indicated that the absence of the proline residue destabilized all ring interactions, not just R1-R2 interactions, suggesting that changes in sequence could impact the entire conformational space of the peptide rather than just the local region interactions. There was a noticeable shift towards R1-R3 as the most likely interaction.

In pep^{10}_{mim}, the elongation of the peptide sequence led to a recovery of some ring interactions. This increase in stacking may be attributed to greater

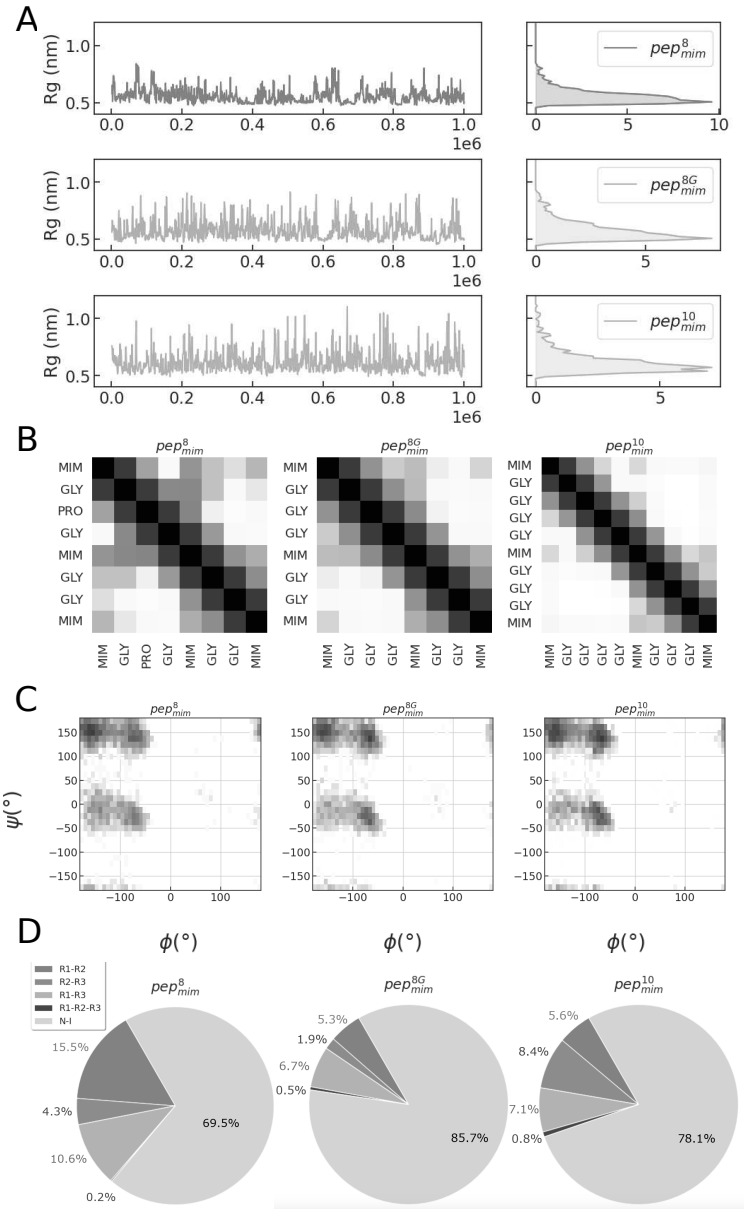


Figure 3.3: Analysis of the mimosine peptide and its variants $\text{pep}^{\text{8G}}_{\text{mim}}$ and $\text{pep}^{\text{10}}_{\text{mim}}$. (A) Time series data for R_{g} —left— and distributions from the simulation trajectories —right—. (B) Contact-maps. (C) Density plots for the Ramachandran angles —ϕ and ψ— for the corresponding ring-containing residues of each peptide. (D) Pie charts for the fraction of different ring stacking states.

flexibility, which facilitates better contact formation between rings. Here, the most likely ring interactions were R2-R3 (8.4%) > R1-R3 (7.1%) > R1-R2 (5.6%). These findings suggested that a three-residue gap between mimosines was more favourable than gaps of two or four amino acids. Short distances, as seen in pep^{8G}_{mim} with only a two-residue gap between R2 and R3, seemed not to permit adequate ring interactions. Conversely, larger gaps, such as the four-residue separation between R1 and R2 in pep^{10}_{mim} also led to reduced stacking interactions.

3.3.3 Metal chelation drastically alters the conformational space of mimosine peptides

We investigated the effects of metal binding on the conformational preferences of our mimosine-containing peptides, shown in Figure 3.4. For aluminium-bound mimosine peptides, we observed significant reductions in polymer dimensions, with $\langle R_g \rangle = 0.53 \pm 0.00$ nm for Al.pep^{8}_{mim} and $\langle R_g \rangle = 0.54 \pm 0.00$ nm for Al.pep^{8G}_{mim} —see Figure 3.4A—. This effect was somewhat less pronounced for the extended backbone peptide, Al.pep^{10}_{mim}, which showed $\langle R_g \rangle = 0.61 \pm 0.00$ nm, indicating that some conformational freedom was regained due to the additional residues.

The contact maps —depicted in Figure 3.4B— demonstrated that the metal centre anchored the mimosine residues at specific positions. This anchoring prevented further collapse of the structure into more compact conformations while simultaneously restricting the chain from extending. The fixed positions of residues due to the metal ion limited the conformational exploration, as evidenced by the Ramachandran space —Figure 3.4C— when compared to their unbound counterparts. For the shorter peptides, Al.pep^{8}_{mim} and Al.pep^{8G}_{mim}, we observed a reduced exploring area rela-

tive to the free peptide. This effect was mitigated in the case of the longer Al.pep$^{10}_{mim}$ peptide. Notably, the β-sheet region was less frequently accessed by the mimosine residues, further highlighting the constraints of the metal on extended conformations.

3.3.4 Effects of sequence length, composition, and metal binding in the configurational entropy of peptides

We assessed the impact of structural constraints on configurational entropy across different systems. Figure 3.5 presents the values of $T\Delta S$ for various peptides. As anticipated from the similar Ramachandran plots —Fig. 3.2D—, the presence of mimosine residues did not significantly affect the configurational entropy of the unbound peptides as evidenced in Figure 3.5A. Although the general distribution of ring interactions was notably higher for pep$^{8}_{mim}$, the resulting entropy difference was approximately 0.5 kcal/mol when comparing pep$^{8}_{phe}$ and pep$^{8}_{tyr}$. Thus, from an entropic perspective, the inclusion of mimosine residues in a sequence was only slightly less favourable than that of standard amino acids. Conversely, the substitution of proline with glycine had a more pronounced impact on configurational entropy, resulting in a substantial entropy gain of 2.6 kcal/mol —see Fig. 3.5, cyan —, aligning well with the previously noted changes in conformational flexibility.

We evaluated the effects of metal binding on configurational entropy by calculating it using *backbone dihedrals* DOF only and *all dihedrals* DOF —this is, the backbone and the side chains dihedrals—. Results are depicted in Figure 3.5C and D. In both scenarios, we observed a significant entropic penalty upon metal binding, ranging from –4.6 kcal/mol to –8.6 kcal/mol for backbone dihedrals alone, and increasing to about –10.5 kcal/mol to –16.5 kcal/mol when considering all dihedrals.

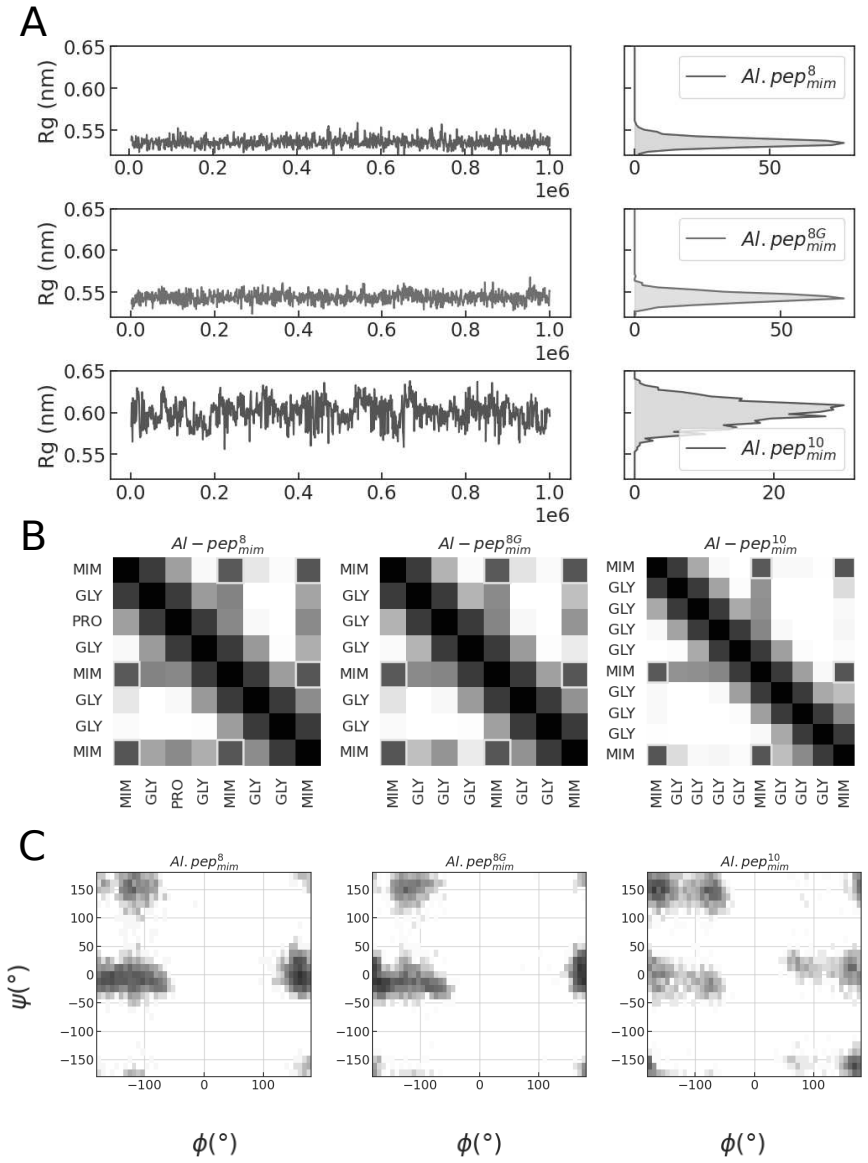


Figure 3.4: Analysis of the aluminium-bound mimosine peptides. (A) Time series data for R_g —left— and distributions from the simulation trajectories —right—. (B) Contact-maps. The metal coordination is highlighted in cyan. (C) Density plots for the Ramachandran angles —ϕ and ψ— of the corresponding ring-containing residues of each peptide.

There were notable differences among the three peptides, suggesting that sequence variation could modulate and reduce the entropic penalty associated with metal binding. The entropic penalty was larger for the shorter peptides, Al.pep$^{8}_{\mathrm{mim}}$ and Al.pep$^{8\mathrm{G}}_{\mathrm{mim}}$, compared to the longer Al.pep$^{10}_{\mathrm{mim}}$, where the metal interacted with residues that were more distantly separated. This indicated a partial recovery of backbone movement due to the additional glycine residues introduced between the mimosines. This effect was significant, as Al.pep$^{10}_{\mathrm{mim}}$ exhibited nearly half the entropic penalty for metal binding compared to the shorter peptides.

Additionally, if we compare the entropy contributions from *all dihedrals* and *backbone dihedrals* DOF, we can determine the entropy associated with the side chains only. We found that the trends for the *backbone dihedrals* and *all dihedrals* were consistent, but the values were offset by approximately 8 kcal/mol. This suggested that the primary contribution to entropy arose from backbone dihedrals, while side-chain contributions remained comparable across systems. Notably, pep$^{8}_{\mathrm{mim}}$ experienced a slightly lower entropic penalty than its proline-free counterpart, reinforcing the notion that backbone effects were the dominant factor.

3.3.5 Determination of the metal binding affinity using quantum calculations

To evaluate the metal-binding capacity of the mimosine coordination environment and compare it with established bio-chelation systems, we utilized quantum mechanical calculations with simplified models. This approach allowed us to determine whether the coordination shell formed by mimosine residues could effectively compete with well-known bio-chelators, such as citrate, for metal binding.

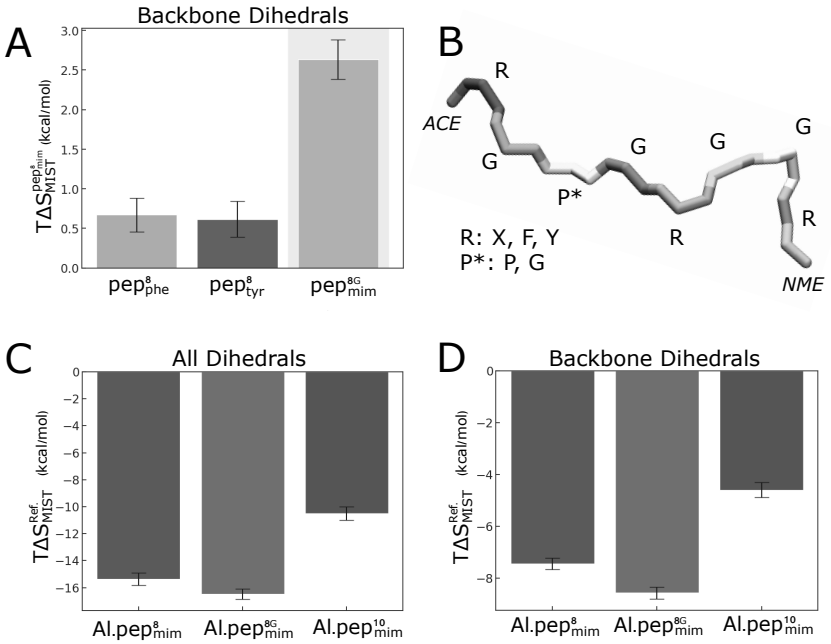


Figure 3.5: Configurational entropy $T\Delta S_{\mathrm{MIST}}$ values calculated with reference to the entropy of the $\mathrm{pep}^{8}_{\mathrm{mim}}$ system. Positive values imply that the structure is favoured entropically, while negative values imply an entropic penalty. (A) $T\Delta S^{\mathrm{pep}^{8}_{\mathrm{mim}}}_{\mathrm{MIST}}$ values for unbound systems with equivalent backbone. (B) Representation of the backbone of 8-residue peptides as used in the entropy calculation. *R*: mimosine (X), phenylalanine (F), or tyrosine (Y). *P**: proline (P) that gets substituted by a glycine (G); (C, D) $T\Delta S^{\mathrm{REF}}_{\mathrm{MIST}}$ values for systems upon metal-binding. The corresponding unbound systems were used as references —REF— for in each case.

Table 3.2: Results of interaction energies of the coordination shells obtained from QTAIM calculations. All energies are shown in kcal/mol.

System	$\Delta E_{\text{Int}}^{\text{Al}}$	$\Delta E_{\text{f}}^{\text{CS}}$	$\Delta E_{\text{Int}}^{\text{T}}$
$\text{Al.pep}_{\text{mim}}^{8}$	-1881.4	399.2	-1482.3
$\text{Al.pep}_{\text{mim}}^{8\text{G}}$	-1884.2	402.2	-1482.1
$\text{Al.pep}_{\text{mim}}^{10}$	-1888.4	406.7	-1481.7

We calculated the *local* metal binding affinity —$\Delta G_{\text{aq}}^{\text{Phys}}$— using DFT. This free energy reflects the favouring in the formation of the complex between the three mimosine residues and aluminium —AlL_3—. The structures of $\text{Al.pep}_{\text{mim}}^{8}$, $\text{Al.pep}_{\text{mim}}^{8\text{G}}$, and $\text{Al.pep}_{\text{mim}}^{10}$ were optimized using TURBOMOLE. The resulting structures are depicted in Figure 3.6.

We defined reduced models of the coordination spheres —see Figure 3.7— and estimated the interaction energies of these models using QTAIM. The results, summarized in Table 3.2, indicated that all coordination spheres exhibited nearly identical interaction energies between the mimosine residues and the metal. Furthermore, they showed equivalent spatial conformations, suggesting that the stabilization from the interactions in the coordination shells did not vary significantly across the three peptide variants.

Next, we calculated the formation energies of the AlL_3 complex associated with various substitution reactions —see reactions 3.5 and 3.10—. For this, the coordination sphere depicted in Figure 3.7 was fully optimized using quantum mechanical methods.

Building on a previous work [223], we considered two types of equilibria. In the first, we assessed the complexation energies from deprotonated mimosine residues, incorporating a correction for deprotonation at pH 7.4 based

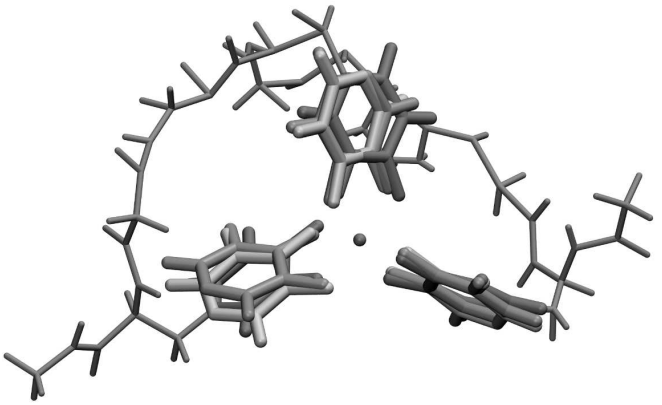


Figure 3.6: Overlapping of the three mimosine rings for the bound systems $Al.pep^{8}_{mim}$, $Al.pep^{8G}_{mim}$ and $Al.pep^{10}_{mim}$, as optimized with TURBOMOLE. Structures are centred on the metallic cation and only the backbone of one is displayed for visual guidance —grey —. The colour code is equivalent to the previous usage.

on the pK_a of mimosine. In the second equilibrium, the mimosines were treated as protonated, with the quantum reaction energies reflecting the deprotonation of these residues, resulting in the formation of three hydronium ions. The results of these calculations are presented in Table 3.3.

In both scenarios, all reaction energies were highly negative, indicating a strong affinity of mimosine for aluminium. These values were compared to the reaction's free energies for the well-known metal chelation reactions, such as $[Al(H_2O)_6]^{3+} + 3LH \rightarrow AlL_3 + 3H_2O + 3H_3O^+$, for chelators like deferiprone –84.0 kcal/mol, citrate –91.8 kcal/mol, and EDTA –121.9 kcal/mol —from [223]—. Although these energies were calculated using a different theoretical framework, our findings suggest that the binding of mimosine to aluminium is expected to be very strong, further supporting the potential of mimosine peptides as effective metal chelators.

Table 3.3: Results of the local affinity energies evaluated by quantum calculations according to the formation energies of AlL_3. Calculations were done at ωB97XD/Def2TZVPP level of theory and PCM solvation. ΔG_{aq}^{Phys} values for Reaction 3.10 can be compared to those of well-known chelators reported by [223]. Energies in kcal/mol.

Reactions	ΔH_{aq}^{Comp}	ΔG_{aq}^{Comp}	ΔG_{aq}^{Phys}
Reaction 3.5	-144.8	-154.9	-148.1
Reaction 3.10	-54.6	-71.7	-102.2

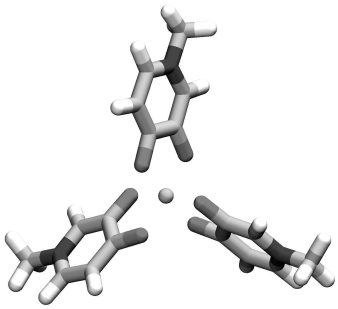


Figure 3.7: AlL_3 coordination sphere shared among all the bound systems. The mimosine ligands L are truncated at the C^{β}.

3.4 Discussion: To be considered in the design of adequate mimosine-containing peptide chelators

The exploration of non-standard peptides with biological activities in medicine is an active research area [224]. Mimosine compounds, known for their metal-chelating abilities, have shown significant antimicrobial properties. Thus, investigating mimosine-containing peptides, where the peptide chain can influence metal affinity, is essential for designing more bioactive peptides. Identifying the most effective chelating peptide requires pairing side chains with

high metal affinity with a suitable polypeptide backbone. A key challenge is ensuring that the peptide backbone can maintain the correct geometry for the chelating groups in the coordination shell around the metal. This constrained geometry incurs a configurational entropy penalty —especially in flexible IDP/IDR-like systems such as these *de novo* mimosine-containing peptides—, making it difficult to evaluate peptide-metal binding properties [225].

Optimizing the binding efficiency of mimosine-containing peptides involves two main aspects:

1. Selection of High-Affinity Side Chains: Using the 20 standard amino acids, side chains such as histidine, cysteine, or aspartate can provide strong metal-binding properties. Non-proteinogenic amino acids like mimosine broaden these possibilities, but their inherent metal affinity must be carefully assessed, potentially using a cluster-continuum quantum mechanical approach [225].

2. Conformational Compatibility of the Peptide Chain: The spatial arrangement around the metal may significantly reduce flexibility, translating into a configurational entropy penalty. This is especially relevant for the present IDP/IDR-like peptides that do not adopt fixed stable structures —such as α-helices or β-sheets— and remain disordered. The entropic penalty can be estimated through long-timescale classical MD simulations.

Our quantum calculations revealed that peptides containing three mimosine residues —Al.pep$^{8}_{\mathrm{mim}}$, Al.pep$^{8\mathrm{G}}_{\mathrm{mim}}$, and Al.pep$^{10}_{\mathrm{mim}}$— exhibit similar coordination environments around aluminium. All peptide chains maintain sufficient flexibility for interaction within an ideal octahedral configuration.

Furthermore, the binding energies are comparable to well-established aluminium chelators like deferiprone and citrate, although slightly lower than EDTA [223]. Thus, these peptides demonstrate promise as effective chelators, particularly due to the local affinity provided by the three mimosine residues [194]. When a mimosine-containing peptide binds to aluminium, the mimosine side chains become highly ordered, forming a hexa-coordinated shell around the metal. Each mimosine ring engages in a bi-dentate binding mode to aluminium, resulting in significant entropic penalties ranging from 10-16 kcal/mol. While these penalties cannot be directly compared to our quantum calculations, they align with the established configurational entropy loss of 1-2 kcal/mol per residue in protein folding [226, 227]. Importantly, these penalties are considerably lower than the local quantum affinities, suggesting they are insufficient to inhibit metal binding. Some of these peptides have been shown to bind to iron experimentally [194], a metal with similar chelation properties to those of aluminium. Our findings indicate that the entropic penalty for metal binding in these peptides can be adjusted through variations in chain length and composition, without compromising the metal affinity of the mimosine coordination shell. This strategy could enhance the efficiency of this family of metal chelators.

To better understand the origin of configurational entropy losses upon metal binding, we evaluated the free and bound states of the peptide. Systems with free mimosine residues encourage ring stacking interactions in a sandwich arrangement, a phenomenon not observed in analogous phenylalanine or tyrosine-based peptides [196]. This difference stems from the larger dipoles of the mimosine rings compared to standard amino acids. However, these changes result in only a modest reduction in configurational entropy —around 0.5 kcal/mol—. The presence of proline in the peptide

backbone seems to have a more pronounced effect —as it restricts rotational flexibility—, promoting more compact conformations and facilitating interactions between mimosine rings. Our results demonstrate that substituting proline with glycine increases the configurational entropy by 2.5 kcal/mol, five times greater than replacing mimosine with phenylalanine or tyrosine. This suggests that backbone torsion effects are dominant in the entropy differences observed for the free peptide. Modulating configurational entropy through proline substitution may be a valuable design strategy for therapeutic peptides.

There is a significant entropic penalty that can also be modulated by variations in sequence length. Longer peptides exhibit a reduced overall entropic penalty due to compensatory effects from regions outside the metal centre, which balance the restrictions imposed by metal binding. Additionally, a crucial aspect is balancing peptide length with constraints for biological assimilation [228]. While our findings indicate that longer peptides can reduce the entropic penalty related to metal binding, their size must be managed to ensure effective biological uptake and compatibility. Therefore, optimal peptide design requires careful consideration of these trade-offs.

3.5 Conclusions

In this study, we conducted a series of 1 μs MD simulations to investigate the conformational ensembles of mimosine-containing peptides and how these change upon binding to the multivalent metal Al(III). Mimosine, a non-proteinogenic amino acid, is recognized for its multitude of beneficial properties due to its ability to chelate metals, making our findings significant for peptide chelator design and optimization. To facilitate this, we developed

parameters for aluminium-mimosine bonded models based on quantum calculations. Our simulations allowed us to evaluate the configurational entropy of these peptides, a crucial yet challenging thermodynamic property. This evaluation is essential for estimating the entropic penalty associated with the binding of IDPs/IDRs-like peptides to a metallic centre. We also calculated the inherent binding affinities to aluminium expected from the coordination environment formed by three mimosine residues around the metal.

Our results indicate a strong interaction between aluminium and the coordinating mimosine, sufficiently compensating for the entropy loss associated with this metal binding, which is comparable to the entropic changes observed during protein folding —1-2 kcal/mol per residue—. Additionally, we found that different peptides can achieve similar metal-binding affinities through comparable coordination shells. However, the entropic penalties associated with various peptides differ, suggesting that modulating the configurational entropy penalty is a promising strategy for developing peptides with enhanced overall metal-binding affinities. Incorporating proline residues and optimizing sequence length can facilitate ring-stacking interactions in the free peptide, and lastly reduce the entropic penalty upon binding. The length of the peptide is also a critical factor in chelator design, requiring careful balancing to minimize the entropic penalty while maintaining the potential for biological assimilation. Finally, our quantum mechanical calculations of formation energies indicate that these peptides are promising metal-binding chelators, with energies comparable to those of well-known aluminium chelators like deferiprone and citrate.

4

Al(III) & Neuro-peptide Monomers

4.1 Effect of Metal Binding on Peptide Monomers

The general introduction of this thesis has already highlighted the growing interest in IDPs/IDRs and their critical role in disease pathology. The abnormal aggregation of IDPs has been identified as a primary driver of diseases such as AD, PD, and HD [49, 52, 53, 93, 94]. Despite significant research efforts, the lack of effective clinical treatments for these disorders underscores the urgent need for a more precise understanding of IDPs/IDRs conformational ensembles and the factors driving their aggregation.

Over time, extensive studies on the protein aggregation process have revealed its multi-causal nature and the interdependence of the contributing factors [229]. Rather than addressing this inherently multifactorial problem

directly, the scientific community has often sought to simplify it by proposing hypotheses based on specific experimental and theoretical observations. This approach allows researchers to isolate specific steps within the aggregation process and evaluate them under controlled, manageable conditions.

One prominent hypothesis is the Metal Ion Hypothesis (MIH), which links metal ions to protein aggregation [96, 98, 99, 230]. Experimental evidence points to a correlation between metal accumulation in specific brain regions, increased protein aggregation in these regions, and the ability of various metals to promote aggregation or influence related factors. According to the MIH, the presence of endogenous or exogenous unpaired metallic ions may act as a key driver of protein aggregation. However, the precise mechanism and timing of metal-protein interactions remain unclear. Two potential pathways have been proposed, shown in Figure 4.1:

1. Metal-dependent aggregation —Figure 4.1A—: In this pathway, the aggregation-related IDPs are unable to aggregate in the absence of significant metal concentrations. Once the metal concentration surpasses a threshold, metal coordination with IDPs facilitates aggregation, allowing multiple chains to interact via metal complexes and promoting fibril growth. Here, metal ions play an essential role as catalysts for aggregate formation.

2. Metal-assisted aggregation —Figure 4.1B—: In this alternative pathway, aggregation-related IDPs can aggregate independently of metal ions, although the mechanisms enabling this aggregation remain poorly understood. However, this spontaneous aggregation is insufficient to sustain fibril growth, leaving the system in a dynamic equilibrium between free forms and small aggregates —oligomers—. When metal concentrations rise, metal coordination with existing aggregates stabi-

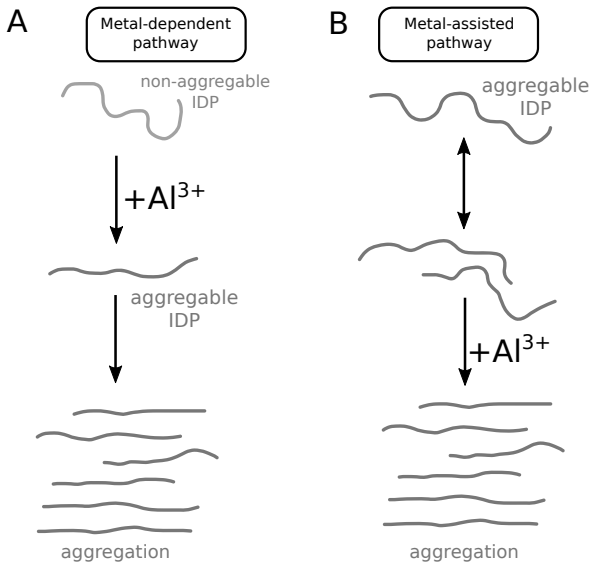


Figure 4.1: The two potential pathways for the metal-protein aggregation in the MIH. A) the metal-dependent aggregation. B) the metal-facilitated aggregation.

lizes them, shifting the equilibrium towards fibril formation. In this case, metal ions play a secondary stabilizing role.

Moreover, emerging evidence highlights the significant role of phosphorylation and hyperphosphorylation in IDP aggregation [231–237]. MIH integrates this relevant condition by assuming that the metal-IDP interactions occur by the active binding of the metal ion to phosphorylated residues.

Among the metals implicated in the MIH, those commonly found in the brain include Zn(II), Cu(II), Fe(II)/Fe(III), Ni(II), and Al(III). Aluminium, in particular, stands out due to its tendency to accumulate in the brain, lack of biological functionality, and well-documented neurotoxic effects [189–193, 238–247]. Based on prior expertise within our research group, Al(III) was chosen as the focal metal ion for this study.

To investigate protein aggregation in AD in the presence of Al(III) ions, we selected a short thirteen-residue peptide from the intrinsically disordered C-terminal region of the Human Mid-size Neurofilament (NF-M) protein, which constitutes an IDR. This has been previously used to evaluate protein aggregation regarding AD [97, 248]. Besides being intrinsically disordered, the C-terminal region presents multiple phosphorylable residues, representing a critical phosphorylation site implicated in protein aggregation in neurodegenerative diseases [249]. Studying a peptide small enough for functional simulations, yet capable of emulating both phosphorylation and hyperphosphorylation states, provides a practical and effective strategy for exploring the mechanisms underlying protein aggregation in the presence of metals.

4.2 Methodology

4.2.1 Model systems

We focused on the thirteen residue peptide KSPVPKSPVEEKG, repeated several times serially in the IDR —C-terminal region— of the Human Mid-size Neurofilament protein —NF-M, Uniprot ID: P07197—. The sequence is considered a major multi-phosphorylation site *in vivo* and has been used to evaluate the metal-ion coordination aggregation promotion previously [97, 248]. Moreover, the small repeated sequence KSPV is present in other proteins related to pathological aggregation in AD, such as Tau [250]. The aluminium (III) cation was bound to the peptide through phosphorylated serines —SerP— or glutamates —Glu—, resulting in three different variants according to the phosphorylation degree of the peptide. We classified the systems into three groups:

(i) The S2 group, where the phosphorylation was only in the serine at position 2 —SerP2—.

(ii) The S7 group, where the phosphorylation was only in the serine at position 7 —SerP7—.

(iii) The S2S7 group, where both serines were phosphorylated — hyper-phosphorylated —.

Figure 4.2A shows a schematic representation of the variants. In Table 4.1 we enumerate all the systems considered. The full nomenclature of the systems is: $NF13Al^{Group}_{Coord}$, where NF13 designates the peptide; Al presence indicates the binding to aluminium; superscript indicates the membership group; and subscript indicates the specific coordination to aluminium. However, we sustain the explanations in the text using the [Group].[Index] code for the sake of clarity —Index defined in the Table 4.1—. Structures were built from scratch using Molden [251], Chimera [197] and VMD [252]. The peptides were manually adjusted to fit the aluminium cation in the right place. Peptides are acetylated in the N-terminus and amidated in the C-terminus.

4.2.2 Molecular simulations

MD simulations were carried out with GROMACS package [154] (version 2018) and CHARMM36 forcefield [253]. Non-bonded parameters of Al(III) from Faro *et al* [254] were included. Dodecahedral boxes were used leaving a 1 nm distance to the box edge and the CHARMM-TIP3P [255] water model was used for solvation. Sodium and chloride ions were added to neutralize the charge when needed. The simulation protocol was equivalent for all systems, with an initial energy minimization using the steepest descent method, followed by two 500 ps equilibrations, in the NVT and NPT en-

Group	System	Index
Reference	NF13	1
	$NF13^{S2}$	2
	$NF13^{S7}$	3
	$NF13^{S2S7}$	4
S2	$NF13Al^{S2}_{S2}$	5
	$NF13Al^{S2}_{S2E10}$	6
	$NF13Al^{S2}_{S2E11}$	7
	$NF13Al^{S2}_{S2E10E11}$	8
	$NF13Al^{S2}_{E10E11}$	9
S7	$NF13Al^{S7}_{S7}$	10
	$NF13Al^{S7}_{S7E10}$	11
	$NF13Al^{S7}_{S7E11}$	12
	$NF13Al^{S7}_{S7E10E11}$	13
	$NF13Al^{S7}_{E10E11}$	14
S2S7	$NF13Al^{S2S7}_{S2}$	15
	$NF13Al^{S2S7}_{S2E10}$	16
	$NF13Al^{S2S7}_{S2E11}$	17
	$NF13Al^{S2S7}_{S2E10E11}$	18
	$NF13Al^{S2S7}_{S7}$	19
	$NF13Al^{S2S7}_{S7E10}$	20
	$NF13Al^{S2S7}_{S7E11}$	21
	$NF13Al^{S2S7}_{S7E10E11}$	22
	$NF13Al^{S2S7}_{S2S7}$	23
	$NF13Al^{S2S7}_{S2S7E10}$	24
	$NF13Al^{S2S7}_{S2S7E11}$	25
	$NF13Al^{S2S7}_{E10E11}$	26

Table 4.1: List of peptides considered in this chapter. The first column indicates the membership group in terms of the phosphorylation state. The second column indicates the system. The last column indicates the index used to refer to the system in the main text, combined with its membership (ej: S2S7.25). Selected systems for the discussion are highlighted in boldface.

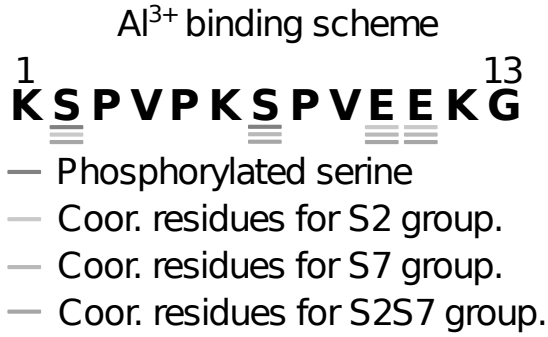


Figure 4.2: Schematic representation of the different systems considered in this study by coordination and group.

sembles respectively, with the leap-frog algorithm using 2 fs time step and 300 K temperature. Production was done in the NPT ensemble, running 1 μs and extended up to 2 μs when necessary to achieve convergence. All H-bonds were constrained using LINCS [207]. Electrostatic interactions were evaluated using the PME method [208] with a 0.8 nm cut-off for short-range Van der Waals interactions. The Bussi-Donadio-Parrinello thermostat [205] and Berendsen barostat [206] were used. We saved frames every 10 ps for data analysis.

4.2.3 Entropy and electronic structure calculations

Entropy was calculated using the MIST scheme. Similar to the previous Chapter 3, we used only the soft DOF from the backbone to determine the configurational entropy of the systems, along with a bootstrapping method. Details about soft entropy and bootstrapping are described in Appendix A. Calculation of the configurational entropy with all DOF is shown in Appendix B.

In an initial step, the inherent stabilities of each aluminium coordination

shell were determined based on a previous work [256]. We considered the First Coordination Shell (FCS) and the Second Coordination Shell (SCS). In the case of the FCS, only the side chains of the residues directly bound to aluminium were considered. For the SCS, we introduced the side chain of the residues interacting with the FCS via salt-bridge analysis, following the considerations of Dudev *et al* [65]. Calculations were done using DFT, with the CAM-B3LYP functional [131] based on a previous benchmark with aluminium [257], D3 Grimme's version of empirical dispersion corrections [258] and 6-31++G(d,p) basis set. The implicit solvent model was used through the PCM [220] implemented in Gaussian09 [202].

Stability was evaluated considering the following ligand substitution reaction for the FCS,

$$[\mathrm{Al(H_2O)_6}]^{3+}_{(\mathrm{aq,1M})} + \mathrm{mL}^{\mathrm{k}}_{(\mathrm{aq,1M})} \longrightarrow [\mathrm{Al(H_2O)_{6-m}L_m}]^{3+\mathrm{k}}_{(\mathrm{aq,1M})} + \mathrm{mH_2O} \quad (4.1)$$

where L is the monodentate ligand —SerP or Glu—, m is the number of monodentate ligands and k is the total charge of all monodentate ligands involved.

The corresponding formation enthalpies were then calculated as,

$$\Delta H_{\mathrm{f}} = H_{\mathrm{aq}}(\mathrm{Al(H_2O)_{6-m}L_m}) + mH_{\mathrm{aq}}(\mathrm{H_2O}) - H_{\mathrm{aq}}(\mathrm{Al(H_2O)_6}) - H_{\mathrm{aq}}(\mathrm{L}) + \Delta nRT\ln(24.46), \quad (4.2)$$

while the free energies were determined as,

$$\Delta G_{\mathrm{f}} = G_{\mathrm{aq}}(\mathrm{Al(H_2O)_{6-m}L_m}) + mG_{\mathrm{aq}}(\mathrm{H_2O}) - \mathrm{G_{aq}(Al(H_2O)_6)} - G_{\mathrm{aq}}(\mathrm{L}) + \Delta nRT\ln(24.46) + mRT\ln(55.34). \quad (4.3)$$

In these expressions, the term $\Delta nRT\ln(24.46)$ accounts for the volume change from the considered ideal gas at 1 atm to 1 M in solution, and the term $mRT\ln(55.34)$ accounts for the concentration of water as a ligand in liquid water as a solvent, which is 55.34 M.

When considering the SCS, only interactions with the lysines —Lys— were extracted from the salt-bridge analysis. The interaction between ligands from the FCS and SCS was considered already formed previously to the binding to the aluminium. The resultant reaction is then,

$$\begin{aligned}[\mathrm{Al(H_2O)_6}]^{3+}_{(\mathrm{aq,1M})} + \mathrm{mL}^{\mathrm{k}}_{(\mathrm{aq,1M})} + \mathrm{n(L@Lys)}^{\mathrm{q}}_{\mathrm{aq,1M}} \\ \longrightarrow [\mathrm{Al(H_2O)_{6-(m+n)}(L)_m(L@Lys)_n}]^{3+\mathrm{k}+\mathrm{q}}_{(\mathrm{aq,1M})} + (\mathrm{m+n})\mathrm{H_2O}, \quad (4.4)\end{aligned}$$

where L is the FCS monodentate ligands —SerP or Glu— that do not interact with lysines, m is the number of L monodentate ligands, k is the total charge of all FCS monodentate ligands involved, L@Lys is the monodentate ligands that interact with lysines, n is the number of L@Lys monodentate ligands and q is the total charge of all L@Lys monodentate ligands. The corresponding formation enthalpies ΔH_f and Gibbs free energies ΔG_f of the SCS systems were calculated analogously to Equations 4.2 and 4.3. The examined FCS and SCS of the systems are shown in Figures 4.3 and 4.4.

Additionally, DFT calculations using TURBOMOLE [214] with B3LYP functional, m3 dispersion, and Def-SV(P) basis set were carried out to evaluate the stabilization of the whole peptide-metal complexes. Determining the stabilization energy of the complexes by considering all possible conformations is computationally prohibitive. Therefore, our strategy consisted of clustering the trajectories of the systems regarding the backbone conformations. For each cluster, we selected an arbitrary system and performed

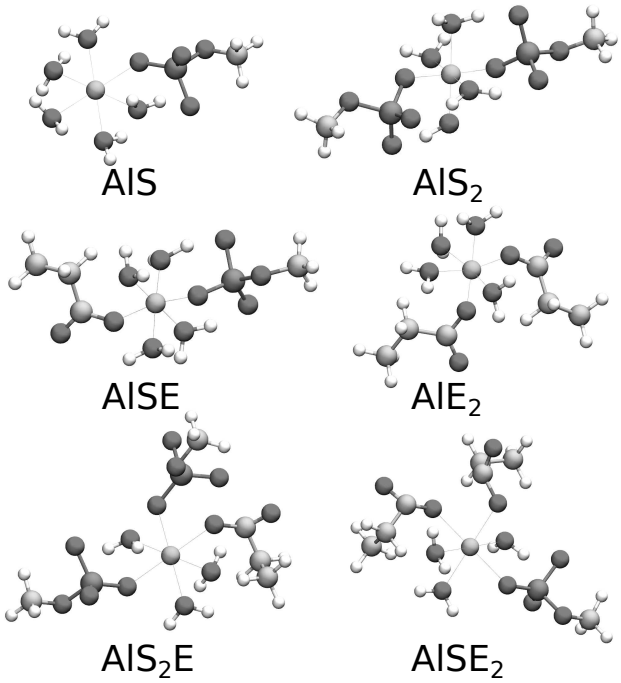


Figure 4.3: First Coordination Shells —FCS— structures. Binding of the Al^{3+} with the peptide occurs through phosphoserines —SerP2 or SerP7— or glutamates —Glu-10 or Glu-11—.

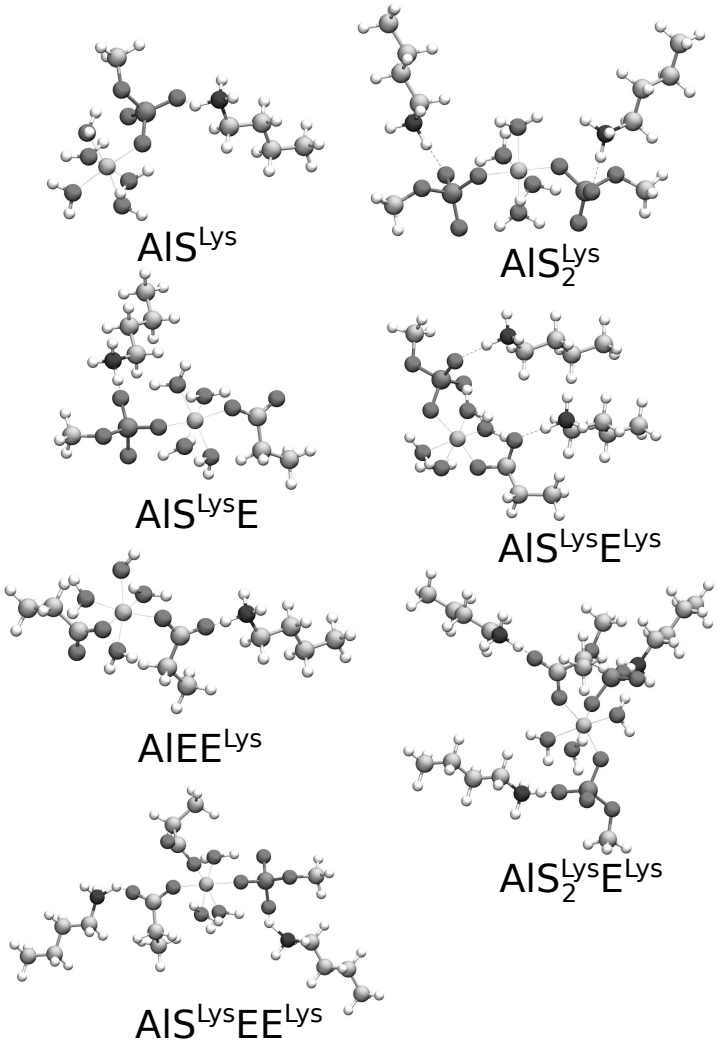


Figure 4.4: Second Coordination Shells —SCS— structures. Addition of salt-bridge forming lysine —Lys— residues, interacting with the FCS.

the single-point DFT calculations. The resultant mean *interaction* energy $\langle \Delta E \rangle$ was determined then by computing the weighted arithmetic mean of all clusters for each system, as shown in the following equation,

$$\langle \Delta E \rangle = \frac{\sum_{i=1}^{n} w_i E_i}{\sum_{i=1}^{n} w_i} \tag{4.5}$$

where E_i is the interaction energy of the i-th cluster, w_i is the contribution of each cluster depending on its population, and n is the number of clusters. With this strategy, the resultant interaction energy is more accurate to the energy profile and describes the behaviour of the disordered peptide with higher fidelity. However, there is still a limitation: as the selection of the conformations that represent a cluster for each system is arbitrary, the choice of the most stable conformation inside each cluster cannot be ensured. One should trust that, since structures corresponding to the same cluster are similar, the picked one is a good representative of the whole group.

4.3 Results

4.3.1 Inherent stabilities of the coordination shells

To evaluate the inherent stability of the coordination shells, we carried out QM calculations with FCS and SCS, determining the formation enthalpies ΔH_f and Gibbs free energies ΔG_f according to Equations 4.2 and 4.3. The Table 4.2 shows the systems considered. In Figure 4.5 these energetic contributions are shown.

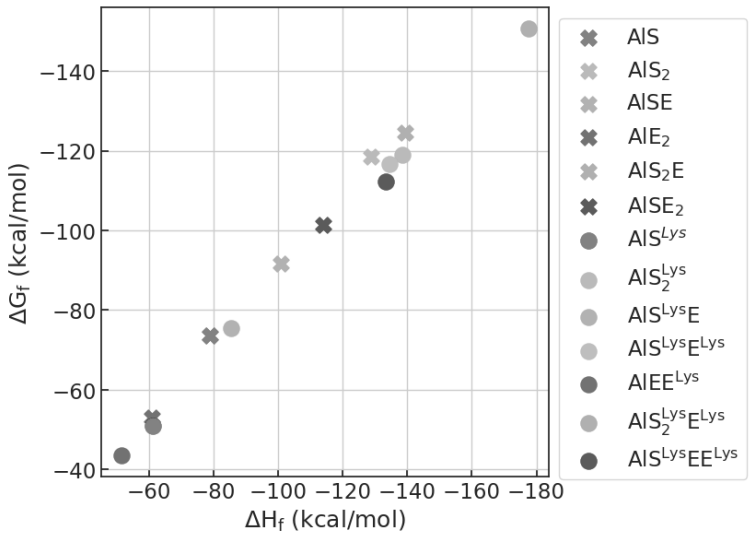
The energies of the FCS models exhibited the expected increase in the stabilities of the systems with the increased number of negative charges around the aluminium. The less stable compound corresponded to AlE_2 with

Coord. Shell	System	Molecular Formula
FCS	AlS	$[\ Al(H_2O)_5(SerP)]^+$
	AlS_2	$[Al(H_2O)_4(SerP)_2]^-$
	AlSE	$[Al(H_2O)_4(SerP)(Glu)]^0$
	AlE_2	$[Al(H_2O)_4(Glu)_2]^+$
	AlS_2E	$[Al(H_2O)_3(SerP)_2(Glu)]^{2-}$
	$AlSE_2$	$[Al(H_2O)_3(SerP)(Glu)_2]^-$
SCS	AlS^{Lys}	$[Al(H_2O)_5(SerP)(Lys)]^{2+}$
	AlS_2^{Lys}	$[Al(H_2O)_4(SerP)_2(Lys)_2]^+$
	$AlS^{Lys}E$	$[Al(H_2O)_4(SerP)(Glu)(Lys)]^+$
	$AlS^{Lys}E^{Lys}$	$[Al(H_2O)_4(SerP)(Glu)(Lys)_2]^{2+}$
	$AlEE^{Lys}$	$[Al(H_2O)_4(Glu)_2(Lys)]^{2+}$
	$AlS_2^{Lys}E^{Lys}$	$[Al(H_2O)_3(SerP)_2(Glu)(Lys)_3]^+$
	$AlS^{Lys}EE^{Lys}$	$[Al(H_2O)_3(SerP)(Glu)_2(Lys)_2]^+$

Table 4.2: Systems considered to evaluate the inherent stabilities of the coordination shells. The System column in the SCS coordination shell shows where the Lysine residues are interacting through salt bridges.

a ΔG_f of -53.1 kcal/mol and two negative charges surrounding the metal. The most stable one corresponded to AlS_2E with a ΔG_f of –124.5 kcal/mol and five negative charges surrounding the metal. Moreover, we observed that at the same total charge, the condensation of the charge in fewer ligands improved the stability of the systems. For example, the compound AlS_2 that presents two non-water ligands —SerP— and a total charge of –1 provides a higher stabilization than compound $AlSE_2$, which presents three non-water ligands —SerP and Glu— with the same total charge.

As shown by Dudev and co-workers, the stability of metal coordination shells is profoundly affected by the presence of charged second shell residues



4

Figure 4.5: Inherent stabilities —ΔG_{f} vs ΔH_{f}— of aluminium coordination shells. FCS are cross markers. SCS are circle markers. Images of the systems —FCS and SCS— can be found in Figure 4.3 and Figure 4.4 respectively. The most stable systems are shown at the top right. Charges are indicated in the molecular formulas of the Table 4.2.

[65]. To explore this, we included the positively charged side chains of lysine residues —Lys, from C^{β} included—. The results, shown also in Figure 4.5, indicated that the addition of these Lys had a different stabilization effect depending on the charge of the first coordination shell. Thus for compounds AlS, AlSE, and AlE_2, with a total FCS charge of 0 or +1, the inclusion of positively charged Lys side chains —systems labelled as AlS^{Lys}, $AlS^{Lys}E$ and $AlEE^{Lys}$— had a destabilizing effect. In contrast, for negatively charged FCS systems AlS_2, AlS_2E, and $AlSE_2$, there was a substantial stabilization of the resultant structures —systems AlS_2^{Lys}, $AlS_3^{Lys}E^{Lys}$ and $AlS^{Lys}EE^{Lys}$—. This situation was especially notable for compound AlS_2E, where the inclusion of the SCS residues —$AlS_2^{Lys}E^{Lys}$— increased the ΔG_{f} from –124.5 kcal/mol to –150.6 kcal/mol. This seemed to indicate that the negatively charged residues in the FCS had a significant Coulombic repulsion that could be

effectively reduced upon the addition of positively charged residues in the second coordination layer. As demonstrated by Dudev et al. [65], the existing upper bound of q+1 acetate-type ligands around a given q-charged metal due to these repulsive effects can be shifted to q+2, due to the stabilization of the negative charge of FCS by the positive charge of SCS. Our results are in agreement with these statements, and also with the well-known metal preference of aluminium for phosphate groups obtained in previous works [259–261], with the most stabilized systems involving phosphorylated serines.

4.3.2 Molecular simulations of neuropeptide systems

For the sake of clarity, we present in the main text only the results for a selected number of complexes, along with their references, that are representative of the conclusions. These systems are depicted in Figure 4.6. For each system, we analyzed the integrated Ramachandran plots, the contact maps, and the radius of gyration, and performed K-means clustering with the principal components of the Principal Component Analysis (PCA). The analysis of each system and further details on the PCA and the clustering are available in Appendix B.

4.3.3 Phosphorylation effects in the absence of metal cation

The effect of the phosphorylation modification was evaluated in the unbound systems by comparing the non-phosphorylated Ref.1 and its phosphorylated variants: Ref.2, Ref.3, and Ref.4. A similar conformational exploration for the four systems was found evidenced by the distribution of conformations along the clustering, as shown in Figure 4.7. In all cases, the most vis-

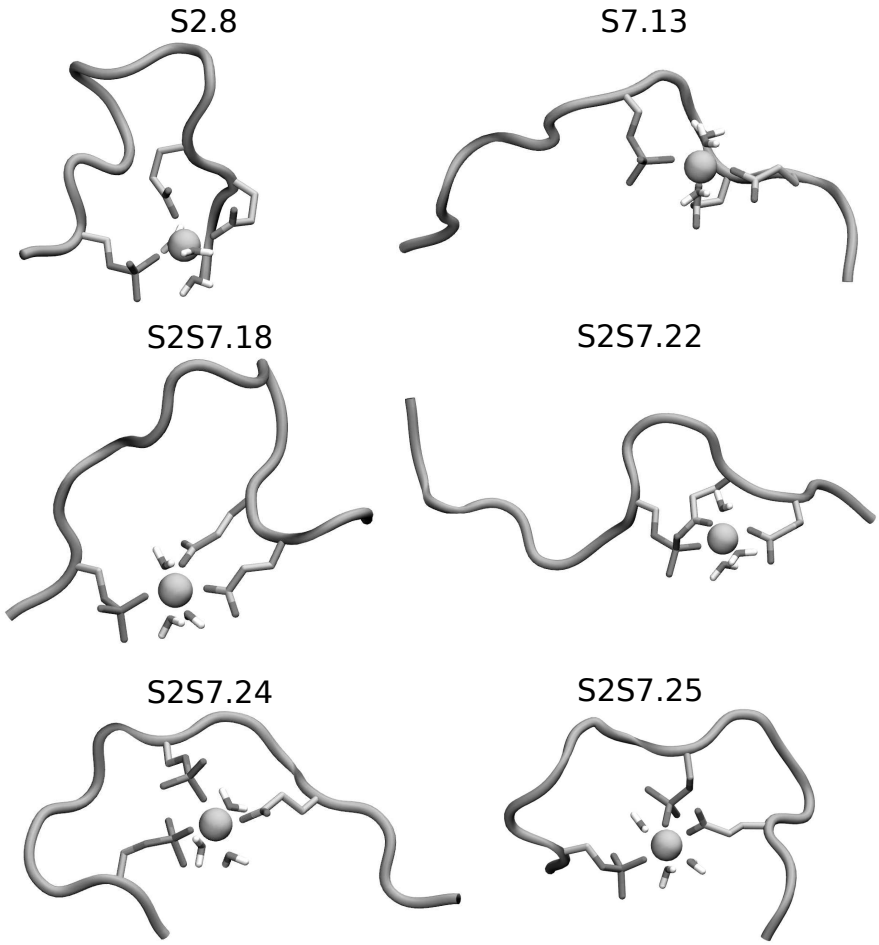


Figure 4.6: Representation of the peptide backbone and metal-coordinated residues of the specific structures discussed throughout the following section. The presence of aluminium results in two main spatial arrangements: extended conformations —S7.13, S2S7.22— and contorted conformations —S2.8, S2S7.18, and S2S7.25—.

ited clusters corresponded to the extended forms, namely clusters C5 to C8, whereas more contorted clusters C1 to C4 had significantly lower populations. Results also indicated a significant response to phosphorylation, which resulted in even more extended forms for partially or fully phosphorylated systems. For Ref.1, the population of clusters C5 to C8 was 74%, which increased to around 82% upon phosphorylation at SerP2 and to 91% upon phosphorylation at SerP7. It seemed that phosphorylation of the central serine SerP7 forced more extended conformations, in contrast with the more conservative distribution of extended conformations when phosphorylation affected the SerP2.

The non-phosphorylated Ref.1 structure showed a significant population of 10% of the most contorted cluster C1, which was absent in the phosphorylated systems. This was probably because the absence of phosphorylation allowed structural arrangements that were inaccessible when phosphorylation was present due to steric hindrance. This possibility increased conformational space exploration. Consequently, post-translational phosphorylation did not fundamentally modify the general structural propensity of Ref.1. However, there were subtle effects on the structure, favouring more extended conformations and avoiding the most contorted forms.

In Figure 4.8, we show the analysis of the secondary structure of the four reference systems without aluminium. Analysis of the integrated Ramachandran plots, contact maps, and radius of gyration of these four structures showed a similar picture for all of them, irrespective of the degree of phosphorylation of the peptide. All four structures presented notably extended conformations with a large population of dihedral angles in the β-sheet region of the Ramachandran map —see Fig. 4.8A—. Long-range interactions were rare in the contact maps, evidencing a tendency to take on extended

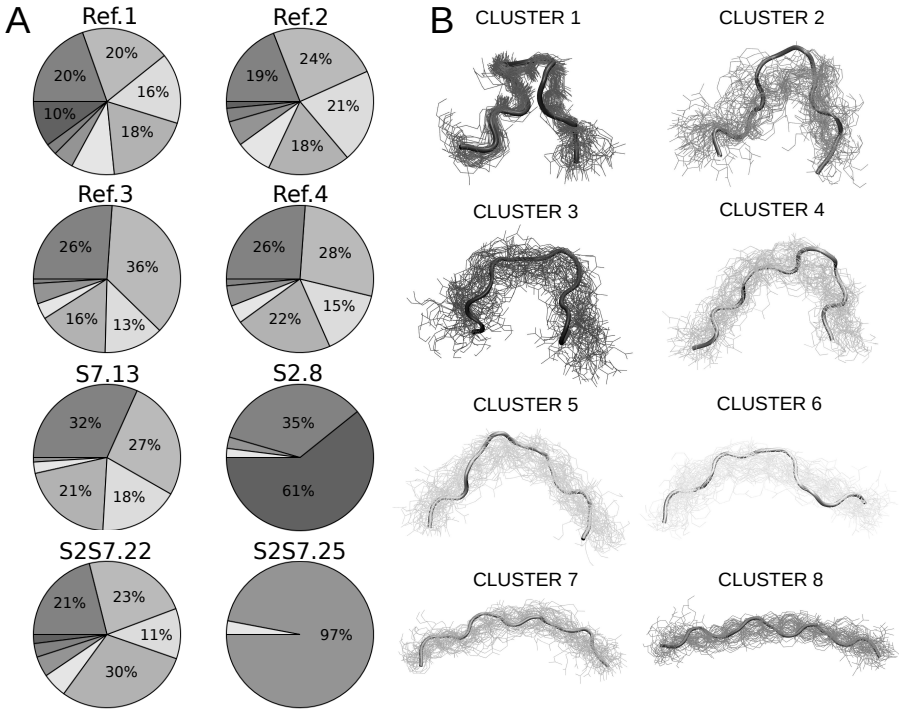


Figure 4.7: (A) Pie chart representation of cluster exploration for each system. (B) Clusters recovered from the PCA analysis. The use of a general trajectory which includes all the trajectories of the systems ensures that similar spatial conformations are assigned to the same cluster regardless of the trajectory of origin. Clusters are of a great variety, from contorted structures —cluster C1— to extended structures —cluster C8—.

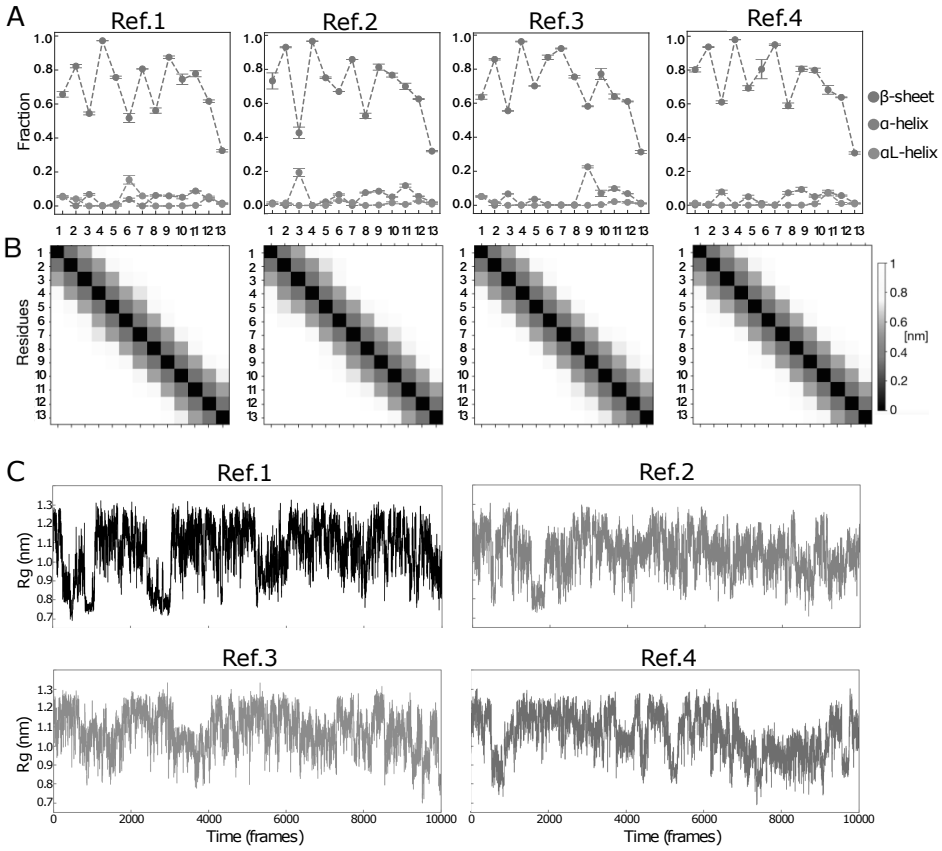


Figure 4.8: Analysis of the conformational ensembles from MD simulations for the reference systems. (A) Fraction of SS as determined by integration of Ramachandran plots. (B) Contact maps between residues. (C) Radius of gyration. Panels A and B correspond to columns of systems.

conformations —Fig. 4.8B—, supported by their relatively long R_g, with mean values between 1.04 and 1.08 nm —Fig. 4.8C—.

4.3.4 Effects of aluminium in conformational states

In contrast with the modest effects of phosphorylation in the conformational propensities of the peptide, the introduction of aluminium cation profoundly affected the secondary structure and the peptide landscape. The effects

depend on the specific coordination site and the nature of the coordination sphere.

Some aluminium-peptide complexes, for instance S7.13 and S2S7.22, exhibited similar conformational properties compared with their reference. Some significant interactions took place in the surroundings of the metal cation, as could be visualized from their contact maps, shown in Figure 4.9B. Generally, in these cases, aluminium-bound residues were located mainly in the middle-to-final part of the chain, which implied a phosphorylated SerP7. Correspondingly, and similar to the unbound systems, these structures presented a generally extended nature, with extended secondary structures such as β-sheets. The population of different regions of the Ramachandran plot —Fig. 4.9A— evidenced this.

In addition, contact maps —Fig. 4.9B— and R_g values —Fig. 4.9C—, with mean values of 1.10 nm and 1.04 nm for S7.13 and S2S7.22 respectively, resembled those of the aluminium-free references. However, in these structures, the presence of aluminium was not without consequence, and the cluster analysis shown in Figure 4.7 revealed subtle changes in the cluster distribution upon aluminium binding. For instance, in the case of S7.13, the most extended structure was more populated.

On the other hand, when SerP2 was phosphorylated, and there was the possibility of aluminium binding more distant residues, the metallic cation imposed heavy constraints on the backbone of the peptide, and aluminium profoundly altered the secondary structure of the peptide. Two paradigmatic cases are analyzed here in detail: S2.8 and S2S7.25. In these cases, the aluminium cation bound residues at the N- and C-termini, limiting the movement of the chain. The different analyses performed reflected this behaviour. For instance, β-sheet-like structures were depleted in the integrated

Ramachandran plots, shown in Figure 4.9A. Contact maps —Fig. 4.9B— revealed remarkable distant contacts, with distances of around 4-6 Å between separated residues —C^{α}-C^{α}—, even at six-to-eight residues apart. The anchoring of the metal to distant residues provoked a dramatic decrease in the spatial extension of the peptide, revealed by the small radius of gyration obtained in comparison with reference systems or even previous metal-peptide complexes. The mean radius of gyration was 0.74 nm for S2.8 and 0.80 nm for S2S7.25, shown in Figure 4.9C. The metallic ion induced an important rotation of the peptide, which translated into an easier formation of the turns required to form α-helices. The population of different regions of the Ramachandran map —Fig. 4.9A— shown that the fraction of α-helices structure increased significantly compared to the reference.

The restraining effects of binding to the metal cation were well-captured by the clustering analysis. As depicted in Figure 4.7 a predominance of the most contorted clusters existed, which contrasts with the balance seen for previously extended conformations. In this sense, S2.8 showed 61% of the most contorted cluster C1, whereas S2S7.25 showed a 97% population of a single cluster C3, classified as a contorted structure. In summary, in the absence of the metal centre or when the coordination was middle-to-end, the effect of the cation was less dramatic and the complexes tended to be more extended, whereas systems with end-to-end coordination of the aluminium tended to be more contorted. Phosphorylation affected the ability of systems to contract or extend but the effect of the presence of the metal seemed to dominate.

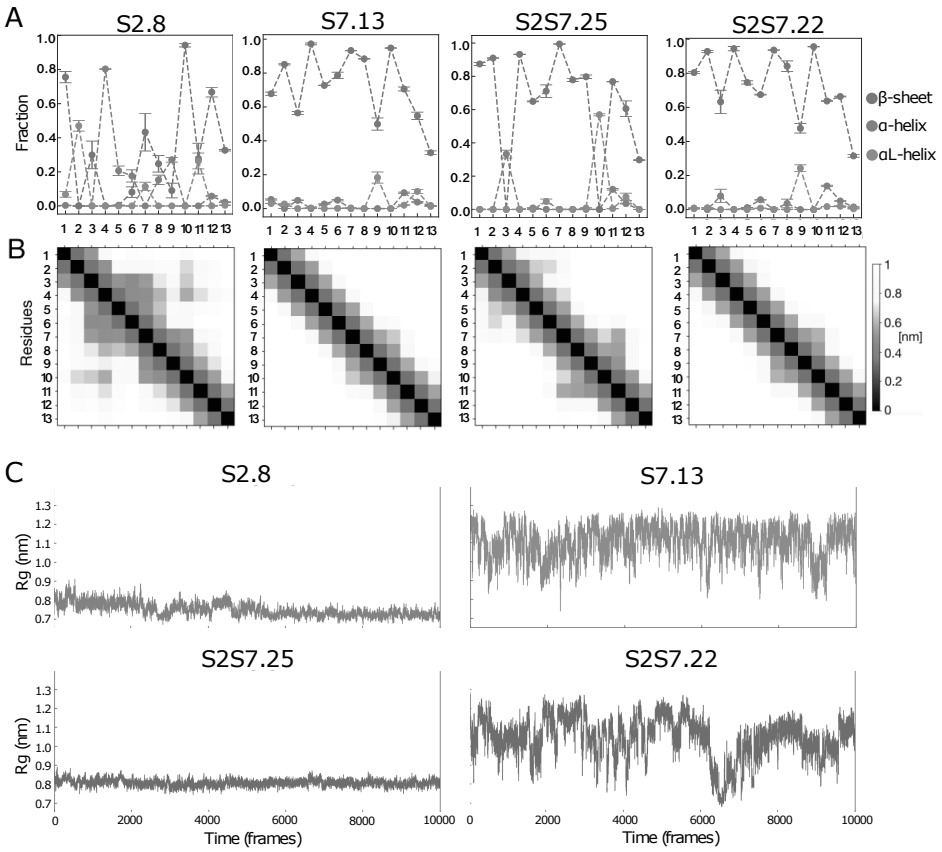


Figure 4.9: Analysis of conformational ensembles from MD simulations for the Al-peptide. (A) Fraction of SS as determined by integration of Ramachandran plots. (B) Contact maps between residues. C) Radius of gyration. Panels A and B correspond to columns of systems.

4.3.5 Configurational entropy loss upon aluminium binding

Results from the configurational entropy calculations are shown in Figure 4.10. As expected, the largest entropy value corresponded to the unbound non-phosphorylated peptide Ref.1. Upon phosphorylation, the entropy decreased slightly. This was in agreement with the reduction in the cluster variety reported in Figure 4.7.

Regarding the formation of metal-peptide complexes, our calculations clearly showed a reduction in the entropy value upon binding of aluminium to the different phosphorylated peptides. In general, the binding process entailed an entropic penalty $T\Delta S_{\mathrm{MIST}}$ that ranged from almost negligible to a 15 kcal/mol penalty. Systems from the S2 group —coloured in cyan in Figure 4.10— shown a larger loss of entropy than systems from the S7 group —coloured in green—, with the S2S7 group —coloured in magenta— showing the largest loss of entropy upon binding. The entropic penalty was highly modulated by the specific coordination of the metal. For instance, in group S2S7, the entropic penalty of the complex S2S7.16 was negligible, whereas the complex S2S7.25 led to an entropic penalty of –16.2 kcal/mol. The largest loss in entropy for each group corresponded to tri-coordinated complexes, namely S2.8 with –10.4 kcal/mol, S7.13 with –5.9 kcal/mol and S2S7.25 with –16.2 kcal/mol. On the contrary, the entropic penalties of mono-dentate complexes as S2.5 or S7.10 were almost negligible relative to their reference peptides.

In summary, metal binding introduced inherent constraints and was highly modulated by metal coordination. When the binding affected a local environment or a few residues, the entropic penalty was small or even negligible

in some cases. In these cases, the presence of aluminium did not significantly alter the flexibility and movement of the rest of the chain. However, when the metal acted as a bridge for end-to-end residues, leading to more contorted structures and significantly impacting the movement of the whole chain, there was a high restriction of the accessible spatial arrangements, resulting in a remarkable entropic penalty.

In terms of contorted vs extended structures, we could say that extended structures tended to have lower entropic penalties, as they maintained a considerable level of movement and spatial exploration. In contrast, contorted structures had considerable entropic penalties due to the movement restrictions.

Because of the trade-off between stabilization energy and entropy loss, the interplay between both contributions is key in determining which are the most populated Al-peptide complexes.

4.3.6 Configurational entropy loss and affinity trade-off

To fully understand the effect of the opposing interaction energies of complexes and their associated configurational entropy, we performed additional QM calculations with the whole peptides, evaluating the mean interaction energy $\langle \Delta E_{\text{int}} \rangle$ for the conformations of each system. The average energy determined for each system manifested a clear correlation between these results and the previous FCS/SCS QM results, reproducing similar qualitative trends: higher stabilizations for the tri-coordinated complexes and lower stabilizations for the mono-coordinated complexes.

In Figure 4.11, we show the combined results of the interaction energy and configurational entropy of each system. The majority of the complexes

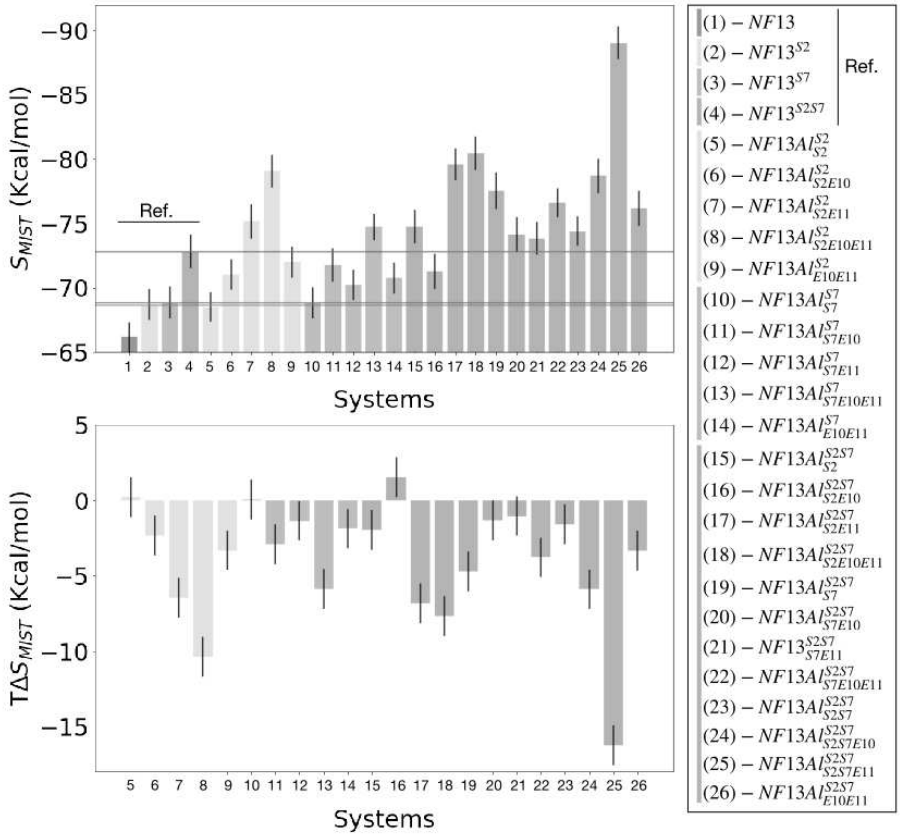


Figure 4.10: (Top) Absolute entropy S_{MIST} values for systems using dihedrals DOF. (Bottom) $T\Delta S_{\mathrm{MIST}}$ values using dihedral DOF —calculated with the corresponding phosphorylated references—. Standard deviations are depicted as black lines over bars. Colours indicate group membership: Red - non-phosphorylation; Cyan - SerP2 phosphorylation; Green - SerP7 phosphorylation; and Magenta - SerP2 and SerP7 phosphorylation.

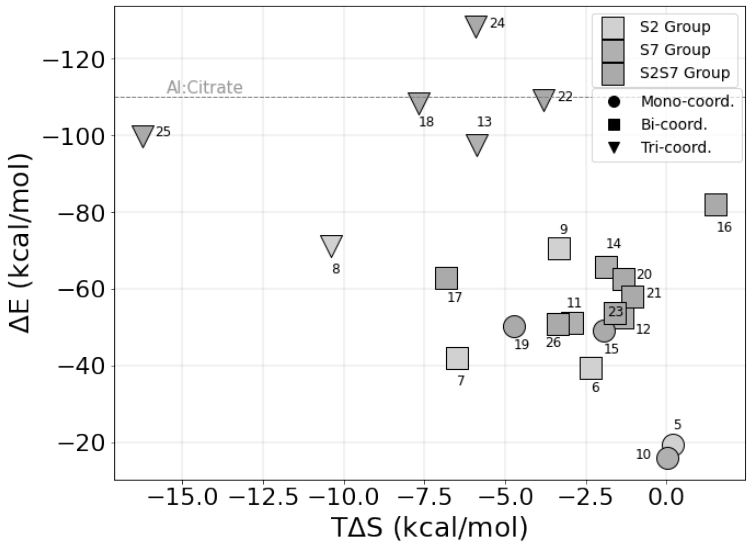


Figure 4.11: Results from $T\Delta S$ and ΔE_{int} calculations. The horizontal red line corresponds to ΔE_{int} value for 1:1 aluminium citrate. Most interesting systems correspond to those with ΔE_{int} comparable to aluminium citrate and modest $T\Delta S$, namely S7.13, S2S7.18, S2S7.22 and S2S7.24). Colour code similar to Figure 4.10.

appeared at the bottom right part of the plot, a region of low entropic penalty but also weak metal-peptide interactions and therefore low interaction energies. This explained the predominance of mono- and bi-coordinated complexes in this region. Opposed to this, in the top part of the diagram, we found an area with a significant ΔE_{int}, with interactions of strength comparable to those of 1:1 ratio aluminium-citrate complexes, calculated at the same level of theory. Because citrate is the main aluminium chelator in blood serum [262–264], its interaction energy provided a realistic reference. We found that only tri-coordinated complexes were able to compete with citrate, represented by a red horizontal line in the graphic. The five more stabilized structures —according to the ΔE_{int}— were: S7.13, S2S7.18, S2S7.22, S2S7.24 and S2S7.25. The entropic penalty found among these complexes was comparable except for complex S2S7.25 which was one order of magnitude above the rest, with a value of –16.2 kcal/mol. We also observed that the level of phosphorylation facilitated access to regions with strong stabilization interactions, with doubly phosphorylated systems positioned in the upper part of the diagram.

Overall, the most promising structures seemed to combine a significant ΔE_{int} capable of competing with citrate, with a relatively low entropic penalty, i.e. systems S7.13, S2S7.18, S2S7.22 and S2S7.24. In Figure 4.12 representative conformations of these complexes are shown. Despite all systems being tri-coordinated complexes bound through equivalent residues, the adopted secondary structure, driven by the location of the metal centre, differed significantly. Thus, S7.13 and S2S7.22 displayed a more extended ensemble of conformations, while S2S7.18 and S2S7.24 displayed a more contorted one. Therefore, although these structures showed similar interaction energies and entropic penalties, the preferential spatial arrangement differed

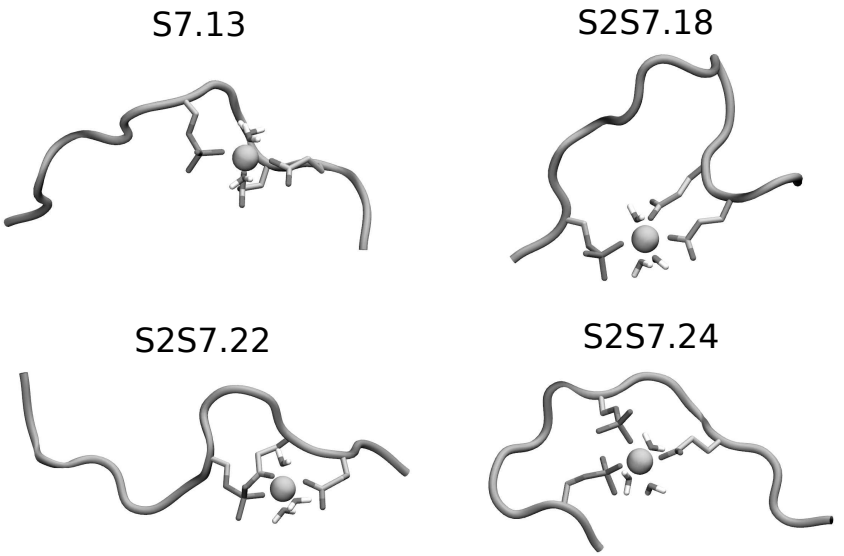


Figure 4.12: Species found in the region of stabilization energies comparable to 1:1 aluminium citrate and low entropic penalty. S7.13 and S2S7.22 were considered extended systems while S2S7.18 and S2S7.24 were considered contorted ones. Notice how three-quarters of the systems corresponded to group S2S7, which agrees with the often-mentioned relevance of hyperphosphorylation in the aggregation processes.

substantially from one situation to another, modulated by the binding site.

4.4 Discussion: biological implications in the context of Metal Ion Hypothesis

Computational studies regarding aggregation formation in the MIH context are not common. However, the experiments by Exley et al. [265–267] demonstrating the co-localization of aluminium and Neurofilament Tangles (NFT) *in vivo*, along with the established induction of abnormal Tau phosphorylation by aluminium [268] and the promotion of Tau oligomer formation

by trivalent metals [269] highlights the potential of investigating the effects of trivalent metals on the conformational landscape of the phosphorylated peptides linked to AD.

Hollender et al. [270] demonstrated —using multi-nuclear NMR and Circular Dichroism (CD) spectroscopy applied to an octapeptide with one phosphorylated serine site— that phosphorylation enhances the affinity of peptides for aluminium, affecting their secondary structure. Computational studies in our group [271] confirmed this increased metal affinity following the phosphorylation. In this chapter, we have focused on a more complex system —NF13— and its phosphorylated derivatives. Our work aims to describe a possible mechanism in the MIH during the early stages of pathological NFT formation by characterizing how aluminium interacts with these key phosphorylation site sequences and how it alters their conformational landscape. However, the short MD simulations from our previous studies did not adequately capture the secondary structure changes caused by the interaction of the metal [271]. In addition, we did not assess the significant impact of changes in configurational entropy, which could be of paramount importance for peptides displaying intrinsic disorder. Therefore, the present long MD simulations provide valuable insights that are difficult to obtain through experimental techniques and complement the previous computational studies.

As discussed in previous sections, the specific coordination location of the aluminium interactions can significantly alter the secondary structure of these peptides. The strength of these interactions generally decreases in the following order: tri-dentate > bi-dentate > mono-dentate. Tridentate complexes exhibit the strongest interactions, with some showing interaction energies comparable to well-known aluminium chelators like citrate. However, aluminium acts as a bridge between different side chains to attain these

multi-dentate structures, anchoring specific residues. The formation of tri-dentate structures typically incurs the highest configurational entropic cost.

Additionally, one might consider that such multi-dentate coordination could promote the formation of contorted systems, which would hinder the development of extended structures. Protein aggregation is expected to proceed through tangential chain stacking, a spatial arrangement not easily accessible for contorted systems. Achieving this arrangement would likely require significant activation energy and spatial redistribution, making it a less probable pathway.

We propose three essential conditions for aluminium to serve as a trigger for oligomer formation and further aggregation: (i) the interaction energy of aluminium with the peptide should be comparable to well-known aluminium bio-chelators like citrate, the primary blood serum chelator for this metal; (ii) a low configurational entropic cost that facilitates significant exploration of the conformational space, even in the presence of aluminium; and (iii) a preference for extended or elongated conformations that are more conducive to aggregation. QM calculations of various aluminium coordination shells can assess the first condition. These calculations have demonstrated that phosphorylated peptides exhibit greater interaction energies than their non-phosphorylated counterparts [256, 261, 271]. However, only a limited number of structures display interaction energies comparable to citrate [256]. Accurately determining the last two conditions necessitates long MD simulations to generate well-converged ensembles of these systems, a challenging task that this chapter accomplishes for the first time.

Given these three criteria, only two of the selected Al-peptide complexes qualify as potential candidates for protein aggregation: S7.13 and S2S7.22. These systems correspond to tri-coordinated structures that exhibit interac-

tion energies similar to a 1:1 aluminium-citrate complex, display relatively low entropic penalties, and favour extended conformations. Thus, only a small fraction of the structures meet all three necessary conditions, suggesting that the presence of aluminium is more likely to hinder the formation of potential oligomerization seeds.

However, it is important to note that aluminium can still form structures that reconcile the seemingly contradictory properties of strong multi-dentate interactions and the ability to create a flexible, extended secondary system with minimal configurational entropy loss. In these cases, the metal coordinates with a localized region of the sequence, which does not significantly disrupt the dynamics of the peptide. These conformations could serve as potential candidates for oligomer growth, where aluminium might act as an additional anchor point for interactions with other peptides. Although these findings are based on Al-NF13 systems, they are likely transferable to other multivalent metals, such as Fe(III), which share similar coordination properties with aluminium.

Alternatively, aggregation may occur through the interaction of preformed oligomers with metals, aligning with a metal-assisted pathway corresponding to an auxiliary role of the metals in the aggregation process. This mechanism would have the advantage of a lower configurational entropic penalty for incorporating the metal into a pre-organized oligomer, where aluminium would interact mainly with phosphorylated residues from different chains. Such multi-dentate interactions would likely not significantly impact the flexibility of a system that is already more rigid than a free peptide. It is worth noting that even in our most favourable cases —S7.13 and S2S7.22— the configurational entropy amounts to –5.9 and –3.8 kcal/mol, respectively. These values, while not negligible, could favour a mechanism

in which oligomerization occurs before metal incorporation.

4.5 Conclusions

In this chapter, we conducted simulations lasting up to 1-2 μs, enabling us to evaluate the conformational ensembles of Al-peptide systems thoroughly. Our simulations also provide converged estimates of the configurational entropy for these systems, allowing us to characterize the entropy loss associated with metal binding. This property is often challenging to assess but is crucial when considering interactions with peptides displaying intrinsic disorder, which exhibit high flexibility and consequently significant configurational entropy.

The interactions between the NF13 peptide variants and aluminium lead to various coordination modes. The nature of these coordinations and the specific residues involved induce different conformational changes that affect the secondary structure. Interaction energy analysis indicates that tri-coordination is favoured over other modes. In tri-coordinated systems, binding to SerP7 results in more flexible structures, while binding to SerP2 produces more rigid systems.

Overall, we can classify the binding into two spatial arrangements: more rigid, contorted structures and more flexible, extended structures. Energetic and entropic analyses of both types show a negative correlation between the two contributions, with the entropic penalty increasing alongside the interaction energy due to steric hindrance from interactions.

Contextualizing these results within protein aggregation allows us to synthesize three necessary, though not sufficient, requirements to identify systems as potential candidates for oligomerization: (i) competitive stabi-

lization energy is essential; (ii) the entropic penalty must be low enough to permit some level of conformational space exploration; and (iii) extended arrangements should be accessible, as they are believed to represent the starting point for aggregation. Most structures do not meet these criteria; however, if aluminium is tri-coordinately bound to a localized region of the peptide, it may form structures that combine strong multi-dentate interactions with a reasonably low configurational entropic cost, favouring extended conformations. These structures could serve as potential seeds for peptide oligomerization and the formation of pathological species.

4

5

Al(III) & Neuro-peptide Dimers

5.1 Effect of Metal Binding on Peptide Dimers

In the previous Chapter 4, the complexation of aluminium and neurofilament peptides linked to the aggregation process showed that modulation of the conformational energy landscape by the metal depended on several factors [225]. The location of the metallic centre in the chain, the coordination type —mono-dentate, bi-dentate, or tri-dentate—, or the coordinated ligands were all relevant aspects whose interplay resulted in a specific regulation of the energy landscape.

In general, the characterized trends indicated that the higher the coordination of the complex, the higher the metal-binding strength, with higher affinities when the charge of the ligand was more negative, leading to con-

densation of negative charge around the metal centre. This led to Coulombic repulsion between the negatively charged ligands around the metal, which limited the number of negatively charged side chains that could be coordinated to an aluminium atom. However, the second coordination sphere of the metal-peptide complex could incorporate positively charged side chains to compensate for the dense negative charge that aluminium tended to accommodate in its surroundings [65].

In addition, there was an essential interplay between establishing stabilizing interactions with multidentate coordination and losing configurational entropy. These two factors tended to point in opposite directions and compensate for each other.

Furthermore, we demonstrated that only a minority of structures satisfied the three necessary conditions to be potential seeds for aggregation, namely: (i) high metal affinity; (ii) low configurational entropy penalty; and (iii) promotion of extended peptide conformation. However, complexes that still reconcile the seemingly contradictory properties of strong multidentate interactions and the ability to create a flexible chain also exist.

However, aggregation is a process that involves multiple copies of peptides. In this regard, although the approach in Chapter 4 was useful in addressing some basic features of aluminium-NF13 interactions, it also had a clear limitation: only one peptide chain was considered. In the present chapter, we increased the complexity of the model by introducing two chains of the NF13 peptide used previously, allowing us to analyze whether previous conclusions still stand in these new conditions.

Moreover, while phosphorylation was acknowledged in the previous Chapter 4, we were unable to emphasize its specific relevance beyond its effect on configurational entropy. Phosphorylation and especially hyperphospho-

rylation situations have gained increasing interest since the beginning of the protein aggregation studies in the context of neurodegeneration, given the amount of evidence showing the existing notable linkage [231, 232]. Phosphorylation is a post-translational modification with a multitude of biological functions, but its presence in aggregates —especially in the hyperphosphorylation abnormal concentrations— seems to place this phenomenon in a central role of aggregation [232, 237]. Serine and threonine are some of the most common phosphorylated residues, and multiple phosphorylation sites with high concentrations of these residues have been identified across aggregable IDPs. Our current objective was to more precisely determine the role of this post-translational modification in the aggregation mechanism, especially integrated with the MIH, considering that the literature identifies it as a key factor.

To achieve the proposed objectives, we considered the aluminium centre to be coordinated with the three most relevant coordination shells previously determined, namely AlS_2, AlS_2E, and $AlSE_2$. For each coordination type, different degrees of phosphorylation were introduced in the peptides. We used REST2-enhanced simulations to force exploration outside the equilibrium, especially for the aluminium-bound peptide dimers.

5.2 Methodology

5.2.1 Models systems

Following the extension of the previous Chapter 4 studies, two identical copies of the same 13 residue peptide —KSPVPKSPVEEKG— from the intrinsically disordered C-terminal region of NF-M protein were considered.

Both were oriented equivalently and maintained with the same initial conformation, only slightly modified to adjust for proper metal coordination when needed. The full nomenclature of the systems is as follows: $NF13^{Phos}_{C.Type}$ — Al — $NF13^{Phos}_{C.Type}$, where NF13 designates the chain, the presence of Al indicates the binding to aluminium, the Phos superscript indicates the phosphorylation and the C.Type subscript indicates the specific coordination to aluminium. Again, we sustain the explanations in the text using the [C.Type].[Index] code for the sake of clarity. Structures were built departing from previous ones and using Chimera [197] and VMD [252] to adjust the metal coordination. Peptides were acetylated at the N-terminus and amidated at the C-terminus.

Equivalently to the previous Chapter 4, three systems without aluminium metal were considered here as reference systems, which differed in their degree of phosphorylation: i) Ref.1 consisted of two non-phosphorylated NF13 chains, NF13—NF13; ii) Ref.2 consisted of a completely phosphorylated chain and a non-phosphorylated chain, $NF13^{S2S7}$—NF13; and iii) Ref.3 consisted of two completely phosphorylated chains, $NF13^{S2S7}$—$NF13^{S2S7}$, a hyperphosphorylation situation.

Peptides were considered to be coordinated to Al(III) cation through phosphorylated serines —SerP— or glutamates —Glu—, introducing water molecules until the octahedral geometry was completed. Three types of aluminium coordination were proposed based on the results of the FCS/SCS energy calculations in Chapter 4: (i) AlS_2, where the metal coordinates one SerP from each chain; (ii) AlS_2E, where the metal coordinates a SerP from one chain and a SerP and a Glu from the other chain; and finally (iii) $AlSE_2$, where the metal coordinates to the SerP from one chain and both Glu from the other chain. A scheme of these coordination spheres is shown in Figure

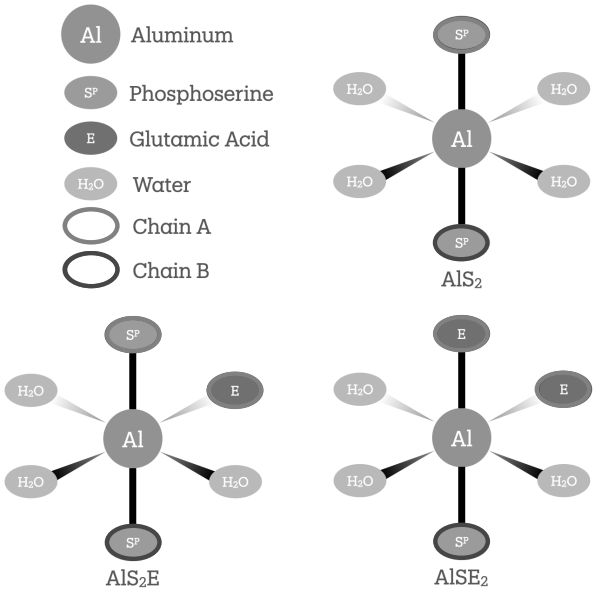


Figure 5.1: Representation of the three coordination sphere types considered in this chapter. The aluminium metallic centre maintains its octahedral coordination, binding both chains and completing the coordination sphere with water molecules.

5.1. The previously determined ΔH_{f} and ΔG_{f} of the different coordination shells are shown in Figure 5.2. The coordinations included in the present chapter were those that exhibit lower formation energies —more negative— and therefore, the most likely to be formed.

For each of the coordination types, at least three systems that differ in the degree of phosphorylation were considered: (i) the non-phosphorylated case; (ii) the partially phosphorylated case; and (iii) the fully phosphorylated —or hyperphosphorylated— case. An additional case was also considered for the $\mathrm{AlSE_2}$ coordination type, where the chain coordinated with both Glu could remain completely non-phosphorylated because none of the serines were used for metal coordination.

Interestingly, the combination possibilities led to a variety of overall

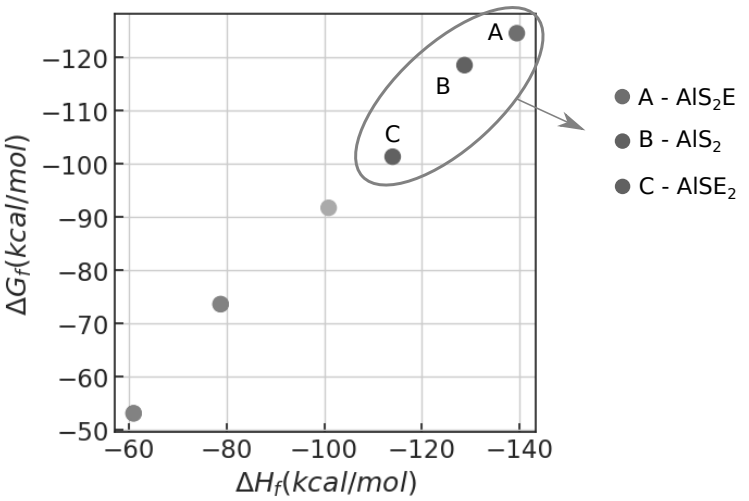


Figure 5.2: ΔH_{f} and ΔG_{f} of formation for all coordination shells considered in the previous Chapter 4. Those selected for the present chapter are highlighted. The energies were determined using QM calculations with DFT, CAM-B3LYP functional, and 6-31++G(d,p) basis set.

5

charges for the dimer systems. The charges of each chain and the resultant overall charge of each system can be visualized in Table 5.1, where all the systems of this chapter are displayed. Due to the large number of elements involved, there was a potentially extensive variety of systems resulting from the high number of combinations. We tried to select the most relevant ones for an appropriate comprehension of the most essential processes.

5.2.2 REST2 Simulations

The systems of this chapter consisted of two short disordered peptides interacting with each other. This type of system presents difficulties in converging its simulations due to its disordered nature. For this reason, we decided to perform REST2-enhanced simulations. These simulations were carried out with GROMACS package [154] (version 2023) with PLUMED (version 2.9.0) and CHARMM36 force field [253]. Non-bonded parameters of Al(III) from

C.Type	Index	*A* Phos	*A* Charge	System	*B* Phos	*B* Charge	Δq
Ref.	1	—	+1	NF13—NF13	—	+1	0
	2	S2S7	–3	$NF13^{S2S7}$—NF13	—	+1	4
	3	S2S7	–3	$NF13^{S2S7}$—$NF13^{S2S7}$	S2S7	–3	0
AlS_2	4	S2	–1	$NF13^{S2}_{S2}$—Al—$NF13^{S2}_{S2}$	S2	–1	0
	5	S2	–1	$NF13^{S2}_{S2}$—Al—$NF13^{S2S7}_{S2}$	S2S7	–3	2
	6	S2S7	–3	$NF13^{S2S7}_{S2}$—Al—$NF13^{S2S7}_{S2}$	S2S7	–3	0
AlS_2E	7	S7	–1	$NF13^{S7}_{S7}$—Al—$NF13^{S7}_{S7E10}$	S7	–1	0
	8	S2S7	–3	$NF13^{S2S7}_{S7}$—Al—$NF13^{S7}_{S7E10}$	S7	–1	2
	9	S2S7	–3	$NF13^{S2S7}_{S7}$—Al—$NF13^{S2S7}_{S7E10}$	S2S7	–3	0
$AlSE_2$	10	S7	–1	$NF13^{S7}_{S7}$—Al—$NF13^{-}_{E10E11}$	—	+1	2
	11	S7	–1	$NF13^{S7}_{S7}$—Al—$NF13^{S7}_{E10E11}$	S7	–1	0
	12	S7	–1	$NF13^{S7}_{S7}$—Al—$NF13^{S2S7}_{E10E11}$	S2S7	–3	2
	13	S2S7	–3	$NF13^{S2S7}_{S7}$—Al—$NF13^{S2S7}_{E10E11}$	S2S7	–3	0

Table 5.1: List of systems considered in this chapter. C.Type indicates the coordination type; Index indicates the numbering of the system; *A* Phos/Charge and *B* Phos/Charge indicates the phosphorylation and charge of the chain *A* —left chain— and chain *B* —right chain—; System indicates the full notation of the system; Δq indicates the charge difference between chains. A graphic representation of all the systems is shown in Appendix C.

Faro et al. [254] were included. Systems were placed in a cubic box leaving 1 nm distance to the edge and solvated with CHARMM-TIP3P [255] water molecules. Na^+ and Cl^- ions were added to reproduce the physiological conditions of 0.1 M. The charge of the systems was neutralized by adding additional Na^+ or Cl^- ions when needed. Energy minimization was carried out using the steepest descent algorithm, followed by two 500 ps equilibrations in the NVT and NPT ensembles, respectively. For these equilibrations, the leap-frog algorithm was used with a 2 fs time step and 300 K temperature. The Berendsen barostat [206] and Bussi-Donadio-Parrinello [205] thermostat were used during equilibrations. For production, the barostat was changed to the Parrinello-Rahman. We ran 8 replicas in each REST2 simulation, with temperatures ranging from 300 to 400 K. The temperature for the solute —including the aluminium cation when present— and the ions was scaled with REST2. Two REST2 simulations were run for each system for a total of 200 ns —100 ns each—. Electrostatic interactions were evaluated using the PME method [208] with a 1.0 nm cut-off for short-range Van der Waals interactions. All H-bonds were constrained using LINCS [207]. Frames were saved every 2 ps for analysis.

5.2.3 Analysis of the simulations

Despite using a non-bonded model for the aluminium-peptide coordination, the metallic centre remained bonded during all simulations. This was not surprising and evidenced the high affinity between a hard cation such as Al(III) and the hard base oxygen atoms from SerP or Glu side chains, resulting in a slow ligand exchange rate of Al(III) cation —around 1.3 s^{-1} [272]—. Given the relevance of the bound state, we decided to characterize these states and explore the conformations adopted by the chain pairs in the

absence and presence of aluminium, with an increasing degree of phosphorylation.

To do so, we used a PCA fed with contact maps of intra-chain contacts and inter-chain contacts, which allowed us to reduce the dimensionality and apply a clustering technique that classified *bound* and *unbound* states. The principal components from the PCA were used for clustering using the Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) algorithm. Moreover, free energy surfaces were also calculated with the PCA results. The parameters used for the clustering are indicated in Appendix C. Each HDBSCAN cluster population was determined as the fraction of frames identified with a given cluster index from the total of frames.

5.3 Results

Firstly, we discuss the results for the peptide dimer systems in the absence of aluminium, highlighting the effect of phosphorylation on the bound/unbound equilibrium of the dimers. Later, we analyze the effect of the presence of aluminium in the bound states of the dimers, characterizing the type of intra-chain and inter-chain interactions in the bound substates.

5.3.1 Phosphorylation favours the bound state of the peptide dimer

Three dimer systems without metal, extensions of the previous Chapter 4 references, were considered to evaluate the effect of the phosphorylation in the dimers —namely, Ref.1, Ref.2 and Ref.3—. These systems were used as

references later to understand the role of the metal in the conformational ensemble of the bound dimers.

The characterization of the *bound* and *unbound* states —Fig. 5.3A— was done using the PCA+Clustering workflow. The identified clusters are shown over the free energy surfaces, along with an energy profile of the energy surface and a schematic representation of the systems Ref.1, Ref.2, Ref.3 and the coordinated system $AlS_2.6$. Intra-chain and inter-chain contacts for an initial characterization allowed us to separate between bound —black— and unbound —white— states.

Ref.1 showed small differences in the population of both states, higher for the unbound state, with a deeper and wider minimum in the free energy landscape, as shown in Figure 5.3B. Since the charge of each chain was +1, it was not strange to see that chains did not have a preference for a bound state. In the case of Ref.2, Figure 5.3C, the minimum in the unbound region disappeared. The fraction of conformations in the bound state was now three times larger than that of the unbound state. This behaviour was consistent with that of opposing charges of both peptide chains, namely +1 for the non-phosphorylated peptide and –3 for the phosphorylated one. Finally, results for Ref.3 are shown in Figure 5.3D. We found a high propensity toward the bound state, with almost all structures explored during the trajectory corresponding to this state. The population of the bound state was ten times larger than the unbound one. This behaviour was particularly interesting and counter-intuitive given that both peptide chains presented the same –3 charge, and therefore, one could in principle expect a strong Coulombic repulsion between both peptide chains. The rationalization of this behaviour is addressed later in this chapter when we characterize the inter-chain interactions present in the bound states of these systems. Fig-

ure 5.3E corresponds to a representative aluminium-coordinated system for comparison, in which the system only explored bound state situations due to the anchoring of both peptide chains to the Al(III) cation.

Given the width of the unbound state region, the conformations located in this area were largely heterogeneous. When the distance between both pairs of chains was longer than that of the cut-off of the different interactions, the chains behaved as isolated. The radius of gyration and the contact maps of the unbound conformations for each chain individually returned results similar to those obtained in the previous Chapter 4 for equivalent systems. As an example, we considered the ensemble of conformations from: (i) the present chapter dimer Ref.1; and (ii) the previous Chapter 4 reference NF13 monomer. In Figure 5.4, the comparison of the radius of gyration and side chain contacts is shown. Notice that the previous NF13 simulation corresponds to a converged 1 μs conventional MD simulation length while Ref.1 dimer simulation corresponds to a 200 ns REST2 simulation length.

5.3.2 Phosphorylation plays a central role in the inter-chain interactions

In Chapter 4, we defined the presence of extended conformations as a pre-requisite for IDP aggregation potential. The reason for this was that these extended conformations allowed interactions with more than a single peptide chain. In contrast, contorted conformations hindered further interactions with other chains as a result of inappropriate orientation because the peptide was saturated by dense intra-chain contacts. In this subsection, we characterized the intra- and inter-chain interactions of reference dimers in the bound state, especially those regarding phosphorylated residues, and we evaluated how they modulate the presence of contorted/extended conforma-

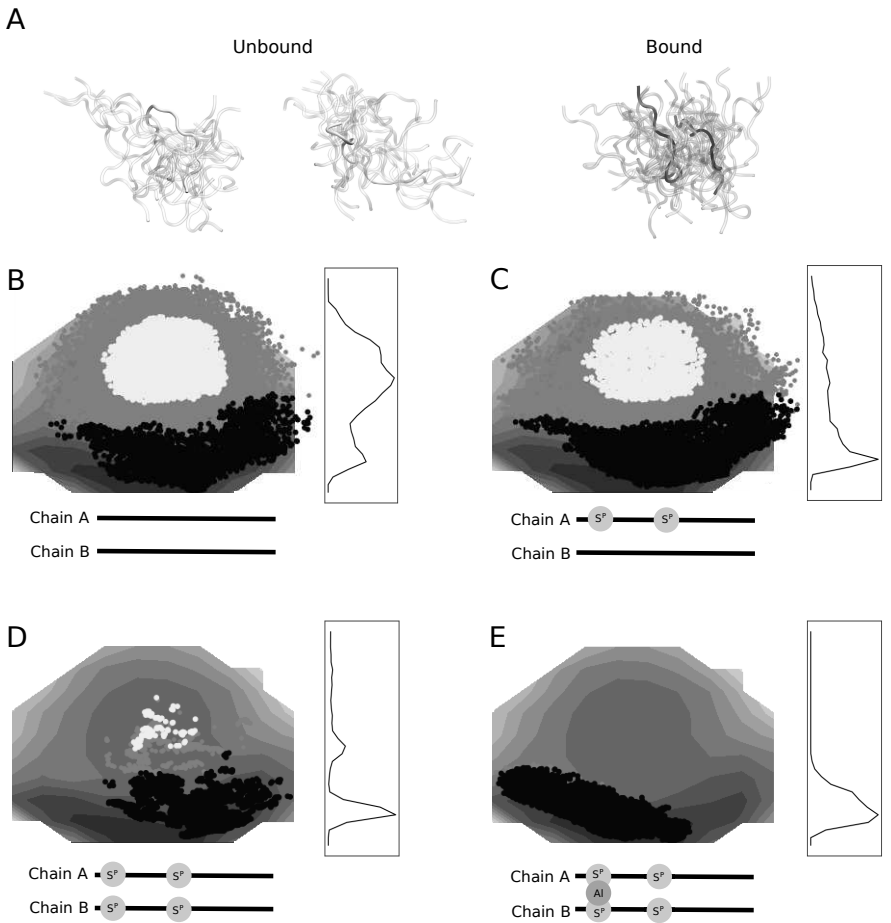


Figure 5.3: HDBSCAN characterized *bound* —black— and *unbound* —white— states labelled from the PCA results of the combined model using intra-chain and inter-chain contact maps. A reduced representation of each corresponding system is shown below each plot. Orange S^P corresponds to SerP; grey Al corresponds to aluminium cation. B) non-phosphorylated Ref.1 reference showed a more or less equivalent population for both states; C) partially phosphorylated Ref.2 reference showed an increased population in the bound state; D) hyperphosphorylated Ref.3 reference evidenced a great tendency towards bound state; E) systems with aluminium remained in the bound state during the whole simulation time.

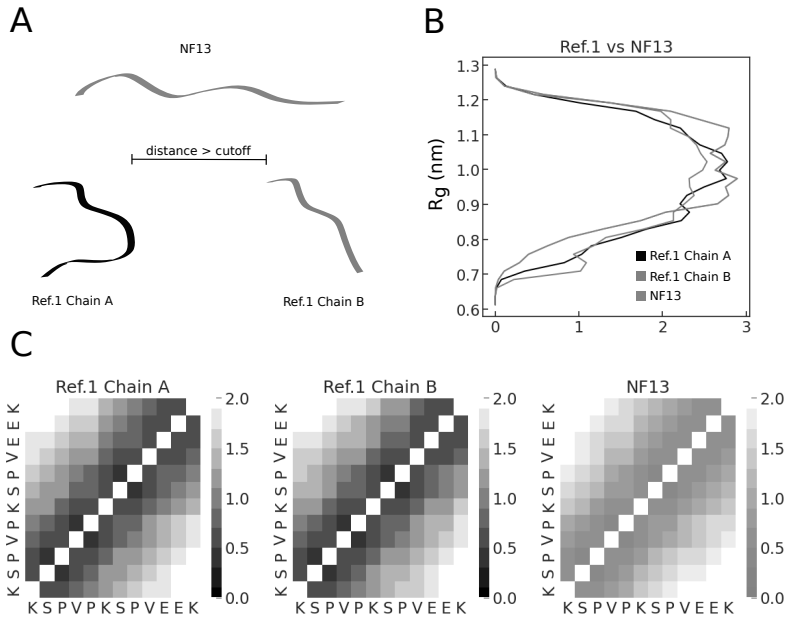


Figure 5.4: Comparison between NF13 monomer systems from the previous Chapter 4 —red— and isolated chains from the present dimer simulations —black and silver—. A) Both chains are considered isolated —equivalent to individual NF13 monomer— when the distance between them was larger than the defined interactions cut-off. B) The radius of gyration for all three chains showed equivalences, with large heterogeneity of extended and contorted conformations. C) Contact maps between side chains of all three systems were also equivalent.

tions according to the radius of gyration.

Results are summarized in Figure 5.5. For each system, the free energy surfaces are shown, with the characterized energy minima identified. Alongside, the contact maps of intra-chain and inter-chain corresponding to the interactions between the side chains of the residues are shown. The radius of gyration is also displayed for each chain. In the three cases, extended and contorted conformations were explored by the systems. However, none of the systems showed a tendency for extremely contorted systems —a radius of gyration below 0.8 nm—. All systems showed a substate that was significantly more populated than the others, indicating a clear tendency for conformations of a specific type of bound substate.

In the case of Ref.1 —Fig. 5.5A—, two minima were characterized. The less populated one —corresponding to magenta C0, 17%— showed extended conformations with contact maps showing the absence of specific contacts. The distribution of the radius of gyration, similar to both chains, was wide in the extended conformation region, indicating flexibility. In the case of the most populated minimum —orange C1, 44%— the intra-chain contact maps showed increased interactions of residues from the N- and C-termini of both chains. The inter-chain contact map also showed more extended contacts along the chains, but again specific contacts were not visible. As a result of the increase of intra-chain contacts, the radius of gyration decreased, although it maintained a notable width.

Regarding the Ref.2 system —Fig. 5.5B—, changes were noticeable with the introduction of phosphorylation in one of the chains. The free energy surface showed a single broad energy minimum —orange C0, 56%— that corresponded to the extended phosphorylated chain interacting with the non-phosphorylated chain, which explored multiple extended and contorted

conformations. The contact maps evidenced this, showing more contacts for the non-phosphorylated —blue— chain. The radius of gyration for each chain showed two differentiated conformational ensembles for each chain, one narrower and extended for the phosphorylated chain and one wider for the non-phosphorylated one.

In the last system Ref.3 —Fig. 5.5C—, both chains were fully phosphorylated. In this hyperphosphorylated situation, two energy minima were identified, although the most important one —orange C1, 61%— was populated more than three times than the other one —magenta C0, 19%—. Focusing our analysis on C1, we detected the appearance of specific stable interactions in both intra-chain and inter-chain contact maps, mainly corresponding to lysine and phosphoserine contacts. The radius of gyration indicated the presence of a notably contorted chain —blue—, with another one displaying more variability with a wider ensemble. Further characterization of the interactions between these two chains and their specificity of contacts revealed the presence of counter-ions —Na^+— in the surroundings of Lys and SerP residues, a situation not observed for any of the previous systems, creating what we labelled as a Charge Condensation Zone (CCZ). In this situation, the positively charged Lys and negatively charged SerP tended to approximate each other in a highly compact structure that forced one of the chains to further contort, resulting in a close positioning of many Lys and SerP residues. The overall structure was further stabilized by the presence of several counterions that neutralized the charge and reduced the Coulombic repulsion of the high density of negatively charged residues. This created an energetically favourable ensemble of interacting peptide dimers. Figure 5.6 shows the ensembles of the most populated energy minimum for each reference system, together with an example of the CCZ formed in the

hyperphosphorylation situation of the Ref.3 system.

In summary, in the absence of phosphorylation, the peptide dimers showed no specificity of contacts and a tendency for a wide range of conformations, evidencing the absence of energetically favourable conformations in the bound state. This was consistent with the apparent similar exploration of both bound and unbound states seen previously. The introduction of phosphorylation in one of the chains increased the number of contacts between chains. When the degree of phosphorylation reached the hyperphosphorylation situation, with both chains completely phosphorylated, both chains interacted with each other using multiple charged residues creating a CCZ of charged Lys and SerP residues, which was further neutralized by the presence of Na^+ counterions. These conformations were energetically favourable, indicating a possible trend for highly phosphorylated regions to interact with each other when positively charged residues —such as lysines— are present and counterions are accessible for charge neutralization.

5.3.3 Aluminium releases residues from the charge condensation zone, promoting more extended conformations

When the peptide chains and the aluminium were bound, we considered at least three different systems with different degrees of phosphorylation for each coordination type. For the sake of clarity, we present here the results for the hyperphosphorylation situation only, as a result of our special interest in these cases. However, results were evaluated in the whole range of degrees of phosphorylation, and a specific analysis for each of the remaining systems is available in the corresponding Appendix C.

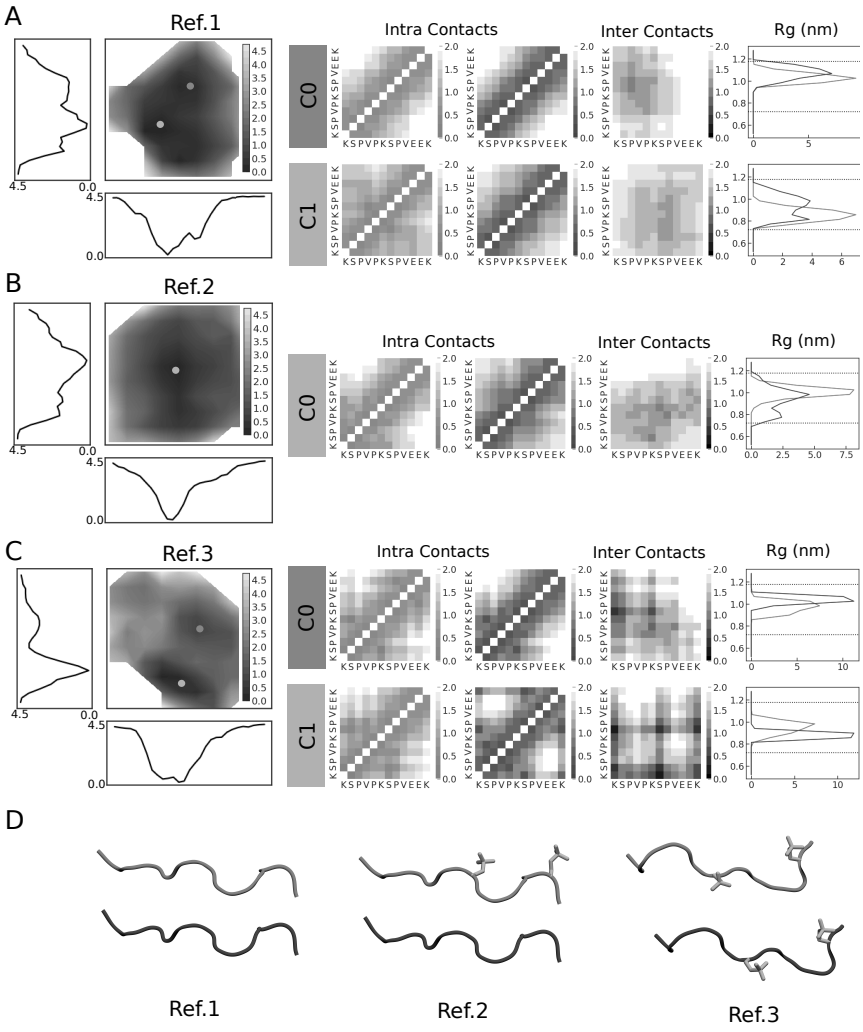


Figure 5.5: PCA free energy surface for bound substate characterization of reference systems. Over the free energy surfaces, the characterized substates —minima— are identified. The orange substate always corresponds to the most populated one. Contact maps for intra-chain and inter-chain side-chain residue interactions are displayed, with the corresponding radius of gyration of each chain. Contact Maps' colour code is consistent with that of the radius of gyration and molecule images.

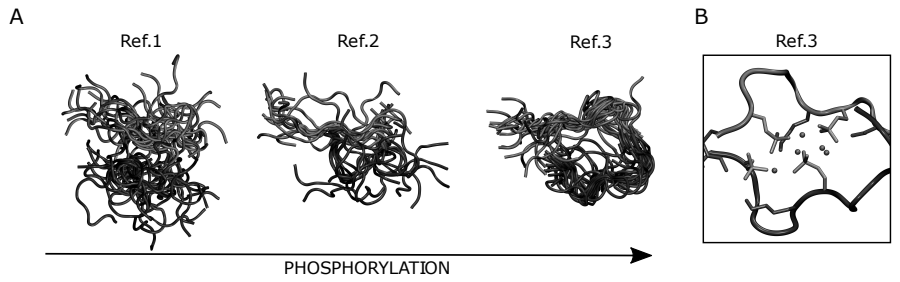


Figure 5.6: Ensemble conformations of each reference system ordered by phosphorylation degree. A) The increase of phosphorylation degree induced more specificity of contacts between chains, moving from an apparently arbitrary interaction conformation to adopting a more particular structure. B) The conformations of the most populated energy minimum of the bound substate of system Ref.3 showed the presence of Na^+ counterions in a highly dense region of charged residues defined as charge condensation zone —CCZ—.

As indicated in the methodology of this chapter, the binding of the aluminium was done by a non-bonded model. However, the metal remained bound to the peptides with the same first coordination shell during all simulations. This evidenced the strong affinity of aluminium to coordinate with ligands such as $-OPO_3^{2-}$ and the slow ligand-exchange rate of this metal —in the order of s^{-1}—. Therefore, all systems bound to aluminium explored only the bound state.

In general, the degree of phosphorylation increased the interactions between the chains coordinated to aluminium, similar to the reference systems. When the chains were in the lowest phosphorylation level, the absence of the highly negatively charged SerP residues implied more flexible chains, especially in regions farther from the aluminium coordination. Although the chains remained bound because of the presence of the metal, there was an absence of specific contacts, and the conformational ensembles were heterogeneous. With an increase in the degree of phosphorylation, positively charged Lys residues started to interact with the now negatively charged

SerP, and more ordered states were found. However, in contrast with the aluminium-free Ref.3 system, no CCZ was formed around the metallic centre.

In Figure 5.7, the resultant free energy surfaces, characterized minima, contact maps, and radius of gyration are shown for the aluminium-bound hyperphosphorylated systems. The first system AlS_2.6 corresponded to the hyperphosphorylated system in which aluminium was coordinated to the SerP2 of both chains. This situation was interesting because of the positioning of the metal at the N-terminus end of both chains, leaving the rest of the peptide free, as depicted in Figure 5.7D. For this system, a broad minimum was found, corresponding to an ensemble where both chains remained roughly extended, as evidenced by the radius of gyration of Figure 5.7A. The inter-chain contact map showed a tendency of Lys and non-coordinated SerP residues to approach the metal coordination shell, like that observed for Ref.3 when the CCZ was formed. However, because aluminium carries three times more charge than a single Na^+ counterion —being, therefore, a more efficient negative charge neutralization agent—, these contacts were not maintained for a long time and rather formed and broken dynamically. Moreover, no counterions were found in the surroundings of the aluminium coordination shell.

The second system $AlS_2E.9$ corresponds to the hyperphosphorylated system in which aluminium was coordinated to the SerP7 of both chains and additionally, to the Glu10 of one chain. This coordination shell corresponds to the most stable one characterized in Chapter 4. This metal coordination locked the central part of both chains, leaving both ends of the chains —especially for the chain coordinated only by one phosphoserine— free to move. The characterization of the bound substates identified a broad minimum in the free energy surface corresponding to an extended chain ensemble.

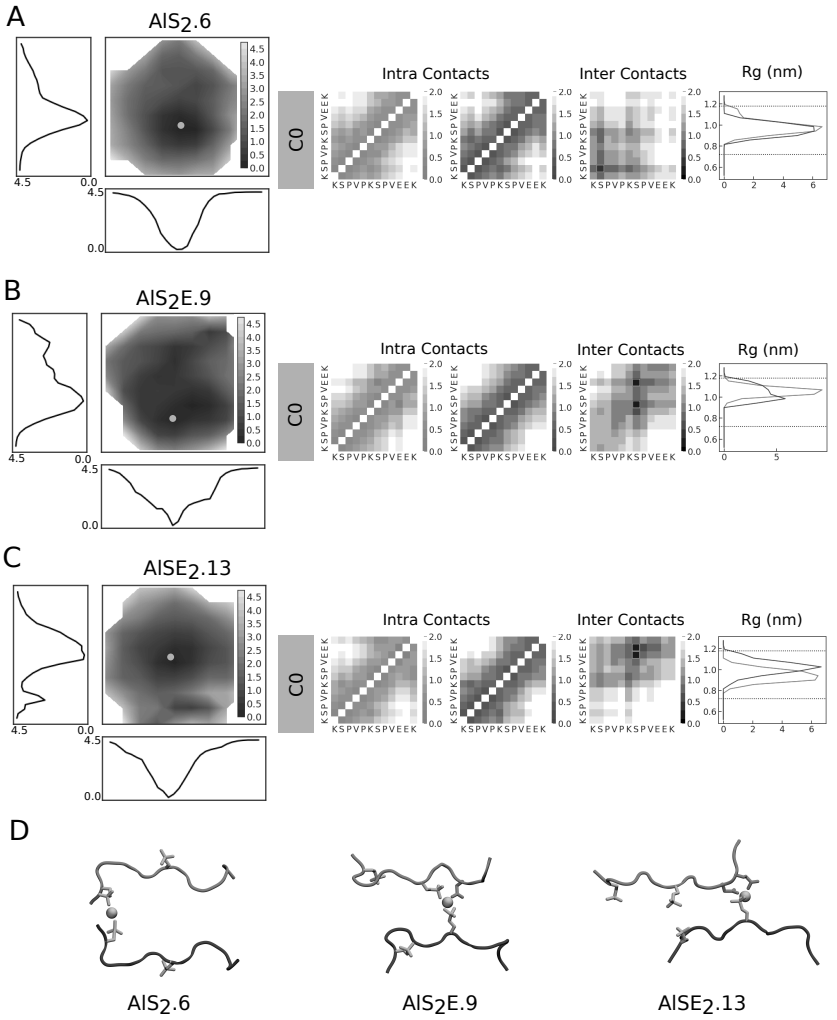


Figure 5.7: PCA free energy surface for bound substate characterization of hyperphosphorylated systems with aluminium. Over the free energy surface, the characterized substates —minima— are identified. The orange substate is always the most populated one. Contact maps for intra-chain and inter-chain side-chain residue interactions are displayed, with the corresponding radius of gyration of each chain. Contact Map's colour code is consistent with that of the radius of gyration and molecule images. Coordination to aluminium is highlighted in red in the Inter-chain contact map.

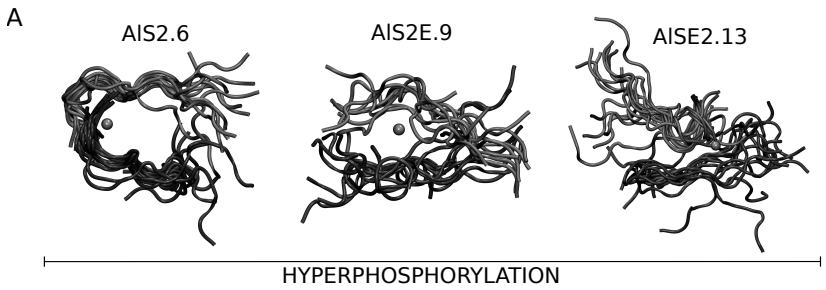


Figure 5.8: Ensemble of conformations of each aluminium coordinated system in its hyperphosphorylation situation. In all three cases, the presence of aluminium releases the residues previously seen to approach the CCZ, promoting extended conformations.

The radius of gyration showed that chains did not tend to contort themselves, although the distribution of the chain coordinated through a single phosphoserine was wider, probably because of the fewer constraints imposed by its mono-coordination to the metal.

Finally, system $AlSE_2$.13 corresponded to the hyperphosphorylated system in which aluminium was coordinated to the SerP7 of one chain and both Glu10 and Glu11 of the other chain. This condition locked the C-terminus end of one chain with the central part of the other one and left both SerP of one chain non-coordinated to aluminium. One ensemble was identified as the general minimum, although another local energy minimum was visible —not labelled—. In agreement with the previous systems, the contact maps and the radius of gyration showed that the extended conformations populated the ensemble of the identified global minimum. A wide distribution of the radius of gyration evidenced the flexibility of the chains to move freely.

In summary, the presence of aluminium metal favoured the bound state of the dimers. Moreover, with a high charge/size ratio, it released residues from the highly ordered and packed CCZ around the SerP residues characterized in the Ref.3 system. Released residues that in Ref.3 were part of

the CCZ were now allowed to explore other conformations, resulting in more extended ensembles than their corresponding aluminium-free hyperphosphorylated references. The type of coordination and location of the metal did not affect this trend. Based on the analysis of these ensembles, it seems reasonable to assume that when several metal cations are present, more than one metal-peptide coordination is possible simultaneously for every chain because the presence of aluminium releases residues that in turn would be able to coordinate other metallic ions. Overall, the charge seemed to play a role of paramount importance in chain-chain interactions in these peptides that lack a well-defined secondary structure. In this sense, phosphorylation and the presence of highly charged metals such as aluminium had a profound effect on the conformational ensembles of these peptides and the equilibrium between bound and unbound states.

5.4 Discussion

Dementia and other neurodegenerative diseases have become a global health issue in recent decades. If predictions are correct and no effective clinical treatment is developed, more than 80 million people will live with dementia globally, with a financial burden of more than US$3 trillion [273].

Although the reason for the development of AD, the presence of protein aggregates of amyloid-β plaques —Aβ— and neurofibrillary tangles —NFT—, was characterized long ago and a large body of literature and work has been done in this regard [274–279], there is a shocking absence of any effective clinical treatment. This situation is a direct consequence of the failure to characterize convincing mechanisms behind the protein aggregation processes. Despite the proven role of the Aβ and Tau proteins as primary

actors in the aggregation process, many other factors may contribute but are still not fully understood in the development of AD pathology: neuroinflammation, oxidative stress, acetylcholine deficiency, metal homeostasis, glutamate imbalance, mitochondrial dysfunction, etc [95]. The fact that so many factors are involved in the pathology makes AD a multifactorial disorder very challenging to address. Currently, there are at least 9 hypotheses proposed that attempt to explain the pathogenesis of AD, some of them being the amyloid hypothesis, the Tau protein hypothesis, the oxidative stress hypothesis, or the metal ion hypothesis. Overall, a comprehensive theory continues to be still out of reach [95].

One of the common factors for the different existing hypotheses is the abnormal phosphorylation or hyperphosphorylation of key peptides [231–237]. Numerous evidences have linked this phenomenon to the protein aggregation process, especially the formation of NFTs by the Tau protein. Ksiezak-Reding *et al* [280] found that AD brains present 3-4 times higher concentrations of phosphorylated Tau than normal brains. Although the exact mechanism of how hyperphosphorylation affects aggregation is still under debate, its relevance has promoted the development of several treatments that target this factor, for example, by inhibiting its formation [281]. Phosphorylation and especially hyperphosphorylation states have been proven to promote the formation of oligomers or protofibrils [233, 236]. These neurotoxic compounds have a wide potential for damage and disruption in cells. Unfortunately, as often happens in the study of the factors that affect AD, both the molecular mechanisms and the causes that lead to the hyperphosphorylation of peptides are still poorly understood.

Our studies have evidenced that hyperphosphorylation of peptides in conjunction with metal coordination is compatible with the Metal Ion Hy-

pothesis. Hard metallic centres such as Al^{3+} have shown a high affinity for binding phosphates and phosphorylated residues [259, 261, 271, 282], modulating their conformational landscape. Evaluation of single metal-peptide interactions pointed out the existence of complexes capable of reconciling the seemingly contradictory properties of strong multidentate interactions and the ability to create a flexible chain. However, these complexes were in the minority, casting doubt on their ability to support a metal-dependent pathway for peptide aggregation.

In this chapter, we demonstrate theoretically that the phosphorylation post-translational modification by itself, without the need for a metal ion, increases the population of bound states for dimers. This is due to a dense network of lysine-phosphoserine interactions, which are further stabilized by the presence of Na^+ counterions, which neutralize excess negative charge in these regions. These CCZs induce more pronounced chain-chain interactions and alter the equilibrium between bound and unbound states toward bound dimer structures.

The presence of metallic centres can further stabilize the bound states of hyperphosphorylated dimers and alleviate the need to co-localize a large number of counterions Na^+. This modulates the conformational ensemble of chain pairs, naturally promoting more extended conformations, as a result of the release of residues from the CCZ. These residues could coordinate with other peptide chains or metals. Notice that when the hyperphosphorylated dimers are in their bound state, several negatively charged residues are concentrated in the same area, which is an ideal environment for metal coordination. In addition, since the rigid CCZ is likely to have a low conformational entropy, the binding of the metal and the release of residues in this region could be even favoured entropically, although this point requires further

analysis. All of these results point out that the metal-assisted aggregation mechanism is a more likely route for the protein aggregation process than the metal-dependent pathway, integrating factors such as hyperphosphorylation and its promotion of oligomerization, the observed accumulation of metals in AD brains, and the apparent correlation between them. In Figure 5.9 a representation of the molecular mechanism proposed for the metal-assisted pathway in the Metal Ion Hypothesis is presented. This mechanism is a further development from the previously proposed mechanism with metal ions [96], and the last depicted scheme is based on Figure 5 of G. Fasman's work [97].

Under normal conditions, the cell presents typical phosphorylation and de-phosphorylation processes, essential for its correct biological functioning [283]. Any anomaly regarding these operations is consequently addressed by the cell using various mechanisms such as enzyme regulation, autophagy, etc. At some point, due to unidentified factors, the phosphorylation of the proteins abnormally increases. It is not yet clear whether the reason for this phenomenon is an increase in phosphorylation capacity, a decrease in the efficacy of the mechanisms that address the problem, or a combination of these or other molecular operations. This situation results in hyperphosphorylated proteins that are not properly handled and begin to interact unexpectedly. The presence of a multitude of charged residues in the major phosphorylation sites of these proteins results in the establishment of highly dense charged residue grouping zones —CCZ—, where the residues neutralize each other while forming interactions. The ions present in the surroundings also approach these zones, further neutralizing the charge. At this point, there is an equilibrium between the bound and unbound states. That is, hyperphosphorylation promotes the presence of bound states, but this

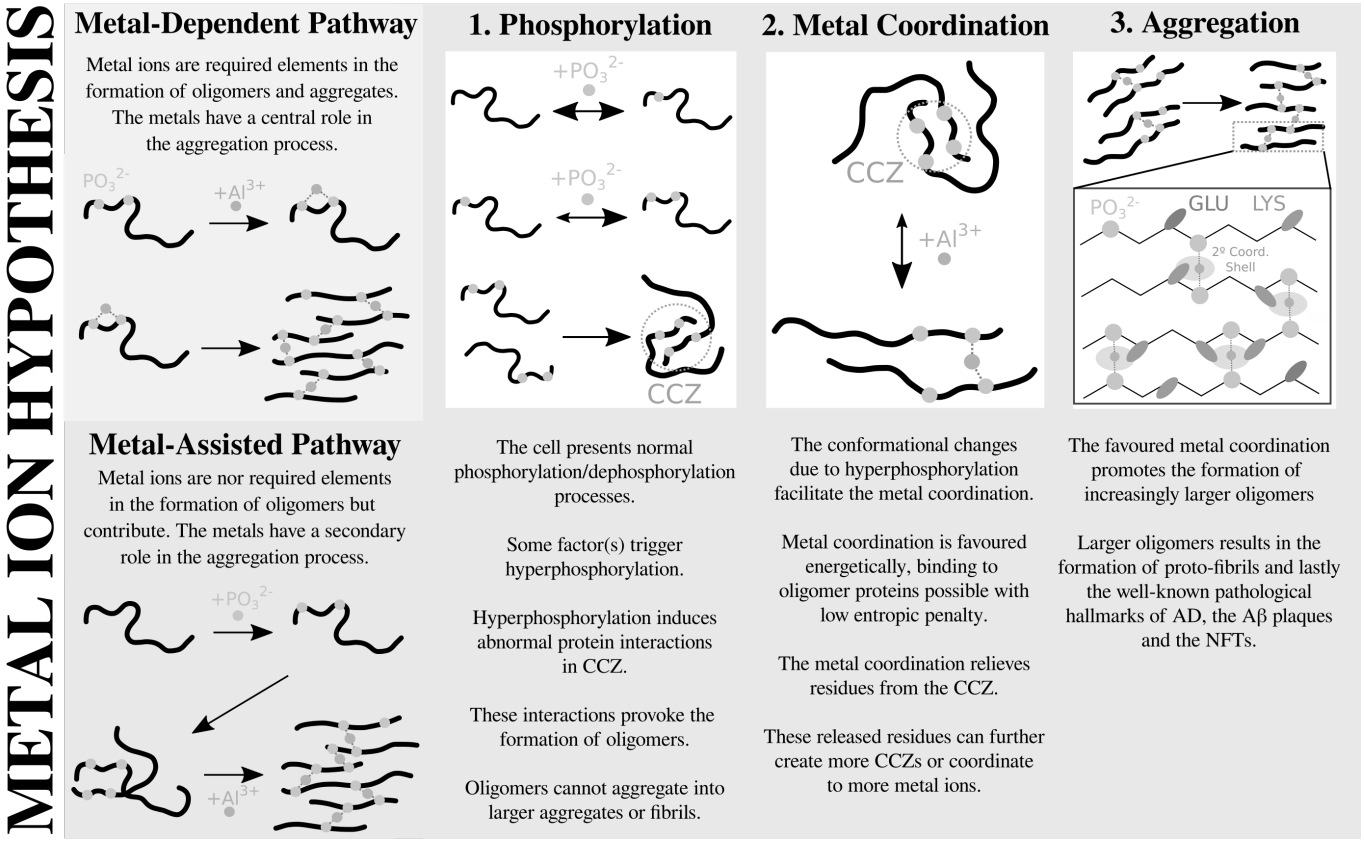


Figure 5.9: The full mechanism proposed for the aggregation of proteins by the metal-assisted pathway. In the first part, the phosphorylation and hyperphosphorylation generate the CCZs that provoke the oligomerization. In the second part, the coordination of the metal takes place, binding oligomers and releasing residues that can further interact. In the last part, assisted by the metal ions, the oligomers aggregate forming larger aggregates.

is not an irreversible process, and therefore, this oligomerization does not lead to the formation of larger fibrils. Notwithstanding, these oligomers are proven to be already neurotoxic and present a wide potential for damage and disruption. In addition, note that the same CCZ that initiates the formation of bound states also prevents the establishment of additional interactions by clustering too many charged residues, which are necessary for the continued growth of the fibril. Interestingly, the formation of these CCZs creates a region that is subject to metal chelation because of the high concentration of negatively charged residues. Metals with a high charge/size ratio would be particularly favoured. However, the interaction with the metal favours irreversible bound states with more extended conformations that can interact with other metals and peptides. This results in the growth of larger oligomers that lastly form the well-known pathological hallmarks of AD, the NFTs.

5.5 Conclusions

Previous studies have shown that metals can modulate the energy landscape and conformational distributions of peptides upon binding. In the context of the Metal Ion Hypothesis, this metal coordination must result in the fulfilment of three requirements to be considered suitable for the protein aggregation mechanism: i) sufficiently favoured energetics; ii) low entropic penalty; and iii) extended conformation exploration. However, previous studies lack multi-chain simulations, a weakness given the nature of an aggregation process. In this chapter, we evaluated the interactions between pairs of chains with different degrees of phosphorylation coordinated in the absence and presence of an aluminium metal cation. The introduction of an additional chain enables the evaluation of aluminium's simultaneous interaction with

multiple peptide chains. Our results show that hyperphosphorylation plays a critical role in establishing inter-chain interactions, consistent with its function as described in the literature. The degree of phosphorylation, especially hyperphosphorylation, increases the interactions between chains, forming regions of high density of charge labelled as CCZz. Phosphorylation shifts the distribution of conformations toward the bound states, compatible with the formation of oligomers. The inclusion of the metal further stabilizes the bound states and has the effect of removing non-coordinated residues from the CCZs. This promotes more extended conformations, enabling the formation of new interactions between the chains. This situation could promote the formation of larger oligomers which in turn would lead to the formation of proto-fibrils, lastly resulting in the well-known pathological hallmarks of AD, the neurofibrillary tangles. In this context, the metal cation is best understood as an auxiliary factor in the aggregation process.

6
Further Work

6.1 Coarse-Grain Models of Neurofilaments

Appropriate modelling of an aggregation process would imply using multiple peptide chains and metal ions. Although REST2 simulations allowed the study of larger aluminium peptide systems in Chapter 5, further inclusion of additional peptides would result in increased difficulties for representatively determining the conformational space of the system, an issue known as the sampling problem. We decided to employ CG modelling to overcome these limitations. This approach facilitates the study of larger systems with several copies of the peptide over sufficiently large timescales, ensuring the collection of reliable distributions [284].

As discussed in the Methodology chapter of this thesis, CG modelling simplifies complex systems, making it possible to study larger systems or simulate longer timescales. This simplification is accomplished by capturing the distributions of various degrees of freedom from all-atom simulations and

projecting them into a CG mapping, typically by merging specific system components. In this work, we employ a CG scheme developed by Prof. Dr. Cecilia Clementi and her lab [72, 172, 173, 285–288]. This method combines a traditional CG framework with an ML approach to construct a transferable CG force field [289]. Transferability in this context refers to the ability of the model to accurately predict the conformational landscapes of proteins not included in its parametrization, similar to the behaviour of transferable force fields of all atoms commonly used in computational studies.

We aim to use this transferable ML-CG strategy to train a model with our short single and dimer peptide systems —from previous Chapters 4 and 5— in the presence and absence of aluminium. The CG modelling would allow us to simulate longer time scales that could enable the evaluation of the aluminium binding and unbinding process. Moreover, the transferability would permit an increase in the number of peptides in the simulation and the evaluation of the already characterized interactions and conformational changes at a larger scale.

6

In this last chapter, I present our latest achievements in the implementation of the mentioned ML-CG. Although we were able to implement our key elements, we are still far from training a suitable model due to our data shortage. Furthermore, the lack of simulations with the aluminium binding/unbinding process —due to its high affinities— prevents its implementation.

6.2 Results

6.2.1 CG representation

The implementation of a system from its all-atom model to the CG model begins by defining the CG particles. Each of the CG particles is called *bead* and the identities of these beads enter the NN as embedding vectors, along with the bead coordinates. In this procedure, the system is redefined and simplified, and the atoms are merged into beads by following a *mapping* scheme that defines the resolution of the CG model. The choice of the best mapping strategy is still a problem under debate [290–292]. For our initial approach, we prioritized capturing the most general processes in the simplest way possible. Therefore, we adopted a straightforward scheme, mapping the atoms of each residue into the C^{α}. This approach involves combining the atoms of each residue in a single α-Carbon (CA) bead and then projecting the all-atom characteristics of the merged atoms of each residue into the corresponding bead. We added embedding vectors for handling the entities of our phosphoserine residues SerP and our aluminium cation.

Figure 6.1 illustrates the resultant CG representation of some peptides after mapping them into their C^{α}: the free NF13 neurofilament system, the phosphorylated $NF13^{S2}$ system coordinated with an aluminium cation, a pair of free NF13 chains, and a system of multiple peptides, all represented by their C^{α} beads. A detailed explanation of the mapping process can be found in the Methodology chapter of this thesis.

6.2.2 Fitting the prior forces

The type of mapping completely defines the interactions between the particles present in the CG system. The sum of these different interactions results

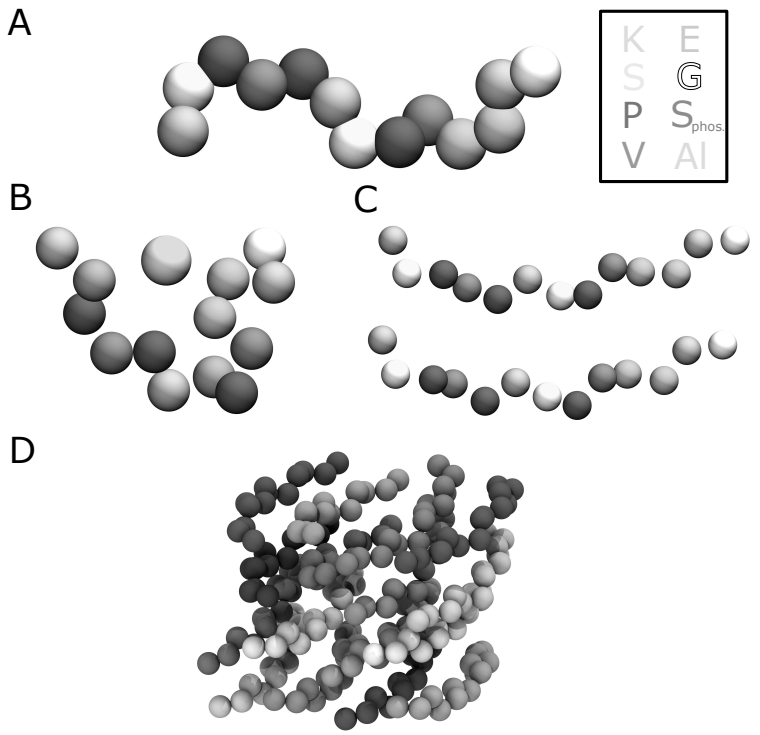


Figure 6.1: A) CG models of the free NF13 neurofilaments; B) $NF13^{S2}$ with aluminium coordinated to SerP2 and both Glu10 and Glu11: C) two NF13 chains; and D) sixteen copies of free NF13 system, using the C^{α} map embedding.

in the total potential energy of the CG model. The forces acting on each bead are then derived from the gradient of the potential energy. The interactions are parametrized using ML techniques by projecting them from the all-atom simulations. However, not all the interactions enter the NN. The covalent interactions between particles —bonds, angles, and dihedrals— and a repulsive non-bonded term are introduced explicitly by statistical fitting to describe them. This avoids sampling of unphysical structures. In addition, since our system integrates an aluminium cation, we included another interaction in these fitted parameters that corresponds to the electrostatic potential. All the functions used to fit interactions are shown in the CG Modelling section of the Methodology chapter. These fitted interactions are called *prior interactions* and the forces resulting only from them are called *prior forces*. Some examples of the results obtained for the parameters fitted between arbitrary CG beads are shown in the following Figure 6.2.

6.2.3 Training the Neural Network

With the prior interactions already defined, the next critical step is training the NN to capture the remaining interactions from the all-atom system and projecting them into the CG model. This step is crucial, as the quality and quantity of data used to train the CG model significantly influence its accuracy. For the training, we used the previously obtained REST2 simulations data: 200 ns length simulations of each system, saving data every 1 ps, resulting in 2.4 TB of data. Figure 6.3 illustrates the optimization process by plotting the validation loss function, which serves as a metric to evaluate the performance of the model in a validation set. During training, the loss function parameters are updated based on the training set, and the resulting model is validated against an unseen validation set. The downward trend

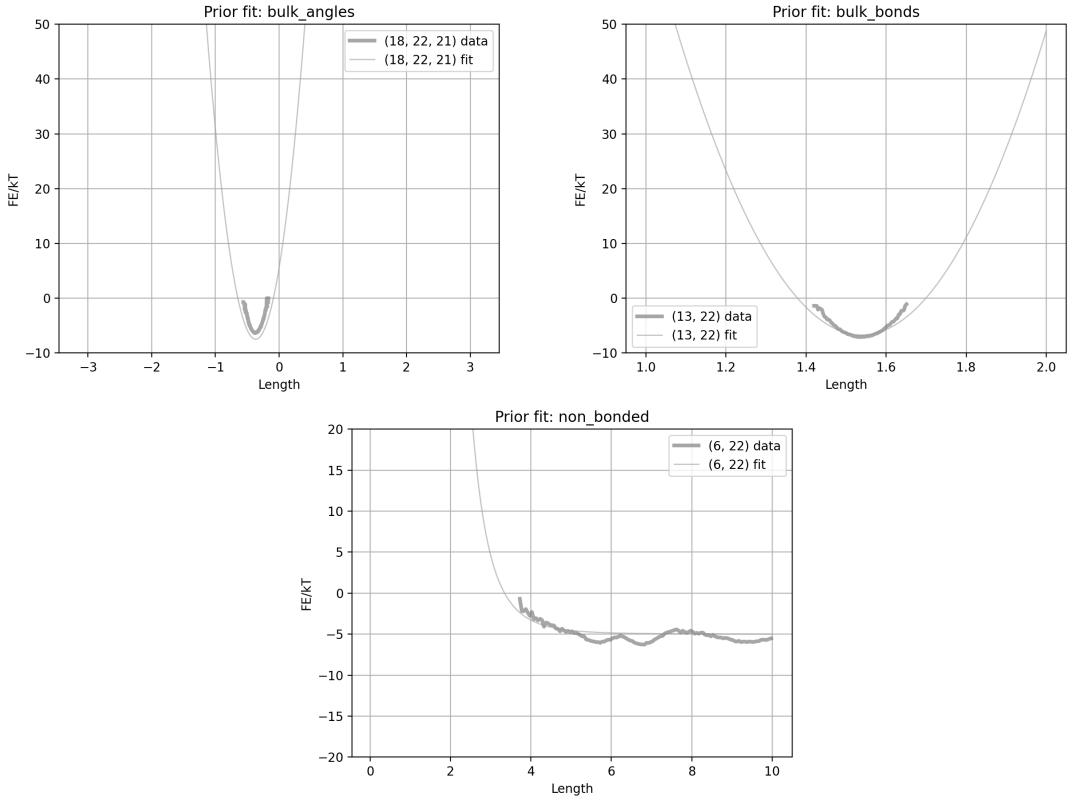


Figure 6.2: Some examples of bond, angle and non-bonded interaction fitting for the NF13 system. The blue lines correspond to the all-atom sampling distribution and the orange line corresponds to the fitting. The numbers in the legend correspond to the internal embedding label of the beads.

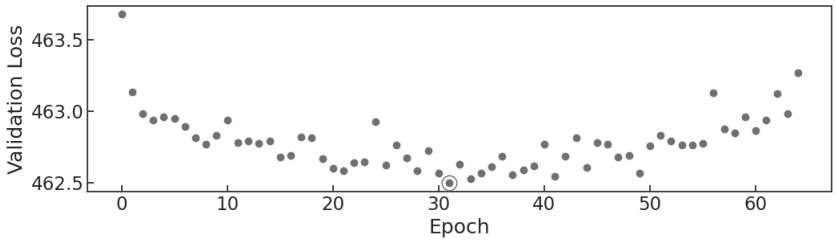


Figure 6.3: The validation loss values are plotted against the training epochs. The best model update is determined based on the minimum validation loss, highlighted in red. It is worth noting that at a certain epoch, the validation loss begins to increase, signalling an overtraining scenario.

of the validation curve reflects the improvements in the model. However, at a certain point, the curve begins to rise instead of maintaining convergence, indicating overtraining. Overtraining occurs when the model shifts from learning general patterns in the training data to over-fitting. As a result, the performance of the model in the validation set deteriorates, leading to an increase in validation loss. Essentially, the model stops generalizing and starts memorizing the training data. After completion of the training process, the updated model with the lowest validation loss value is selected as the best-performing version. The optimized model is then ready for use in simulations.

6.2.4 Some comparisons between CG and All-Atom simulations

Initially, separate models are trained for each system. However, our ultimate objective is to develop a generalized model capable of simulating a diverse range of systems. This stepwise approach allows us to address the unique challenges of each system individually, laying the groundwork for their eventual integration into a unified model.

We can simulate a system using either the complete CG model, which incorporates both the prior statistical fitting parameters and the trained components —Full Model—, or solely the statistical component —Prior Model—. By comparing these models, we can evaluate the extent to which the training process has contributed to the model's performance. Additionally, comparisons with reference all-atom simulations provide insight into how accurately the model captures the target distributions. A larger difference between the distributions of the Full Model and the Prior Model indicates a greater contribution from the trained components. Conversely, a closer match between the Full Model and the all-atom distributions suggests that the model is effectively capturing the reference behaviour.

In Figure 6.4, key observables for the NF13 peptide are presented, including the radius of gyration and intra-chain contact maps for the reference all-atom simulation, the Prior Model CG simulation, and the Full Model CG simulation. The contact maps illustrate interactions between C^{α} atoms or beads in each simulation. Although the AA simulation reveals finer details, the CG simulations primarily capture the general trend of an extended conformation. For the radius of gyration, calculated exclusively for C^{α} atoms or beads, the CG simulations show more compact conformations —lower R_g— and exhibit a narrower distribution compared to the all-atom reference. This discrepancy could be due to insufficient training data or the need for more precise statistical fitting. Addressing these issues could enhance the ability of the model to replicate the reference behaviour more accurately.

Comparison of the CG models —with the Full Model shown in red and the Prior Model in blue— reveals that both exhibit similar distributions and both deviate significantly from the correct distribution of the all-atom reference. With the current dataset, the improvement in the CG model after

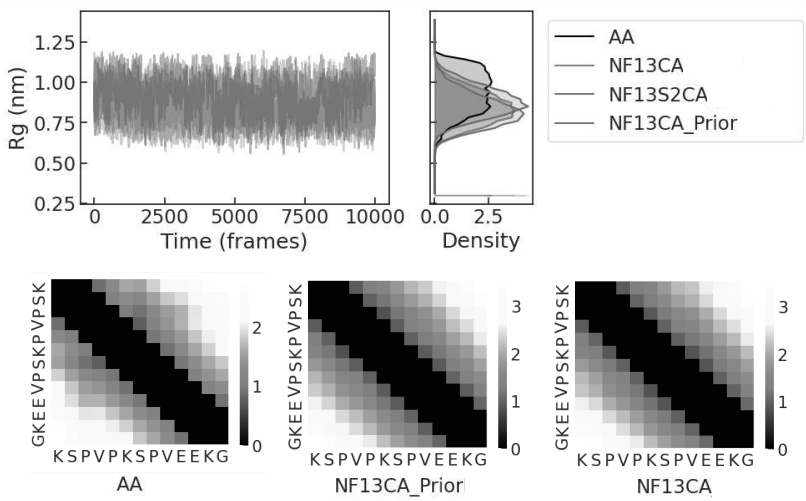


Figure 6.4: Radius of Gyration —top— and Contact Maps —bottom— for different CG models and the All-Atom reference for the NF13. AA corresponds to the All-Atom simulations; NF13C_α corresponds to the Full Model CG; NF13C_α_Prior corresponds to the Prior Model CG; And NF13$^{S2}C_\alpha$ corresponds to another Full Model with phosphorylation introduced.

training is minimal. This indicates that the available data is not sufficient for effective training, and additional all-atom simulation data is needed to accurately capture the reference distributions. Just as an example, consider the work in Majewski *et al.* [172] where model training of 10 residue length Chignolin protein used 186 μs of aggregated time.

6.3 Conclusions

In summary, we have adapted an ML-based CG model to describe NF13 neurofilaments, based on the models developed by the group of Prof. Clementi. A new phosphoserine residue and aluminium cation type were successfully incorporated into the model, and the systems were trained using data from all-atom simulations.

However, the resulting CG simulations demonstrated limitations in capturing the distributions observed in the reference all-atom simulations. A comparison between the Full Model —trained CG model— and the Prior Model —statistical component— indicates minimal learning during training, as both CG models exhibit similar deviations from the all-atom reference. Additionally, the CG simulations display narrower distributions for key observables, such as the radius of gyration, suggesting that insufficient training data or inadequate statistical fitting may be constraining the model's performance. Furthermore, we lack simulations of the aluminium binding and unbinding process, which precludes its implementation.

These limitations can be addressed by incorporating additional all-atom simulation data, refining the statistical fitting process and obtaining aluminium binding/unbinding simulations, which will be the focus of future work.

Conclusions

In this thesis, we evaluated the affinity of aluminium for short disordered peptides, namely a series of *de novo* mimosine-containing peptides and a thirteen-residue peptide —NF13— from the C-terminal IDR of the Human Mid-size Neurofilament protein. We characterized different binding modes and how the binding to aluminium alters the conformational landscape of these peptides. The main conclusions are :

- Due to its small size and high charge (+3), aluminium has a high affinity for interaction with the negatively charged residues of these peptides, namely deprotonated mimosine side chains, and glutamate and phosphorylated serines in NF13.

- Regarding aluminium coordination to mimosine-containing peptides:

  - Mimosine-containing peptides showed strong interactions with aluminium, coordinating three residues in a bidentate mode each. Theoretical calculations revealed that the binding energies are comparable to established chelators such as deferiprone and citrate, sufficient to offset the entropic loss associated with metal binding —1 to 2 kcal/mol per residue—.

  - The length of the peptide, the type of residue, and the intra-chain interactions can modulate this entropic penalty, enhancing the metal-binding affinity. This highlights promising strategies for designing peptides with optimized chelation capabilities.

- Regarding aluminium coordination with NF13 neurofilaments:
  - Aluminium adopts various coordination modes depending on the peptide sites and residue types, showing a significant capacity to influence the conformational landscape of these peptides. Three-coordinated systems are the most energetically favoured, especially when negatively charged residues, glutamate and phosphoserines, are involved.
  - Aluminium binding can either slightly alter the ensemble of accessible conformations or significantly confine the peptide to contorted states, and this is reflected in the changes in conformational entropy upon binding.
  - We have identified three key requirements for potential candidates to undergo oligomerization in the presence of metal ions:
    a The local formation energy of a given coordination shell should be competitive with respect to efficient biochelators of aluminium, such as citrate.
    b The configurational entropy loss upon metal binding should be low enough to allow for sufficient conformational exploration.
    c Extended arrangements must be accessible, as they serve as precursors to aggregation.
  - Our results demonstrated that most aluminium-peptide systems do not meet these three criteria simultaneously. However, there are still some structures that can reconcile the seemingly contradictory properties of strong multidentate interactions and the ability to create a flexible peptide chain.

– The study of systems containing two peptide chains revealed that phosphorylation and especially hyperphosphorylation promote interchain interactions and bound states, consistent with its established role in the literature.

– Aluminium could further stabilize the bound states of two hyperphosphorylated chains and promote extended conformations of the peptides, which, in turn, would favour further oligomerization of these structures, which can ultimately contribute to the formation of neurofibrillary tangles, hallmarks of AD pathology.

Conclusiones

El estudio de la influencia del catión metálico de aluminio Al^{3+} en el paisaje energético conformacional de las proteínas llevado a cabo en esta tesis expone resultados relevantes asociados a las interacciones metal-proteína.

En el Capitulo 3 investigamos los conjuntos conformacionales de los péptidos con mimosina y como éstos cambian tras la coordinación del metal multivalente Al^{3+}. Múltiples estudios han revelado los diversos efectos beneficiosos que tiene el aminoácido no-proteinogénico de mimosina. Estos efectos están estrechamente relacionados con su capacidad para coordinarse y/o quelar metales. Esta situación hace de la mimosina una molécula de gran interés desde el punto de vista del diseño y optimización de péptidos quelantes para el tratamiento de exceso de metales en el cuerpo. Para trabajar con la mimosina primero desarrollamos los parámetros de un modelo coordinado de mimosina-aluminio basándonos en cálculos cuánticos. A continuación llevamos a cabo una serie de simulaciones de dinámica molecular (MD) de 1 μs de tiempo con péptidos de mimosina y otras variantes. Éstas simulaciones nos permitieron evaluar la entropía configuracional de los péptidos. Aunque compleja de estimar, ésta propiedad termodinámica es crucial ya que permite conocer el coste energético asociado a la coordinación del centro metálico. Además, también determinamos las afinidades de coordinación de tres residuos de mimosina al catión de aluminio.

Nuestros resultados mostraron una interacción entre el catión de aluminio y las mimosinas coordinadas lo suficientemente fuerte como para compensar

el coste entrópico asociado a la coordinación, de alrededor de 1-2 kcal/mol por residuo.

Descubrimos que diferentes péptidos pueden presentar afinidades de coordinación similares debido a esferas de coordinación equivalentes. Sin embargo, esto no ocurría para los costes entrópicos asociados a la coordinación, donde hubo mayor heterogeneidad a pesar de las equivalencias en dichas esferas de coordinación. Ésto podría indicar que regular la entropía configuracional es una estrategia prometedora en el desarrollo y diseño de péptidos quelantes, mejorando las afinidades de coordinación de metales. Algunos de los ajustes estructurales que podrían llevarse a cabo son la introducción de residuos de prolina y la optimización de la longitud de la secuencia de aminoácidos. Ambos factores podrían usarse para facilitar las interacciones entre anillos de mimosina y por tanto reducir el coste entrópico de la quelación del metal. El control adecuado de la longitud de la secuencia es un punto crítico, ya que el tamaño del péptido esta inversamente correlacionado con la asimilación biológica de éste. Para acabar, nuestros cálculos mecano-cuánticos de las energías de formación de los péptidos indicaron que los péptidos son potencialmente buenos agentes quelantes, con energías de formación comparables con otros agentes quelantes naturales de aluminio en el entorno biológico, como el citrato u otros agentes quelantes presentes en el mercado, como la deferiprona.

En el Capitulo 4 investigamos los conjuntos conformacionales de una serie de sistemas formados por aluminio y fragmentos de trece residuos pertenecientes a un neurofilamento (NF13). Pacientes de enfermedades como la del Alzheimer presentan en el cerebro agregados patológicos de varias proteínas. A pesar de ésto, el mecanismo especifico por el cual estas proteínas terminan agregando patológicamente es aún hoy desconocido, existiendo

multitud de hipótesis que tratan de integrar las diversas evidencias y detallar el posible mecanismo molecular del proceso. Una de éstas hipótesis es la Hipótesis del Ion Metálico, que establece que la agregación de estas proteínas se debe a la presencia de iones metálicos. Aunque existen pruebas suficientes como para poder vincular a ciertos metales con el proceso de agregación de proteínas, el papel particular que juegan estos metales aún no está establecido con claridad. Hemos utilizado el péptido NF13, perteneciente a la región C-terminal de la proteína NF-M —una región considerada intrínsecamente desordenada (IDR)—, para caracterizar las interacciones entre el catión metálico de aluminio y las proteínas involucradas en el proceso de agregación. Evaluamos la coordinación metal-péptido y determinamos una serie de factores que creemos que podrían ser de utilidad para comprender el papel específico del metal en el proceso. Para ello y continuando con el esquema de trabajo del Capítulo 3, llevamos a cabo una serie de simulaciones de dinámica molecular (MD) de 1-2 μs de tiempo. Éstas simulaciones nos permitieron determinar la entropía configuracional de los sistemas, pudiendo así caracterizar el coste entrópico asociado a la coordinación del catión metálico. Además, llevamos a cabo una serie de profundos análisis de propiedades estructurales —mapas de contactos, radios de giro, análisis de estructura secundaria y demás— para evaluar el impacto del centro metálico en las conformaciones de los péptidos.

El centro metálico se coordinó al péptido NF13 en diferentes variantes según la posición del metal en la secuencia y los residuos coordinados. Los cálculos de energía de formación determinaron que los sistemas donde el metal estaba coordinado de manera tridentada eran los más favorables. Además, dentro de este grupo de sistemas tri-coordinados, aquellos en los que el metal se unía a la serina fosforilada SerP7 tenían estructuras más

flexibles que aquellos en los que se unía a la serina fosforilada SerP2, que presentaban estructuras más rígidas.

Los cambios conformacionales asociados a la presencia del metal pudieron ser divididos en dos grupos, uno con estructuras mas rígidas y contorsionadas y otro con estructuras más flexibles y extendidas. El análisis energético y entrópico mostró que existía una correlación directa entre estos factores, incrementando el coste entrópico al aumentar las interacciones.

Éstos resultados nos sirvieron para establecer un criterio de tres requerimientos para identificar sistemas metal-proteína como posibles candidatos para iniciar el proceso de agregación. Estos tres requerimientos son: (i) tener una energía de formación competitiva a nivel biológico, (ii) tener un coste entrópico modesto asociado a la exploración del espacio conformacional y (iii) explorar conformaciones de estructura extendida que serían los puntos de nucleación del proceso de agregación. A pesar de que la mayoría de las estructuras no cumplían éste criterio, sí que fue posible encontrar estructuras que lo cumplieran, especialmente con el metal coordinado de forma tridentada a la SerP7.

Sin embargo, este Capítulo 4 tenía una limitación importante. A pesar de que el proceso de agregación comprende multitud de proteínas y potencialmente cationes metálicos, solo se utilizó una copia de cada. Para tratar de paliar esta limitación, en el Capítulo 5 incluimos una copia adicional del péptido NF13. De esta forma, evaluamos las interacciones simultaneas que ocurren tanto entre el catión metálico y los péptidos, como entre ambos péptidos directamente. Para ello, llevamos a cabo simulaciones de intercambio de replicas con temperado del soluto (REST2), una técnica que permite explorar el espacio conformacional de forma más eficiente.

Nuestros resultados indicaron que el aumento del nivel de fosforilación

—una modificación postraduccional— aumenta a su vez las interacciones entre los péptidos, especialmente cuando el sistema se encuentra completamente fosforilado. En ésta situación, denominada hiperfosforilación, se llegaron a formar zonas de condensación de carga (CCZ) donde los residuos de diferente carga se aproximan para neutralizarse mutuamente, incluyendo contra iones presentes en el medio como iones de sodio (Na^+). El aumento de interacciones entre péptidos podría ser un indicio de la formación de oligómeros, agregados pequeños que han demostrado ser enormemente patológicos. La adición del aluminio a estos sistemas hiperfosforilados liberó residuos de las CCZs. Esto resultó en estructuras más extendidas y flexibles cuyos residuos liberados podrían continuar formando nuevas CCZ o coordinando más cationes metálicos. Estos dos fenómenos, la formación inicial de oligómeros y la posterior liberación de residuos y extensión de las cadenas por parte del metal podrían indicar que en el proceso de agregación de proteínas, el metal juega un papel secundario de asistencia.

Esta idea fue desarrollada para proponer un posible mecanismo para la agregación de proteínas en presencia de metales, donde la coordinación del metal no es necesaria para la formación de agentes patológicos como los oligómeros, pero sí que resulta necesaria en la posterior formación de agregados presentes en el cerebro de los pacientes con Alzheimer como los ovillos neurofibrilares.

Publications

- Influence of metal binding on the conformational landscape of neurofilament peptides D. Silva-Brea, D. De Sancho, X. Lopez *Phys. Chem. Chem. Phys.* **25**, 26429 (2023)

- Designing mimosine-containing peptides as efficient metal chelators: Insights from molecular dynamics and quantum calculations D. Silva-Brea, J. Aduriz-Arrizabalaga, D. De Sancho, X. Lopez *J. Inorg. Biochem.* **264**, 112807 (2025)

A

S.I. - Chapter 3

Systems

Studied systems are shown in Figure A.1 without the hydrogen atoms.

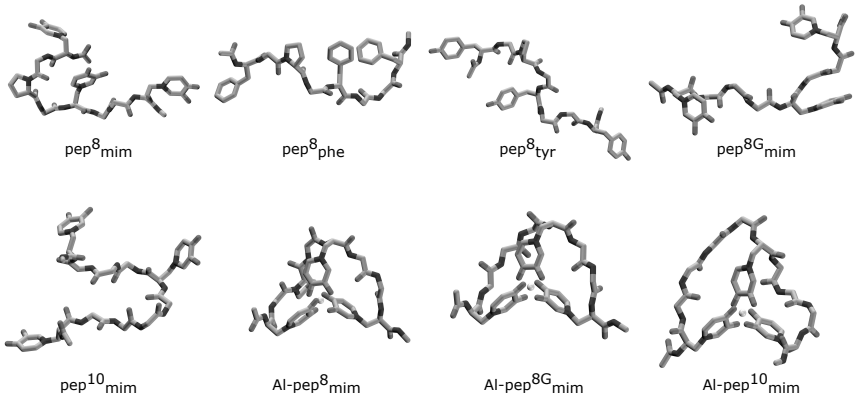


Figure A.1: All eight mimosine-containing peptides or derivatives studied. The last three correspond to the Al(III) bound structures. Arbitrary configurations are depicted.

Methods

R-R Interactions

R-R interactions were characterized by taking into account both the distance of interaction between the rings of the side chains and the angle formed by the two normal vectors corresponding to the planes of the rings, as represented in the following Figure A.2A. Distances were computed as the average distance between all pairs of atoms between rings. Distance and angles are shown in Figure A.2B. The blue squares in the plot represent the cut-off to consider the R-R interactions. The cut-off for the distance was 0.52 nm, and the cut-off for the angle was from 0º—20º or 160º—180º.

Resultant interaction data —pie chart in Chapter 3— was presented as fractions of interaction over the whole trajectory. The R1-R2 interactions were represented by the red colour in the pie charts; the R2-R3 interactions by the blue colour; the R1-R3 interactions by the green colour; the R1-R2-R3 interactions, independent of their spatial ordering, by the purple colour; and the absence of any R-R interactions by the grey colour. The corresponding distance and angle plots for each system can be observed in Figures A.3 and A.4.

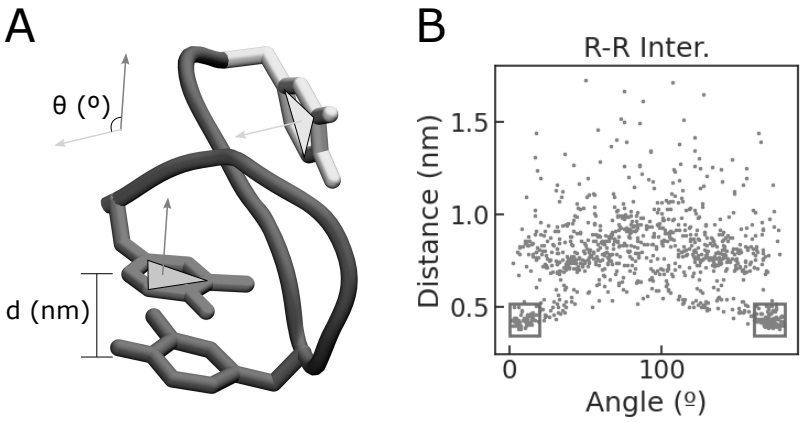


Figure A.2: A) Visual representation of the distance and the angles used to characterize the interactions between rings. The colour scheme is only for clarity. B) Distance and angle plot. The blue squares correspond to the regions considered for accepting an R-R interaction. The limits for these squares were 0.52 nm distance and 0º—20º or 160º—180º angle deviation between planes.

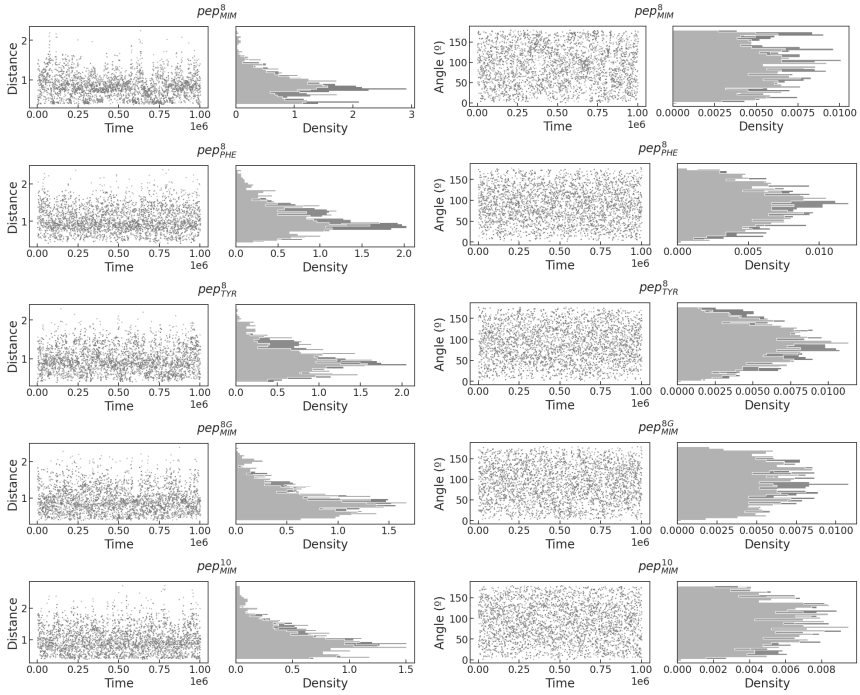


Figure A.3: Temporal series and density distributions of distance and angles for each unbound system. Notice the bimodal shape of the distance plot distributions —left—, showing that a short-distance state existed. The angle distribution plots —right— also shown a more "squared" shape for systems with more sandwich-like interactions between rings —pep^{8}_{mim}, pep^{8G}_{mim}, pep^{10}_{mim}—, while the lack of sandwich-like R-R interaction appeared with a more "rounded" shape —pep^{8}_{phe}, pep^{8}_{tyr}—. The colours correspond to the rings involved in the interaction, following the same code as in the main text: R1-R2 red, R2-R3 blue, and R1-R3 green. R-R-R was not considered here.

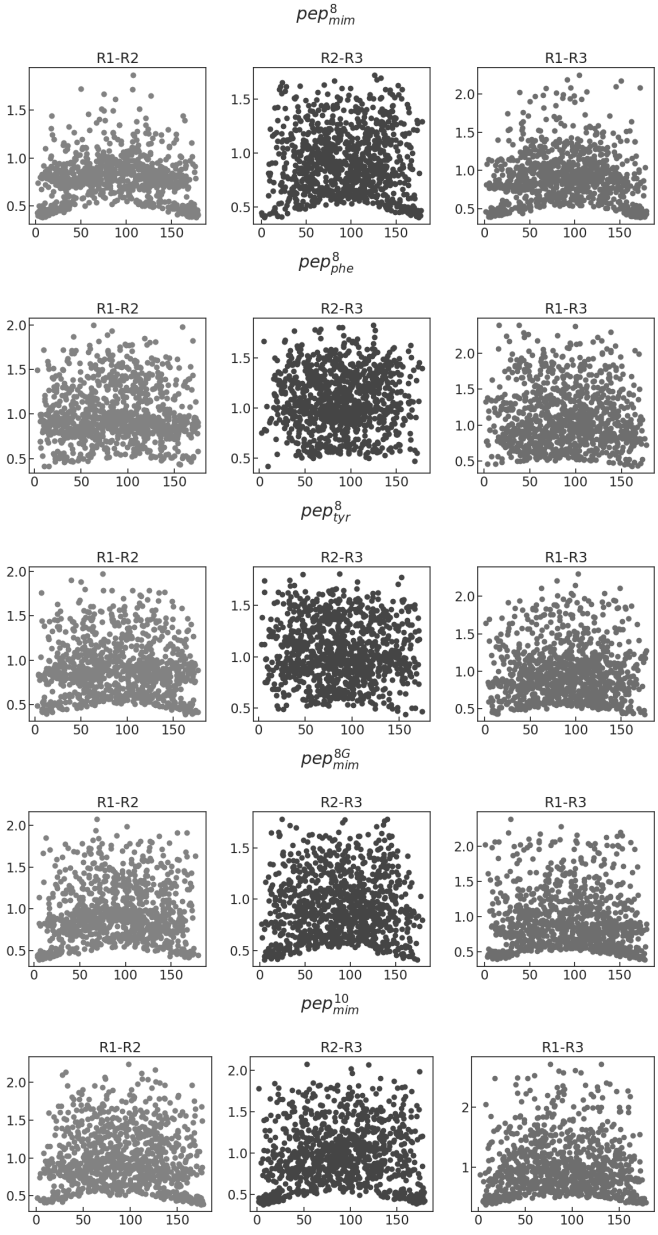


Figure A.4: Distance and angle plots for each unbound system. This figure is a combination of the previous Figure A.3 left and right plots. Notice how the population of sandwich-like interaction region —indicated in Fig. A.2B, blue squares— decreased for systems with phenylalanine —pep^{8}_{phe}— or tyrosine —pep^{8}_{tyr}— residues. The colour code is the same as the previous image.

Results

Convergence of simulations

For checking the convergence of simulations, the R_g of systems was computed at 25%, 50%, 75%, and 100% of the total simulation time. The distributions of R_g indicated that simulations converged, showing negligible differences between the distributions computed for 75% and 100% time. These analyses are shown below for three representative systems, namely, pep^{8}_{mim}, pep^{8G}_{mim} and $Al.pep^{8}_{mim}$, and were equivalent for the rest of the systems.

Radius of gyration

Statistical values for the R_g were determined by block averaging of six blocks. The mean was determined as the mean of means, and the error as the standard error of the mean —SEM—.

Table A.1: R_g of systems calculated using the corresponding utility of MDTraj package.

System	R_g (nm)
pep^{8}_{mim}	0.55 ± 0.02
pep^{8}_{phe}	0.59 ± 0.01
pep^{8}_{tyr}	0.58 ± 0.01
pep^{8G}_{mim}	0.57 ± 0.01
pep^{10}_{mim}	0.61 ± 0.01
$Al.pep^{8}_{mim}$	0.53 ± 0.00
$Al.pep^{8G}_{mim}$	0.54 ± 0.00
$Al.pep^{10}_{mim}$	0.60 ± 0.00

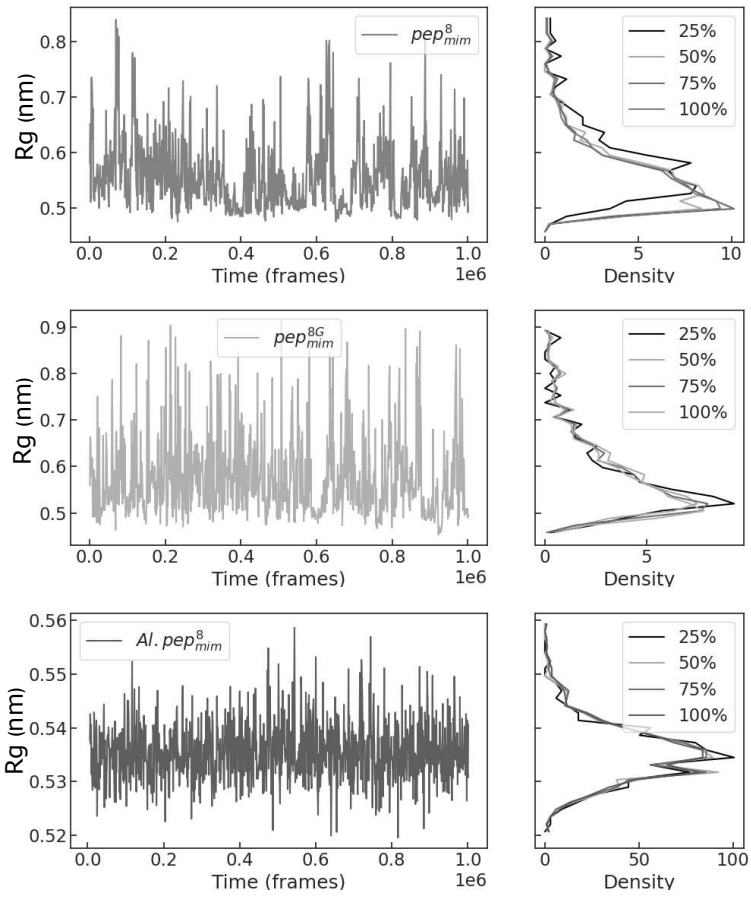


Figure A.5: Convergence plots of R_g for three representative systems. The distributions showed that simulations were converged, with very little changes in the shape for the 75% and 100% time.

Ramachandran Plots

Ramachandran plots of each system for each residue are shown below, calculated using the distributions of ϕ and ψ dihedral angles computed using the corresponding utility of the MDTraj package.

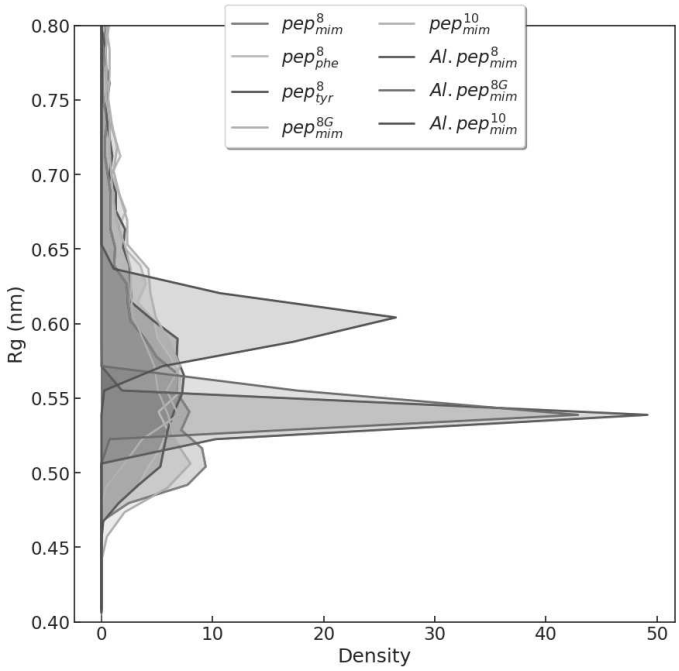


Figure A.6: Distribution of all R_g for all systems side by side. Notice how the extension of the variability of R_g for unbound systems, from 0.45 nm to 0.80 nm, gets decreased upon binding, reduced to a range of 0.50 nm to 0.65 nm.

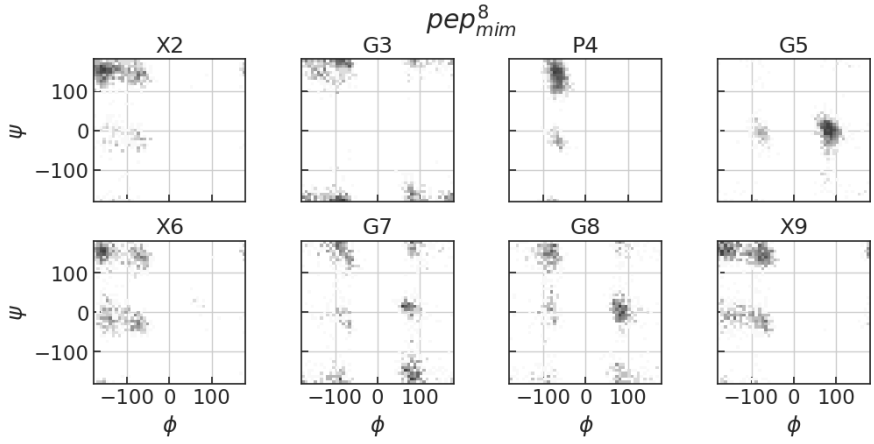


Figure A.7: Ramachandran plots for individual residues of system pep^{8}_{mim}

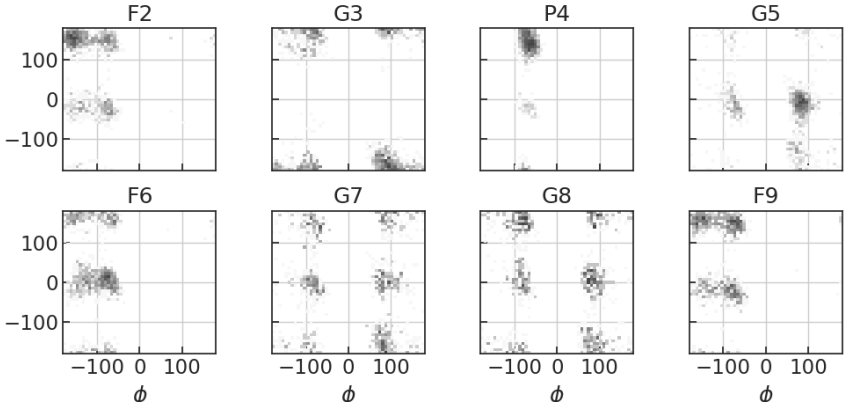


Figure A.8: Ramachandran plots for individual residues of system $\text{pep}^{8}_{\text{phe}}$

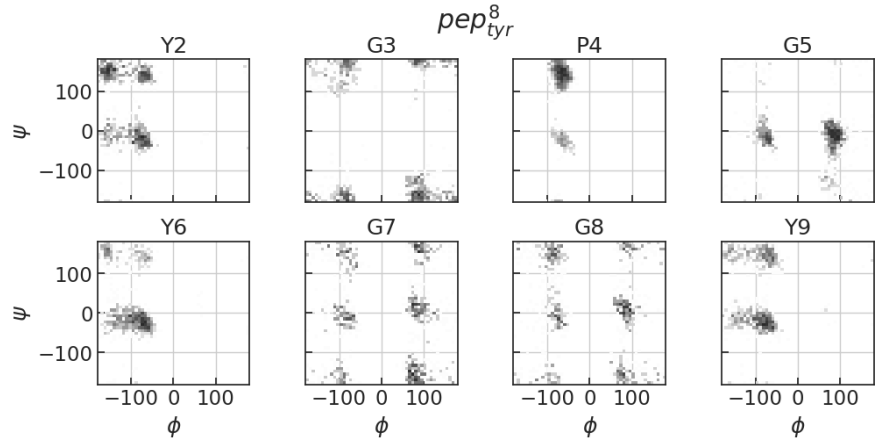


Figure A.9: Ramachandran plots for individual residues of system $\text{pep}^{8}_{\text{tyr}}$

A

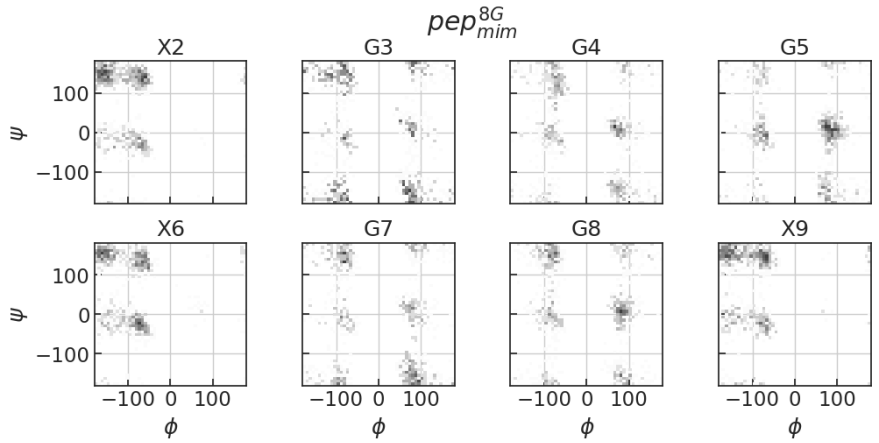


Figure A.10: Ramachandran plots for individual residues of system $\text{pep}^{8\text{G}}_{\text{mim}}$

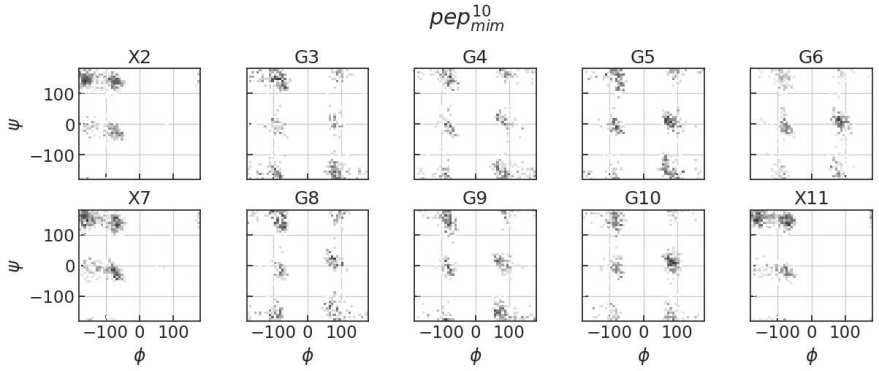


Figure A.11: Ramachandran plots for individual residues of system $\text{pep}^{10}_{\text{mim}}$

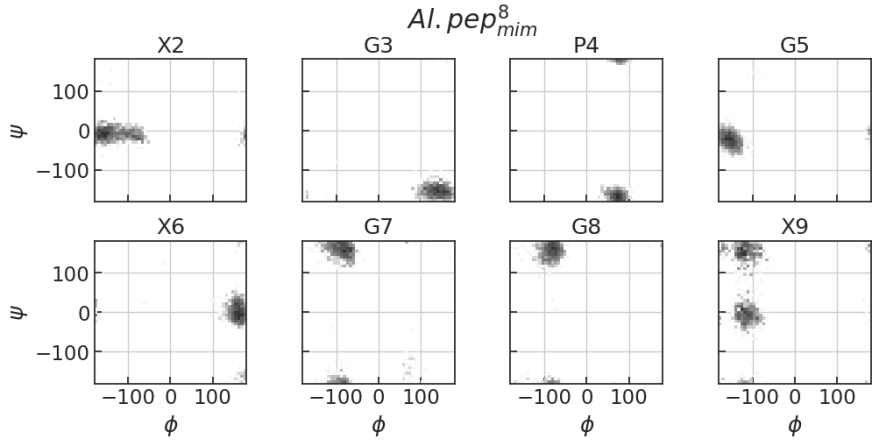


Figure A.12: Ramachandran plots for individual residues of system Al.pep$^{8}_{\mathrm{mim}}$

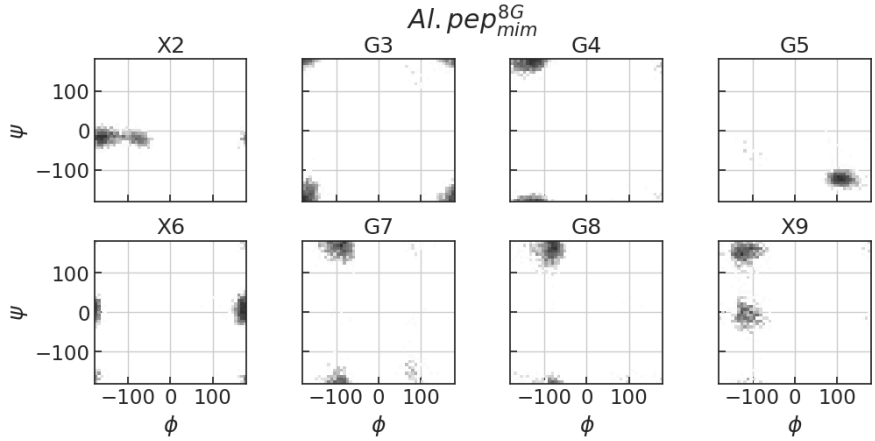


Figure A.13: Ramachandran plots for individual residues of system Al.pep$^{8\mathrm{G}}_{\mathrm{mim}}$

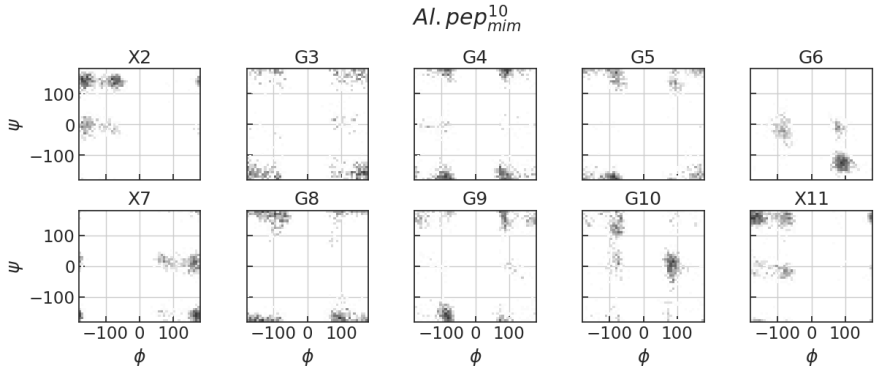


Figure A.14: Ramachandran plots for individual residues of system Al.pep$^{10}_{\text{mim}}$

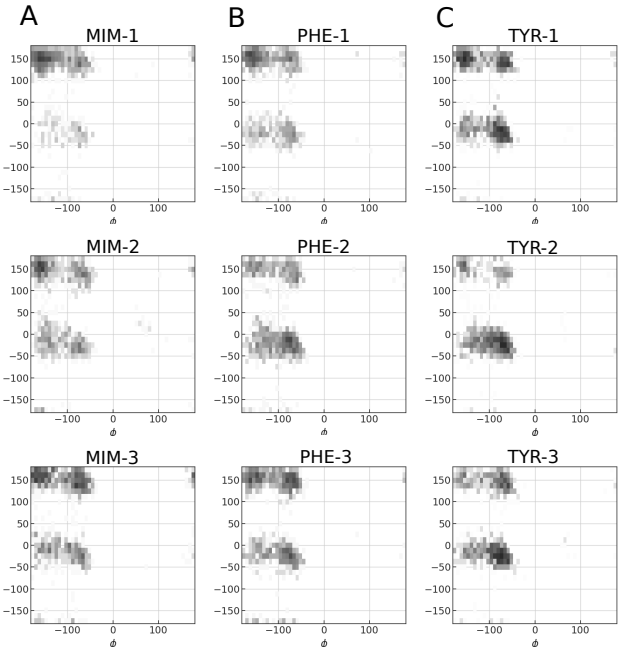


Figure A.15: Ramachandran plots for mimosine or equivalent residues —phenylalanine and tyrosine— for unbound systems. Each column correspond to one system: A) pep$^{8}_{\text{mim}}$; B) pep$^{8}_{\text{phe}}$; C) pep$^{8}_{\text{tyr}}$. Each row shows the equivalent ring containing residue.

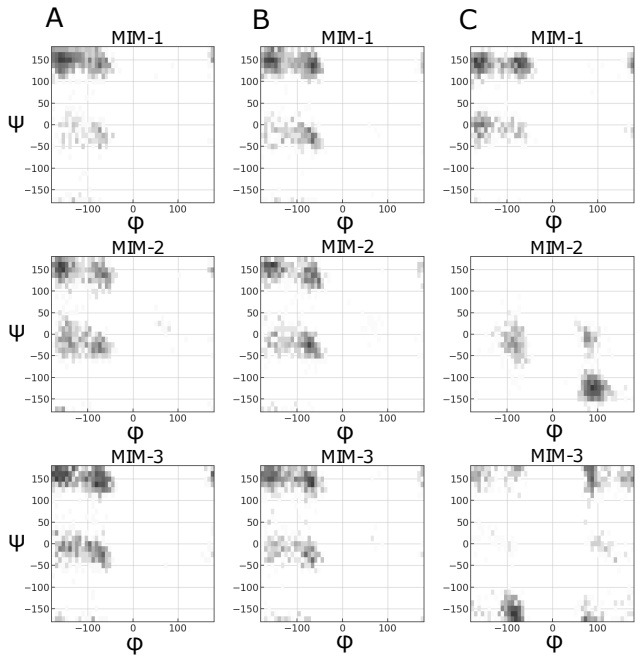


Figure A.16: Ramachandran plots of mimosine residues for mimosine-containing unbound systems. Each column correspond to one system: A) pep^{8}_{mim}; B) pep^{8G}_{mim}; C) pep^{10}_{mim}. Each row shows the equivalent ring containing residue.

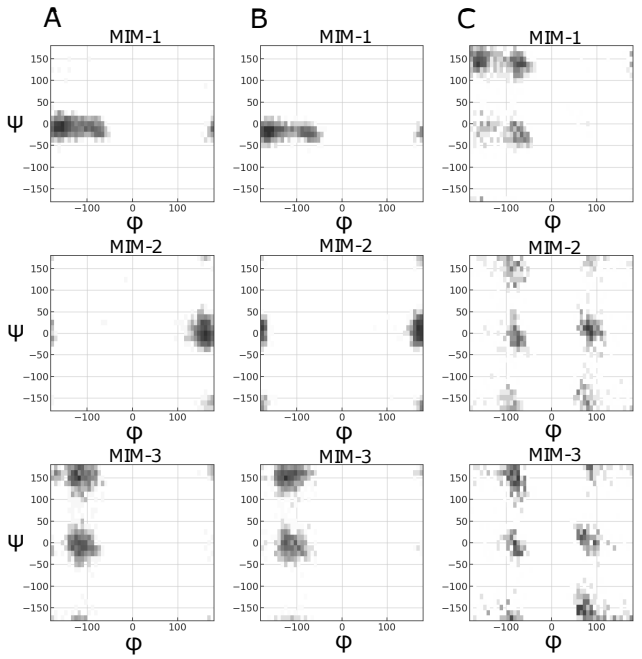


Figure A.17: Ramachandran plots of mimosine residues for mimosine-containing bound systems. Each column correspond to one system: A) Al.pep$^{8}_{\mathrm{mim}}$; B) Al.pep$^{8\mathrm{G}}_{\mathrm{mim}}$; C) Al.pep$^{10}_{\mathrm{mim}}$. Each row shows the equivalent ring containing residue.

Entropy

We used a bootstrapping technique to recover a distribution of configurational entropies for each system, shown in Figure A.18. 500 (M) resamples were done for each system, where the original trajectory was divided into 50 (N) blocks. For each system, the determined mean and error were the mean of means and the combined standard deviation, respectively.

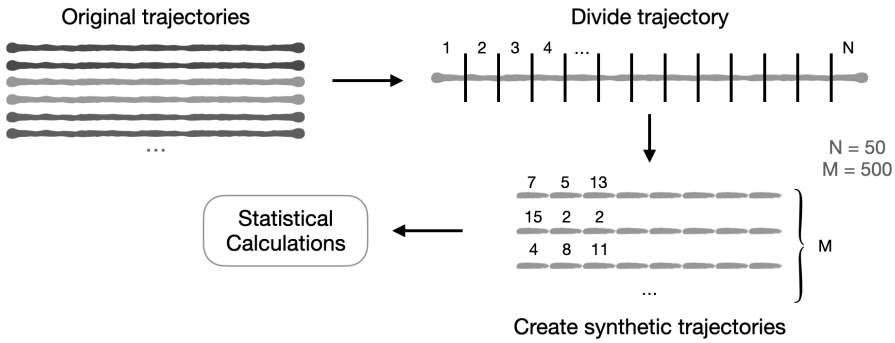


Figure A.18: Scheme followed for the bootstrapping carried out for the S_{MIST} determination. Original trajectories were divided into N pieces (N=50), and then pieces were arbitrarily combined to form M number of synthetic trajectories (M=500). This allowed the determination of the mean value and the standard deviation.

As indicated in the main text, our peptides presented structural differences that, given the extensive property of the entropy, precluded their comparison. Therefore, we addressed this issue by considering only the structural elements common to all the systems, that is, the backbone.

Moreover, the backbone presents three DOFs: bonds, angles and dihedrals; which are grouped under the BAT degrees of freedom. Although the contributions of bonds and angles are larger in the total configurational entropy, they are often very similar between the folded and unfolded or bound and unbound states of systems. For this reason, there are strategies where

these DOFs are neglected and only the dihedrals are used. The two former —bonds and angles— are called *hard* degrees of freedom, while the latter —dihedrals— is labelled as *soft* degrees of freedom [180]. The strategy of determining only the configurational entropy using the dihedrals DOF is known as *soft configurational entropy*. This approximation, which gets rid of the noise, has been supported by other authors [293].

In this particular case, since the mimosine-containing bound and unbound systems shared the same particular structural elements, we computed the soft configurational entropy using two schemes: the *backbone-dihedrals* only —similar to the rest of the systems— and *all-dihedrals*, which was specific for these systems only.

Table A.2: $T\Delta S_{\mathrm{MIST}}$ values for each system. Bound systems have both *all-dihedrals* (All) and *backbone-dihedrals* (BB) values, while unbound systems present only the latter term. Calculations were performed using PARENT software and a bootstrapping technique.

System	$T\Delta S_{\mathrm{MIST}}$ (All) (kcal/mol)	$T\Delta S_{\mathrm{MIST}}$ (BB) (kcal/mol)
$\mathrm{pep}^{8}_{\mathrm{phe}}$	-	$0.67 \pm 0.21^{pep^{8}_{mim}}$
$\mathrm{pep}^{8}_{\mathrm{tyr}}$	-	$0.62 \pm 0.22^{pep^{8}_{mim}}$
$\mathrm{pep}^{8\mathrm{G}}_{\mathrm{mim}}$	-	$2.63 \pm 0.25^{pep^{8}_{mim}}$
$\mathrm{Al.pep}^{8}_{\mathrm{mim}}$	$-15.34 \pm 0.46^{pep^{8}_{mim}}$	$-7.43 \pm 0.21^{pep^{8}_{mim}}$
$\mathrm{Al.pep}^{8\mathrm{G}}_{\mathrm{mim}}$	$-16.47 \pm 0.38^{pep^{8G}_{mim}}$	$-8.56 \pm 0.23^{pep^{8G}_{mim}}$
$\mathrm{Al.pep}^{10}_{\mathrm{mim}}$	$-10.50 \pm 0.49^{pep^{10}_{mim}}$	$-4.59 \pm 0.28^{pep^{10}_{mim}}$

QTAIM calculations

All QTAIM calculations were analyzed making use of in-house Python codes. To ensure the accuracy of these calculations, we looked at the charges of all groups —Figure A.19, Figure A.21 and Figure A.23, summarized in Table A.3—. All three coordination spheres displayed identical and coherent charges. On the other hand, we calculated the total interaction energy and decomposed it into its covalent and electrostatic contributions —see Figure A.20, Figure A.22 and Figure A.24—. As expected, all aluminium-mimosine (Al-Mim) interactions were mostly electrostatic-driven. Similarly, we captured the slight repulsion that mimosine residues experience with each other. By making use of the total interaction energies displayed in Table A.4, we calculated the values displayed in Chapter 3.

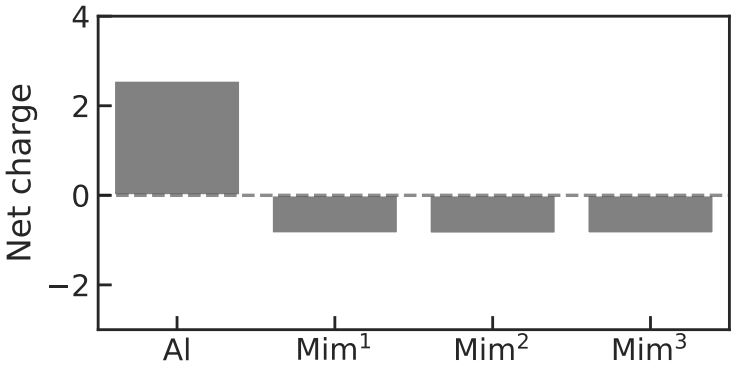


Figure A.19: Charges calculated from QTAIM for system $Al.pep^{8}_{mim}$.

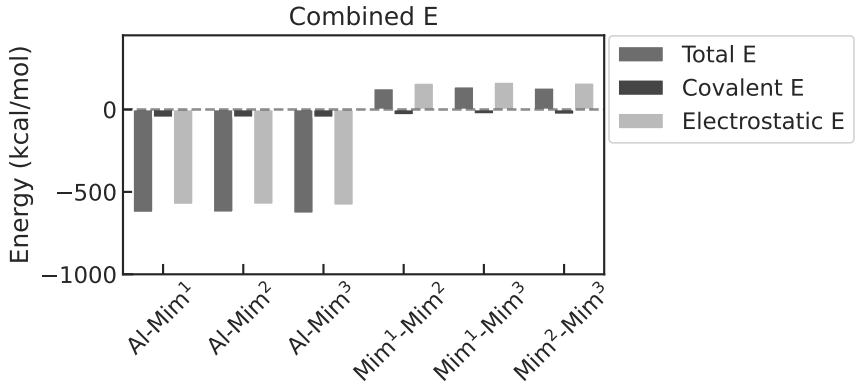


Figure A.20: Interaction energies calculated from QTAIM for system Al.pep$^{8}_{\text{mim}}$. Total interaction energies are shown as green bars, covalent contributions are shown as blue bars and electrostatic contributions are shown as yellow bars.

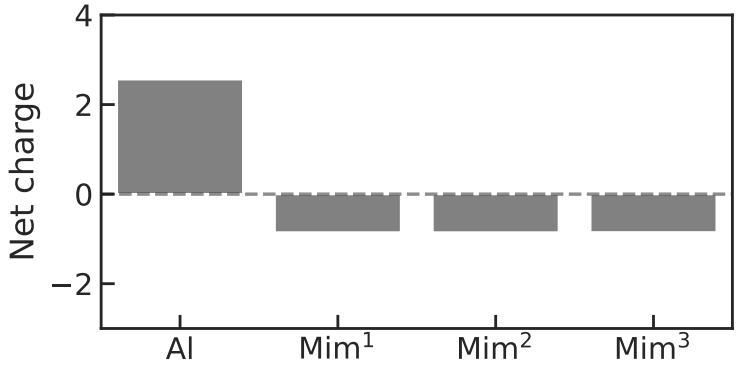


Figure A.21: Charges calculated from QTAIM for system Al.pep$^{8G}_{\text{mim}}$.

Table A.3: Charges of all groups obtained from QTAIM calculations.

System	q(Al)	q(Mim_1)	q(Mim_2)	q(Mim_3)
Al.pep$^{8}_{\text{mim}}$	2.56	-0.85	-0.86	-0.85
Al.pep$^{8G}_{\text{mim}}$	2.56	-0.86	-0.86	-0.85
Al.pep$^{10}_{\text{mim}}$	2.56	-0.85	-0.86	-0.85

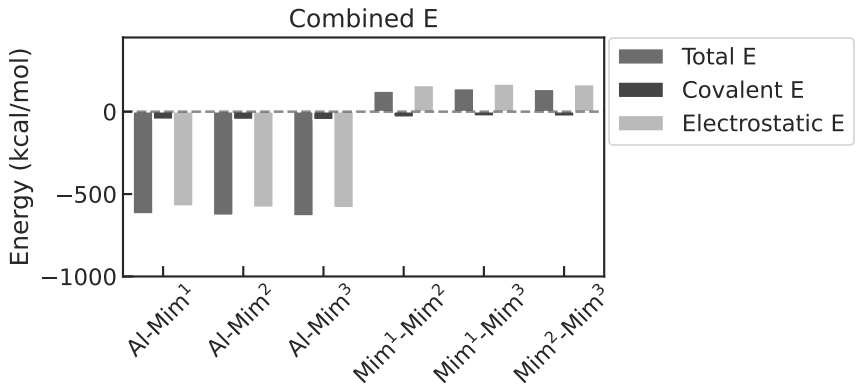


Figure A.22: Interaction energies calculated from QTAIM for system Al.pep$^{8G}_{mim}$. Total interaction energies are shown as green bars, covalent contributions are shown as blue bars and electrostatic contributions are shown as yellow bars.

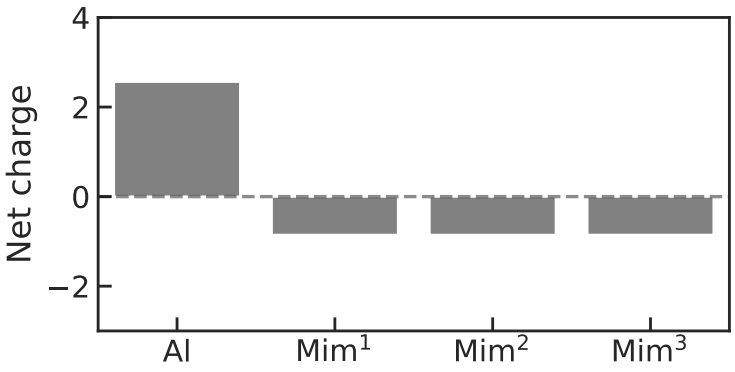


Figure A.23: Charges calculated from QTAIM for system Al.pep$^{10}_{mim}$.

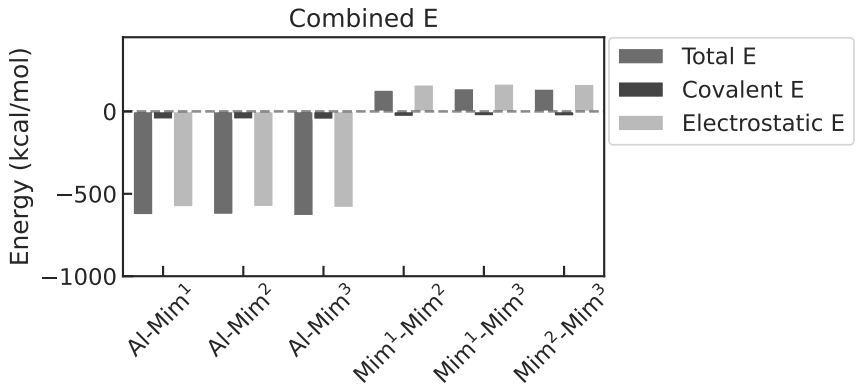


Figure A.24: Interaction energies calculated from QTAIM for system $\mathrm{Al.pep}^{10}_{\mathrm{mim}}$. Total interaction energies are shown as green bars, covalent contributions are shown as blue bars and electrostatic contributions are shown as yellow bars.

A

Table A.4: All interaction energies calculated from QTAIM calculations. All values are shown in kcal/mol.

System	$E_{Int}^{Al-Mim_1}$	$E_{Int}^{Al-Mim_2}$	$E_{Int}^{Al-Mim_3}$	$E_{Int}^{Mim_1-Mim_2}$	$E_{Int}^{Mim_1-Mim_3}$	$E_{Int}^{Mim_2-Mim_3}$
$Al.pep^{8}_{mim}$	-625.94	-624.69	-630.90	127.94	139.18	132.09
$Al.pep^{8G}_{mim}$	-621.05	-629.77	-633.41	125.31	140.56	136.30
$Al.pep^{10}_{mim}$	-628.33	-626.44	-633.66	130.27	139.56	136.86

B

S.I. - Chapter 4

First and Second Coordination Shell calculations

The use of FCS and SCS simplified systems allowed rapid calculations due to the small size, but the absence of the rest of the peptide translated into a considerable lack of interactions. When the entire peptide was considered, some of the resultant structures shared the same FCS and SCS. This is represented in the following Figure B.1 where, for example, the $AlSE_2$ simplified system is converted into two different species from different groups, namely S2.8 and S7.13.

Table B.1: ΔH_f and ΔG_f values for FCS and SCS systems.

Systems	ΔH_f (kcal/mol)	ΔG_f (kcal/mol)
$[Al(H_2O)_5(SerP)]^+$	-78.8	-73.6
$[Al(H_2O)_4(SerP)]^-$	-128.7	-118.5
$[Al(H_2O)_4(SerP)(Glu)]$	-100.9	-91.7
$[Al(H_2O)_4(Glu)_2]^+$	-61.0	-53.1
$[Al(H_2O)_3(SerP)_2(Glu)]^{2-}$	-139.4	-124.5
$[Al(H_2O)_3(SerP)(Glu)_2]^-$	-114.0	-101.3
$[Al(H_2O)_5(SerP)(Lys)]^{2+}$	-61.3	-51.0
$[Al(H_2O)_4(SerP)_2(Lys)_2]^+$	-138.6	-119.0
$[Al(H_2O)_4(SerP)(Glu)(Lys)]^+$	-85.5	-75.5
$[Al(H_2O)_4(SerP)(Glu)(Lys)_2]^{2+}$	-134.6	-116.7
$[Al(H_2O)_4(Glu)_2(Lys)]^{2+}$	-51.5	-43.5
$[Al(H_2O)_3(SerP)_2(Glu)(Lys)_3]^+$	-177.5	-150.6
$[Al(H_2O)_3(SerP)(Glu)_2(Lys)_2]^+$	-133.3	-112.3

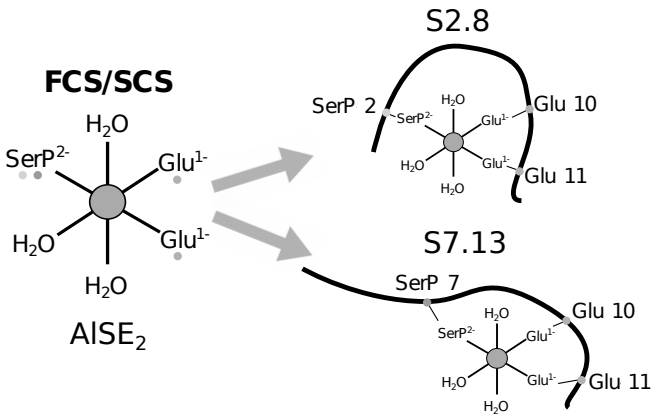


Figure B.1: Representation of the conversion of FCS/SCS systems to entire peptide systems. The grey ball in the centre corresponds to the Al^{3+} cation, binding six ligands. The same FCS/SCS can be observed in the two right whole peptides, but these systems consider the whole peptide instead of only the side chains of the residues.

Images of systems

Images of the Al-peptide systems. Reference systems are shown in Figure B.2, systems from group S2 are shown in Figure B.3, systems from group S7 are shown in Figure B.4 and systems from group S2S7 are shown in Figure B.5.

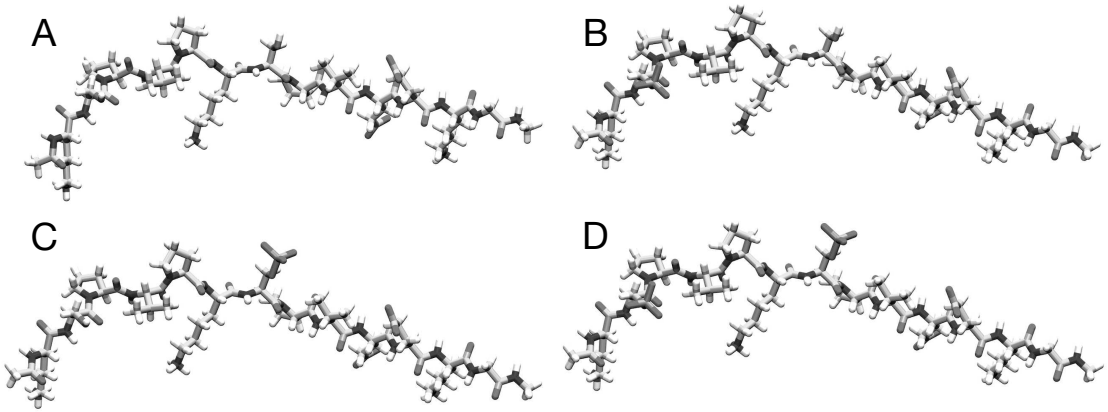


Figure B.2: Reference systems. The peptide was considered non-phosphorylated; partially phosphorylated; and fully phosphorylated. This allowed to characterize the impact of the phosphorylation by itself. Structures were in a configuration of β-sheet. A: Ref.1; B: Ref.2; C: Ref.3; D: Ref.4.

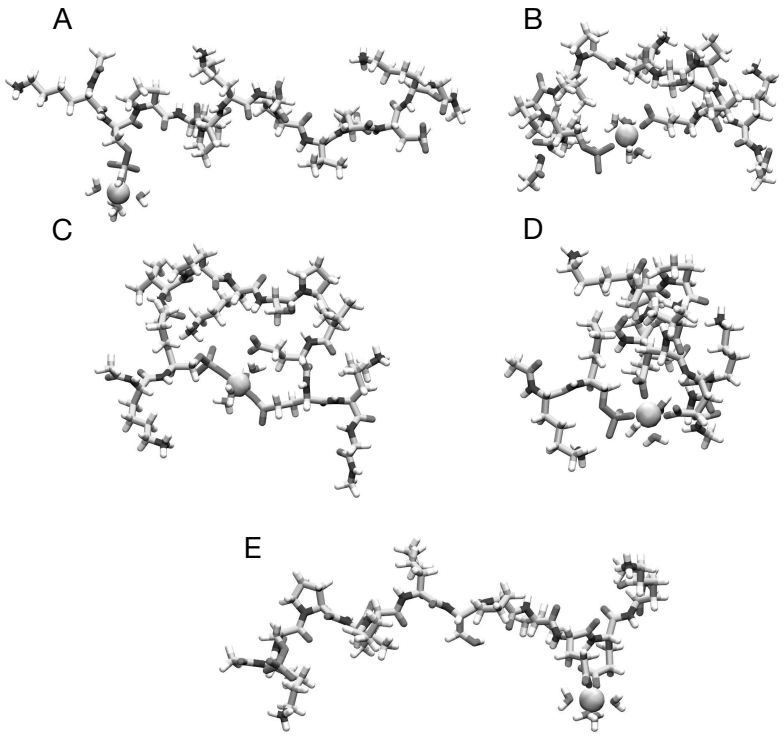


Figure B.3: Systems from the S2 group. In this group, the serine in position 2 was phosphorylated, referred to in the work as SerP2. Aluminium could be bound to the mentioned SerP2, Glu10 and Glu11. A: S2.5; B: S2.6; C: S2.7; D: S2.8; E: S2.9.

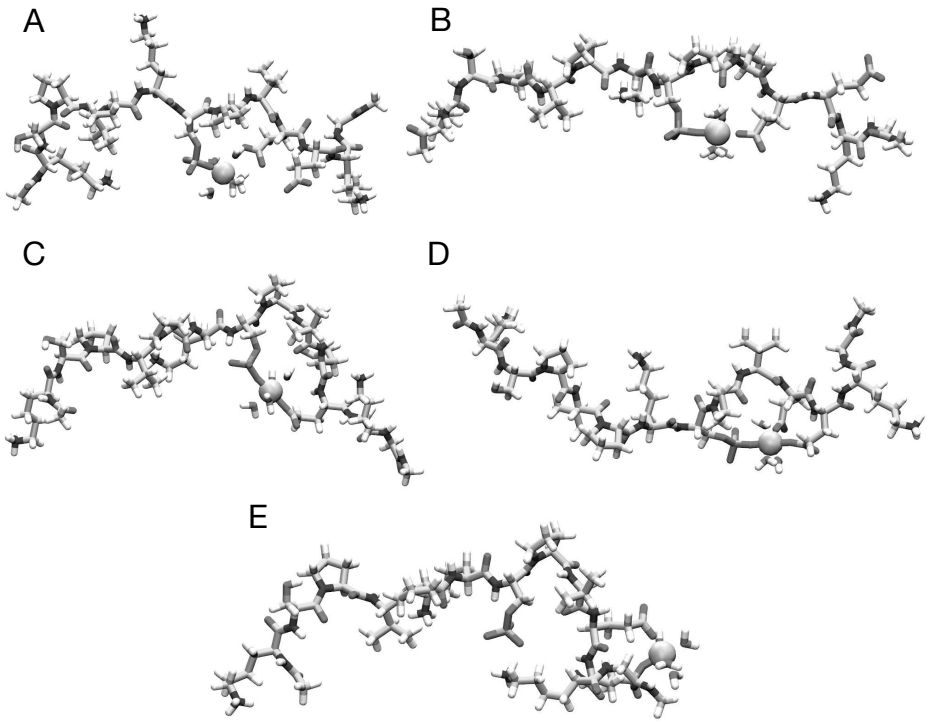


Figure B.4: Systems from the S7 group. In this group, the serine in position 7 was phosphorylated, referred to in the work as SerP7. Aluminium could be bound to the mentioned SerP7, Glu10 and Glu11. A: S7.10; B: S7.11; C: S7.12; D: S7.13; E: S7.14.

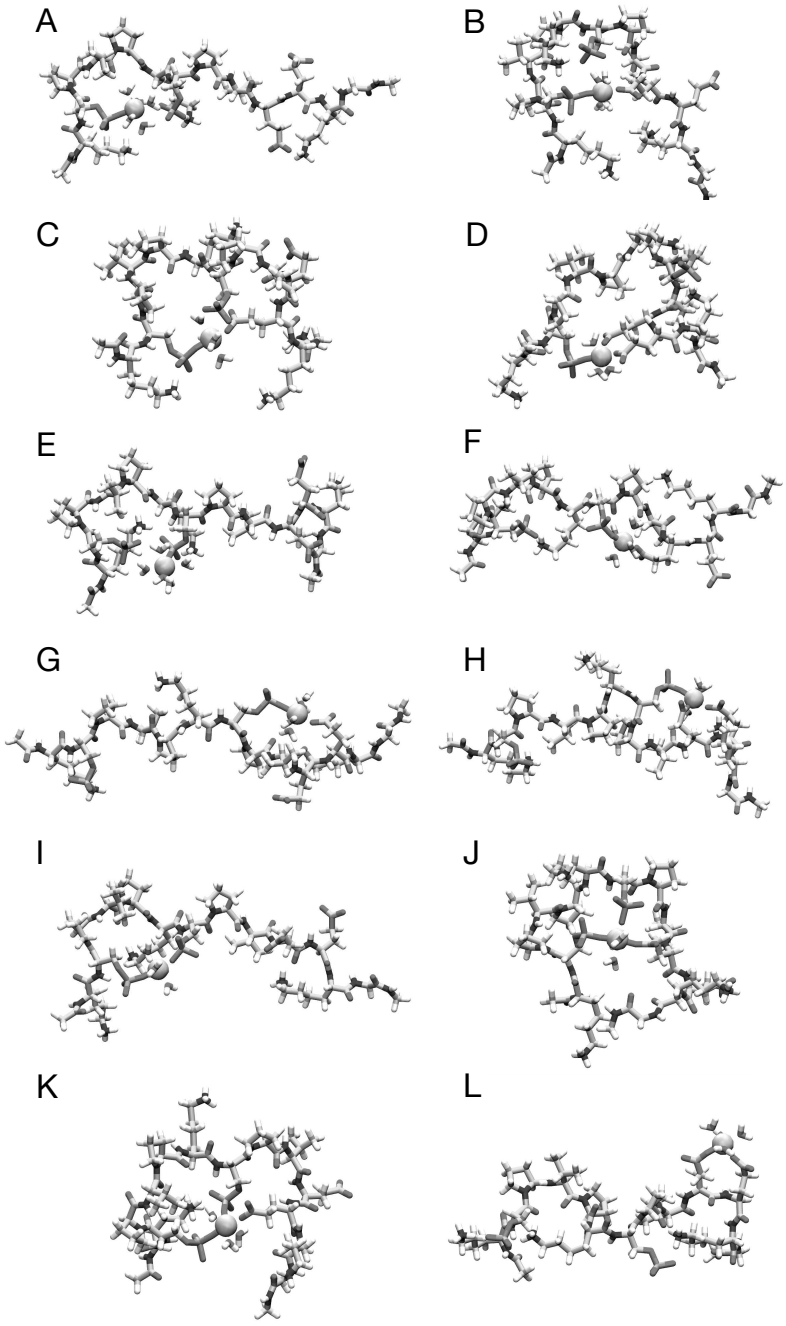


Figure B.5: Systems from the S2S7 group. In this group, both serines in positions 2 and 7 were phosphorylated. Aluminium could be bound to both SerP2 and SerP7 and Glu10 and Glu11. A: S2S7.15; B: S2S7.16; C: S2S7.17; D: S2S7.18; E: S2S7.19; F: S2S7.20; G: S2S7.21; H: S2S7.22; I: S2S7.23; J: S2S7.24; K: S2S7.25; L: S2S7.26.

Entropy

MD simulations were long enough to reach convergence for our systems. Figure B.6 shows the convergence for two systems, namely $NF13Al^{S2S7}_{S2S7E11}$ and its reference $NF13^{S2S7}$.

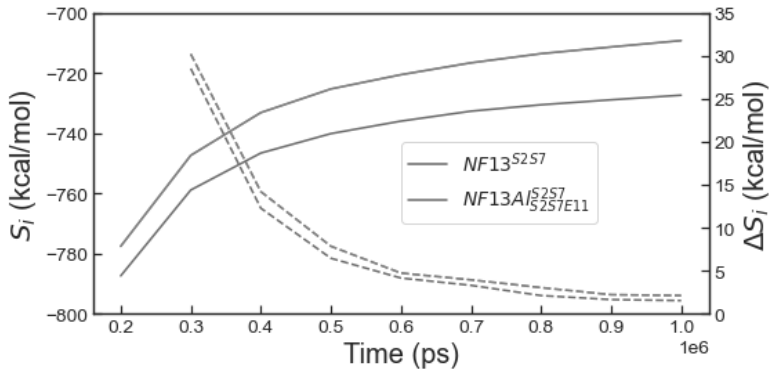


Figure B.6: Convergence of the entropy shown for system $NF13Al^{S2S7}_{S2S7E11}$ and its group reference $NF13^{S2S7}$. The dashed line represents the change of entropy for each system.

Configurational entropy was determined equivalently to Chapter 3, applying a bootstrapping method —explained in Appendix A— with 500 resamples and 50 blocks per original trajectory. Moreover, the same *soft configurational entropy* concept used in the previous Chapter 3 was also applied here for the entropy determination. In the following Figure B.7, the values of S_{MIST} and ΔS_{MIST} are shown, in this case with a direct comparison excluding or not the bonds and angles degrees of freedom. Notice that the qualitative trends displayed are similar to those of the main text Figure 4.10: i.e S2 group entropy loss is greater than S7 group; greatest entropy losses are present in the S2S7 group; three-coordinated systems are the most affected by entropy loss; but the error bars associated —black vertical lines— are much bigger when bonds and angles are not excluded.

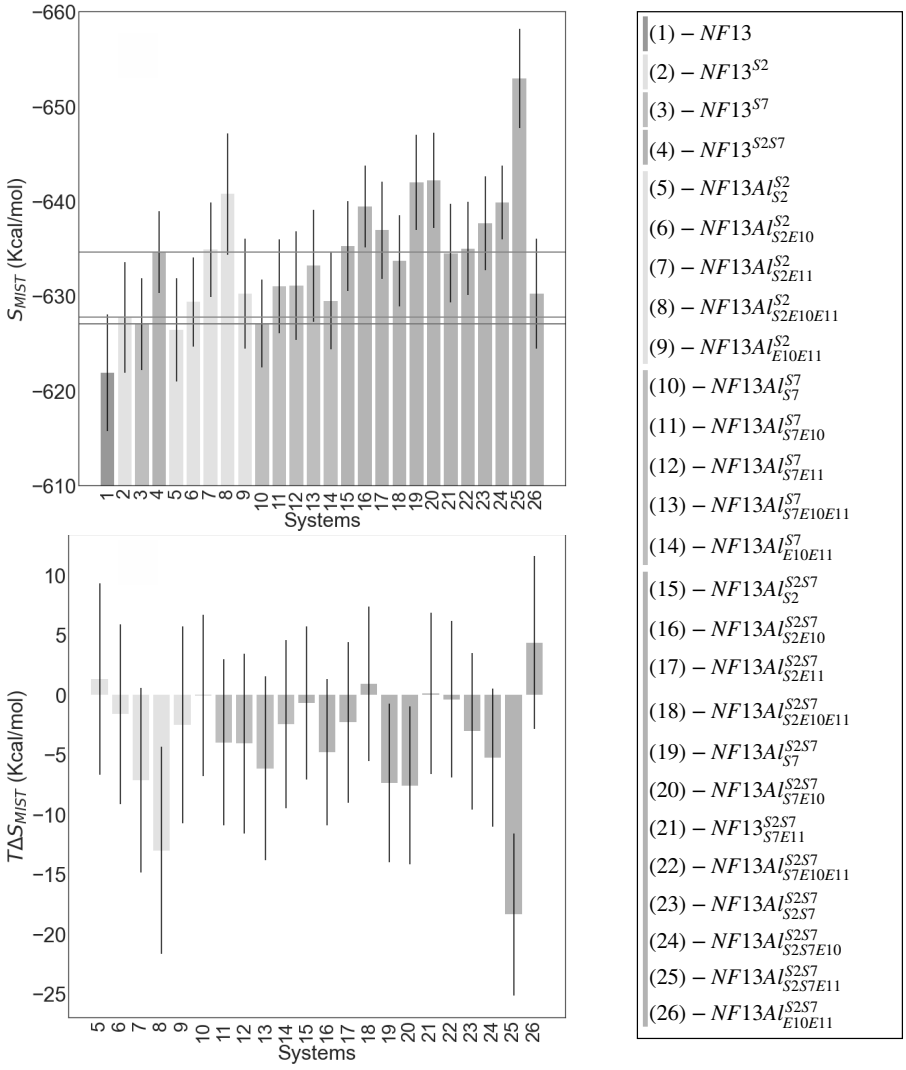


Figure B.7: S_{MIST} and ΔS_{MIST} values obtained for all systems with the MIST scheme when considering all the BAT DOF.

Entropy and Energy tables

In this section, the results of ΔS_{MIST} and ΔE determined for each system using the MIST scheme and TURBOMOLE software are displayed in tables, separated by groups. For each system, the ΔE and ΔS show the mean value and the ΔE SD and ΔS SD the standard deviation. All values are in kcal/mol. Notice again that by the nature of the calculated entropy ΔS_{MIST}, it can not be directly compared to the stabilization energy. For this reason, the determination of the stabilization energies comes exclusively from single-point energy calculations. Therefore, values are not additive and should be used only for qualitative conclusions.

Table B.2: Entropy and energy values for group S2 (kcal/mol).

Index	Systems	ΔE	ΔE SD	ΔS	ΔS SD
S2.5	$\mathrm{NF13Al}^{\mathrm{S2}}_{\mathrm{S2}}$	-19.5	2.7	**0.2**	1.2
S2.6	$\mathrm{NF13Al}^{\mathrm{S2}}_{\mathrm{S2E10}}$	-39.4	6.2	-2.3	1.2
S2.7	$\mathrm{NF13Al}^{\mathrm{S2}}_{\mathrm{S2E11}}$	-42.0	6.8	-6.5	1.3
S2.8	$\mathrm{NF13Al}^{\mathrm{S2}}_{\mathrm{S2E10E11}}$	**-71.6**	15.8	-10.4	1.3
S2.9	$\mathrm{NF13Al}^{\mathrm{S2}}_{\mathrm{E10E11}}$	-70.5	10.5	-3.3	1.2

B

Table B.3: Entropy and energy values for group S7 (kcal/mol).

Index	Systems	ΔE	ΔE SD	ΔS	ΔS SD
S7.10	$\mathrm{NF13Al}^{\mathrm{S7}}_{\mathrm{S7}}$	-15.9	2.3	**0.1**	1.2
S7.11	$\mathrm{NF13Al}^{\mathrm{S7}}_{\mathrm{S7E10}}$	-51.2	6.4	-2.9	1.3
S7.12	$\mathrm{NF13Al}^{\mathrm{S7}}_{\mathrm{S7E11}}$	-52.5	6.3	-1.4	1.2
S7.13	$\mathrm{NF13Al}^{\mathrm{S7}}_{\mathrm{S7E10E11}}$	**-97.6**	12.0	-5.9	1.0
S7.14	$\mathrm{NF13Al}^{\mathrm{S7}}_{\mathrm{E10E11}}$	-65.7	12.3	-1.9	1.2

Table B.4: Entropy and energy values for group S2S7. (kcal/mol)

Index	Systems	ΔE	ΔE SD	ΔS	ΔS SD
S2S7.15	$NF13Al^{S2S7}_{S2}$	-49.3	6.4	-2.0	1.3
S2S7.16	$NF13Al^{S2S7}_{S2E10}$	-82.1	14.6	**1.5**	1.4
S2S7.17	$NF13Al^{S2S7}_{S2E11}$	-62.8	13.2	-6.8	1.2
S2S7.18	$NF13Al^{S2S7}_{S2E10E11}$	-108.4	34.4	-7.7	1.3
S2S7.19	$NF13Al^{S2S7}_{S7}$	-50.3	6.9	-4.7	1.4
S2S7.20	$NF13Al^{S2S7}_{S7E10}$	-62.6	5.4	-1.3	1.3
S2S7.21	$NF13Al^{S2S7}_{S7E11}$	-57.9	7.2	-1.1	1.3
S2S7.22	$NF13Al^{S2S7}_{S7E10E11}$	-109.1	11.3	-3.8	1.1
S2S7.23	$NF13Al^{S2S7}_{S2S7}$	-53.6	8.9	-1.60	1.2
S2S7.24	$NF13Al^{S2S7}_{S2S7E10}$	**-128.3**	27.3	-5.9	1.3
S2S7.25	$NF13Al^{S2S7}_{S2S7E11}$	-99.9	31.8	-16.2	1.3
S2S7.26	$NF13Al^{S2S7}_{E10E11}$	-50.9	5.4	-3.4	1.4

MD Analysis

Ramachandran density plots.

Ramachandran density plots were computed using the MDtraj Python package. Figure B.8 shows the plot for the Ref.1 reference peptide —absence of phosphorylation and metallic ion— with a red bar on top. The other three reference systems —partially phosphorylated Ref.2 and Ref.3, and fully phosphorylated Ref.4— are shown inside each group, to facilitate the analysis. All references appear in black. Colours show the different numbers of coordination and types of residues.

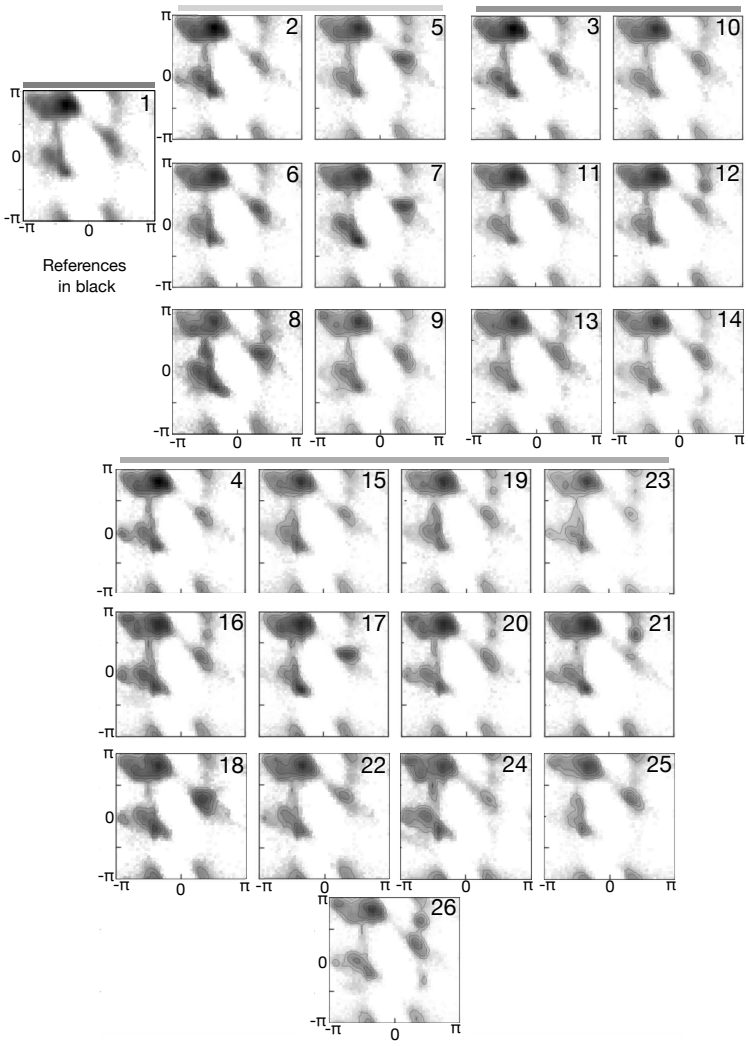


Figure B.8: Ramachandran plots for all systems organized in groups. The S2 group is displayed under the cyan bar, the S7 group under the green bar, and the S2S7 group under the magenta bar. References are shown in black.

Ramachandran based SS Fraction

From previous Ramachandran density plots, we integrated the SS fraction. The top-left panel in Figure B.9 shows the limited regions considered for the integration.

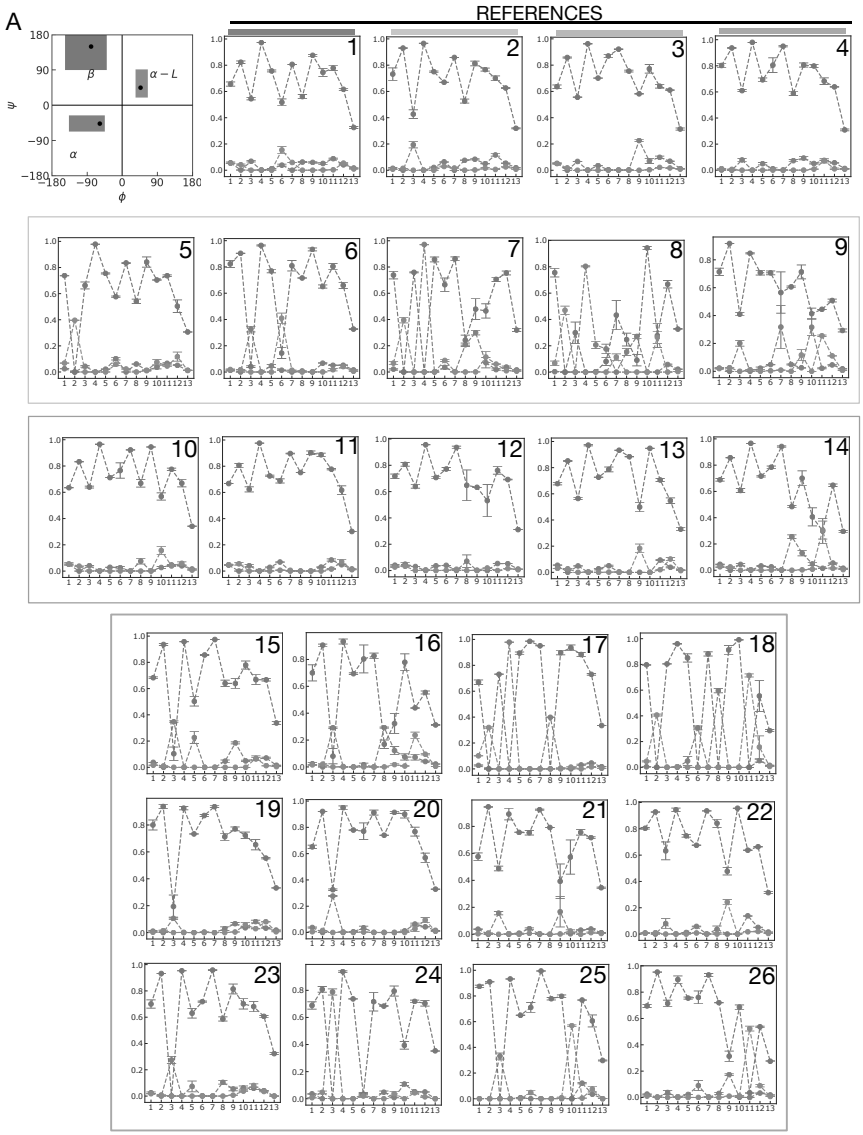


Figure B.9: A) Representation of the regions defined to assign secondary structure fraction from the Ramachandran plots. Black dots represent the most populated point of the regions, expected to concentrate the greatest density. The rest of the plots show the fraction of SS assigned for each residue on each system. References are shown in the top part, with the same colour code as Figure B.8 for each group —red for Ref.1—.

Contact Maps

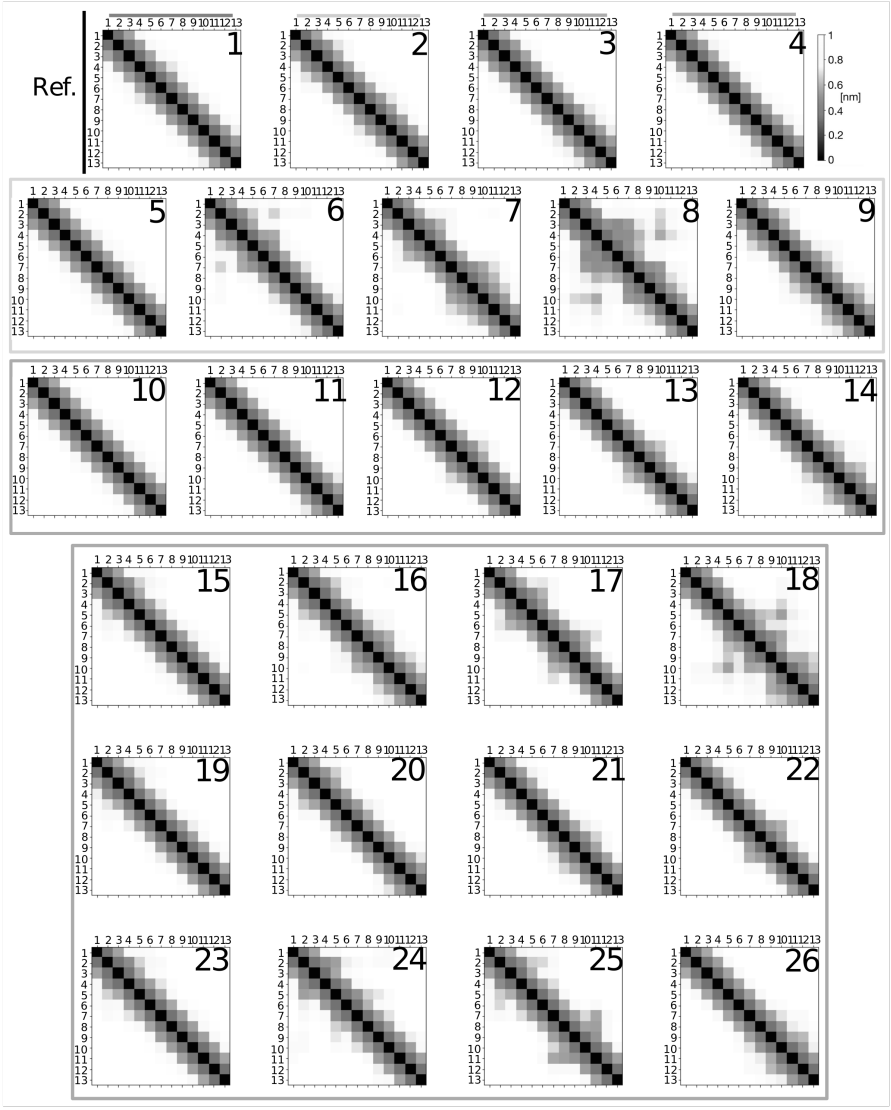


Figure B.10: Contact maps of all systems by group. The top plots are for reference systems. The contacts between residues are shown in a distance range of 0 to 1 nm —the shorter distance the darker colour—. Colour code in groups as in the previous Figure B.8.

Radius of Gyration

Table B.5: Mean values of the radius of gyration —R_g— for systems. Values are calculated from the figures below.

Index	Systems	R_g (nm)
Ref.1	NF13	1.05
Ref.2	$NF13^{S2}$	1.04
Ref.3	$NF13^{S7}$	1.08
Ref.4	$NF13^{S2S7}$	1.06
S2.5	$NF13Al^{S2}_{S2}$	1.06
S2.6	$NF13Al^{S2}_{S2E10}$	0.88
S2.7	$NF13Al^{S2}_{S2E11}$	0.84
S2.8	$NF13Al^{S2}_{S2E10E11}$	0.74
S2.9	$NF13Al^{S2}_{E10E11}$	0.88
S7.10	$NF13Al^{S7}_{S7}$	1.11
S7.11	$NF13Al^{S7}_{S7E10}$	1.13
S7.12	$NF13Al^{S7}_{S7E11}$	1.05
S7.13	$NF13Al^{S7}_{S7E10E11}$	1.10
S7.14	$NF13Al^{S7}_{E10E11}$	1.03
S2S7.15	$NF13Al^{S2S7}_{S2}$	0.98
S2S7.16	$NF13Al^{S2S7}_{S2E10}$	0.84
S2S7.17	$NF13Al^{S2S7}_{S2E11}$	0.87
S2S7.18	$NF13Al^{S2S7}_{S2E10E11}$	0.82
S2S7.19	$NF13Al^{S2S7}_{S7}$	0.99
S2S7.20	$NF13Al^{S2S7}_{S7E10}$	1.01
S2S7.21	$NF13Al^{S2S7}_{S7E11}$	1.03
S2S7.22	$NF13Al^{S2S7}_{S7E10E11}$	1.04
S2S7.23	$NF13Al^{S2S7}_{S2S7}$	1.00
S2S7.24	$NF13Al^{S2S7}_{S2S7E10}$	0.82
S2S7.25	$NF13Al^{S2S7}_{S2S7E11}$	0.80
S2S7.26	$NF13Al^{S2S7}_{E10E11}$	1.03

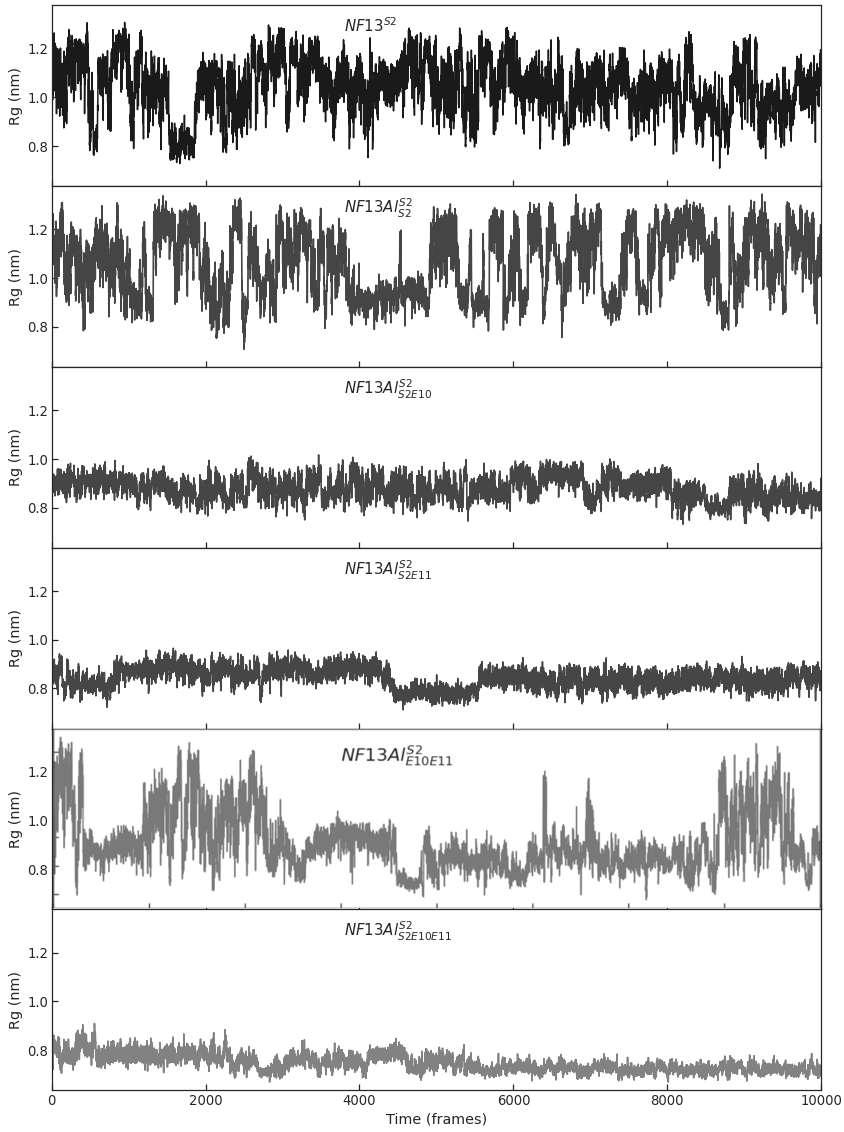


Figure B.11: Radius of gyration for S2 group. The reference system of the group is displayed in black.

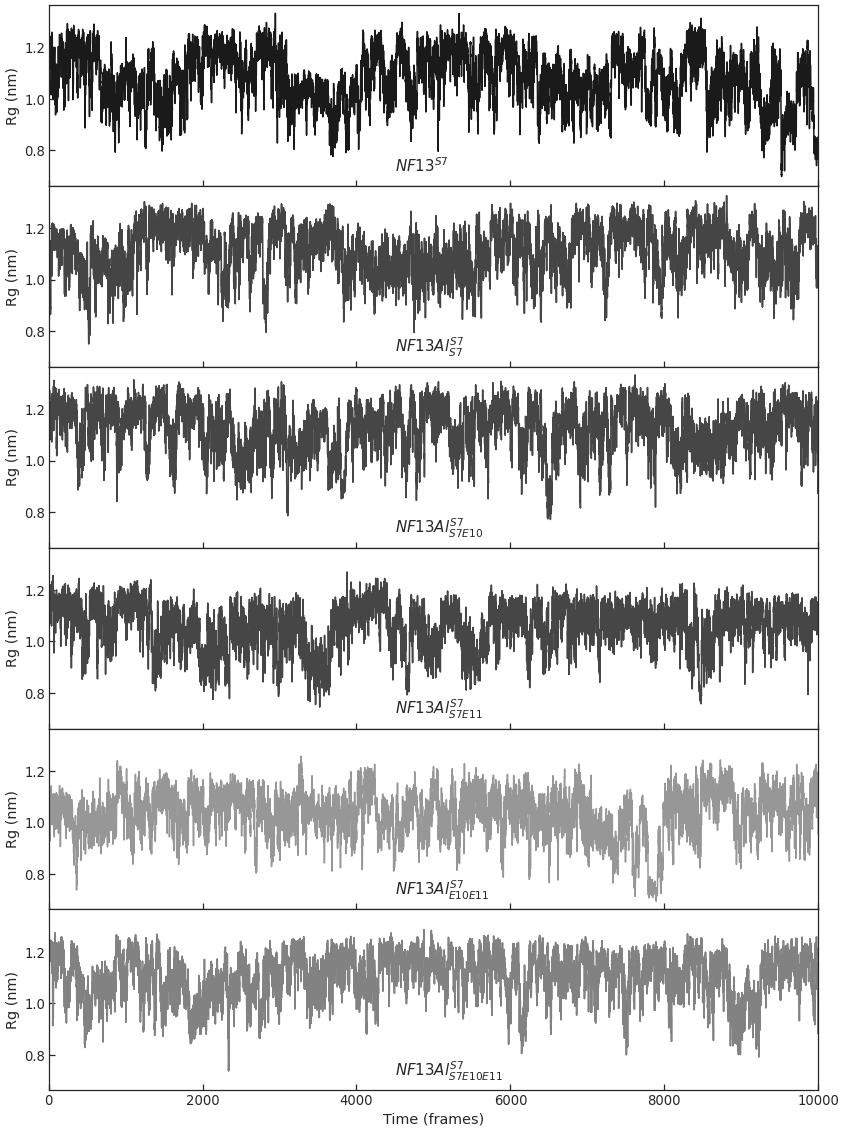


Figure B.12: Radius of gyration for S7 group. The reference system of the group is displayed in black.

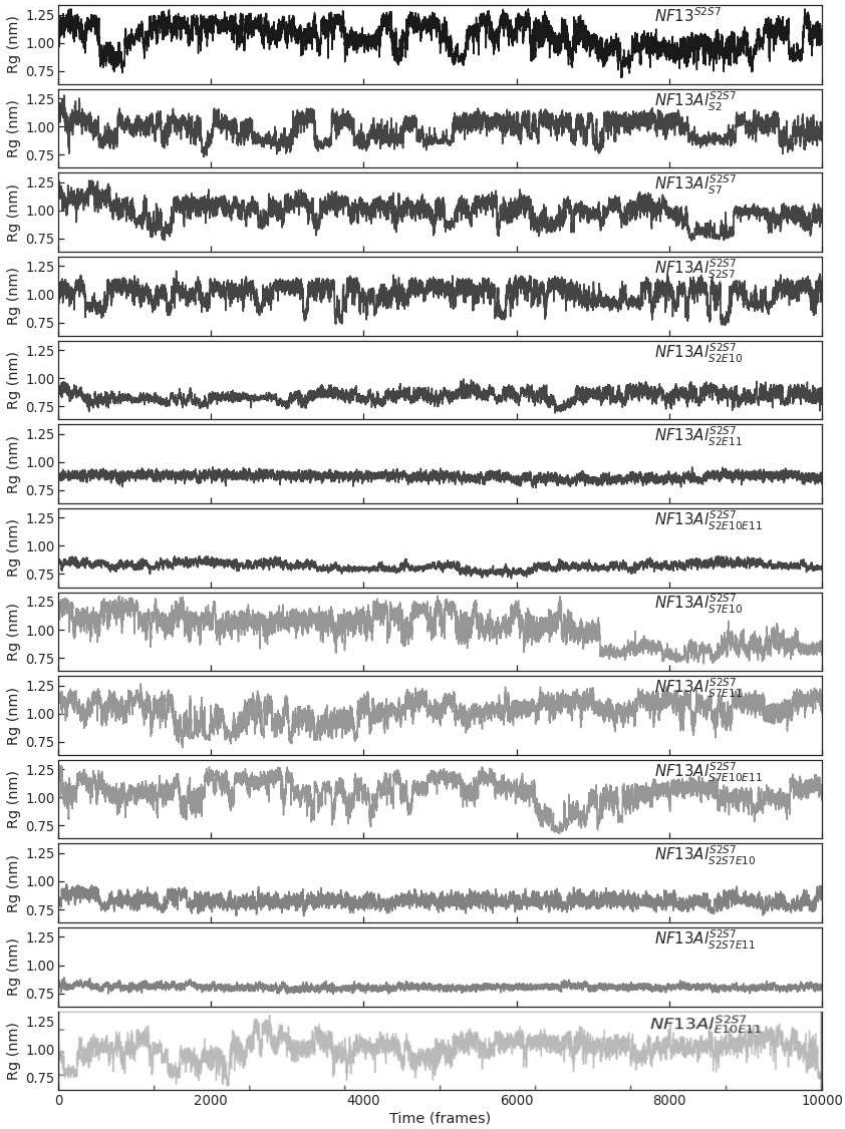


Figure B.13: Radius of gyration for S2S7 group. The reference system of the group is displayed in black.

Clustering Pie Charts

To reduce the dimensionality and handle the data adequately, we used the PCA and the K-means clustering technique. The PCA creates a new set of variables called "components" using linear combinations of the initial variables. These components are created to have as much information as possible but remain uncorrelated. The K-means is a clustering technique that aims to partition N elements into K clusters, where each element belongs to the cluster with the closest mean —the centroid of the cluster—. In this technique, the number of clusters is determined using the *elbow method* where the explained variance is plotted against the number of clusters, selecting the point where diminishing returns of the explained variance are no longer worth a number of clusters.

To ensure the consistency of clustering across systems, we combined the backbone trajectories of all complexes and performed the PCA analysis based on the pairwise RMSD. The >90% of the variance was explained by the first four components of the PCA, as shown in Figure B.14A. The number of clusters determined with the elbow method —using the SEE and ΔSEE combination— is shown in Figure B.14B, with eight clusters corresponding to the selected point of the elbow. The first three principal components, namely PC1, PC2 and PC3 are shown in Figure B.14C.

To ensure the consistency of clustering across systems, we combined the backbone trajectories of all complexes and performed a PCA analysis based on the pairwise RMSD.

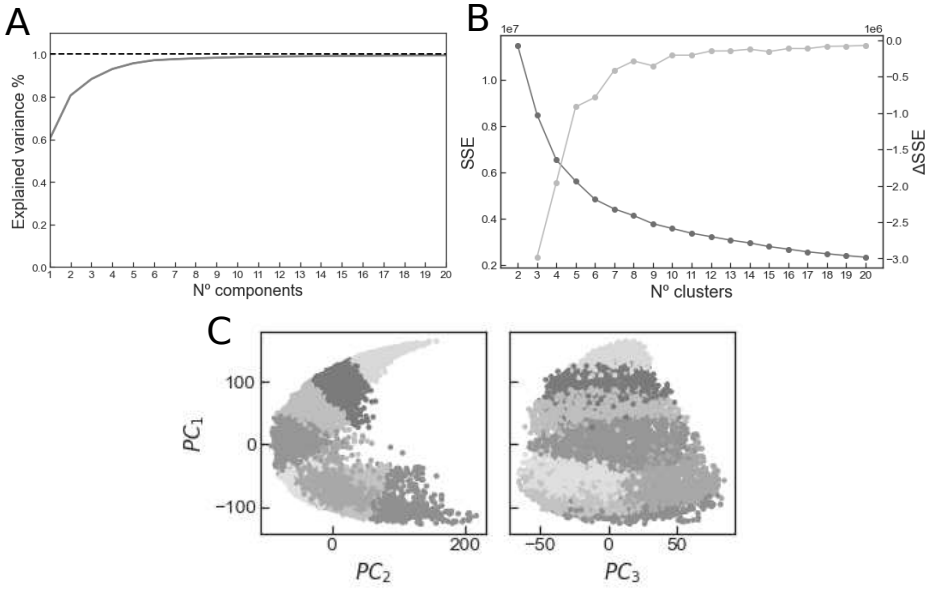


Figure B.14: (A) Explanation of the variance of the data set by the number of components, (B) selection of the number of clusters by the elbow method, (C) representation of the principal components PC1 vs PC2 and PC1 vs PC3 for 8 clusters.

B

C

S.I. - Chapter 5

System Images

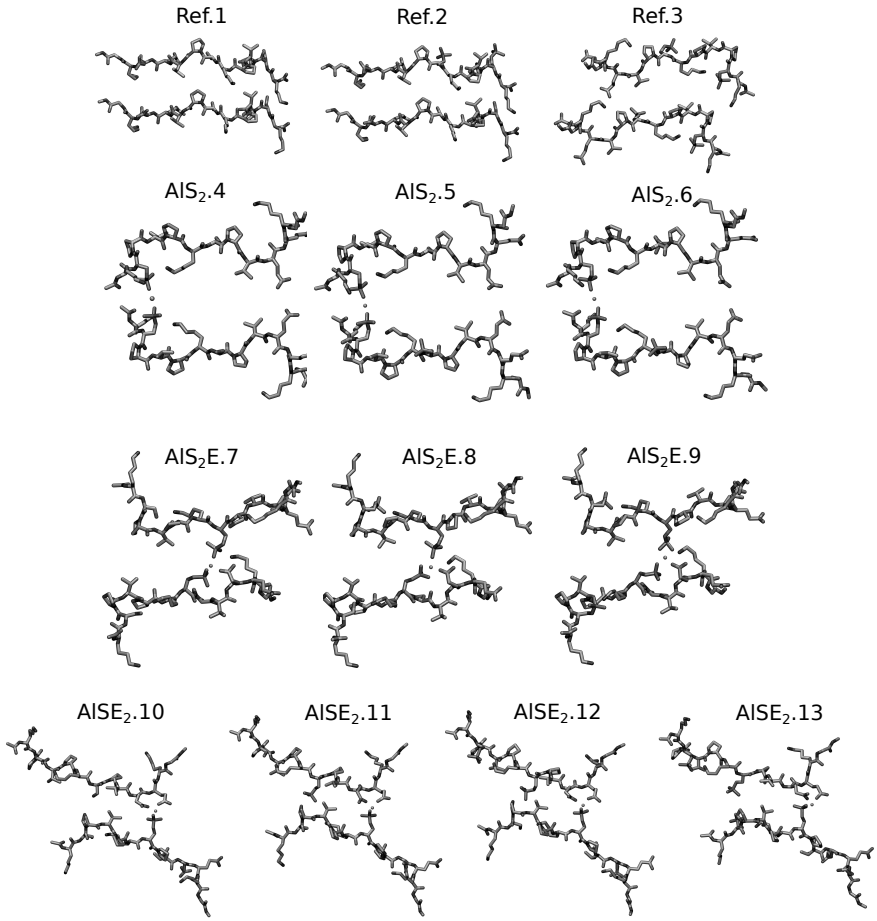


Figure C.1: Images of the studied dipeptide systems, ordered from left to right by phosphorylation degree. The first row corresponds to reference systems without aluminium cations. The second row corresponds to systems with coordination type AlS_2. The third row corresponds to systems with coordination type AlS_2E. The last row corresponds to systems with coordination type $AlSE_2$.

C

PCA and HDBSCAN parameters

Characterization of the *bound* and *unbound* states

For characterizing the *bound* and *unbound* states we computed the contacts between the residue side chains of the peptides using the MDtraj *compute-contacts* utility, with the *closest* scheme. The resultant contacts of each system for each frame were combined and fed into the PCA. We reduced the dimensionality to 4 components that explained >80% of the variance. From the resultant components, the PC1 and PC2 were used to create the free energy surfaces shown in Chapter 5, at a temperature of 300 K.

Moreover, the principal components from the combined PCA were fed into HDBSCAN —from scikit.learn— for clustering. The HDBSCAN method is an unsupervised clustering algorithm that extends the previous DBSCAN algorithm by building a hierarchy of clusters and selecting the most stable ones based on density, allowing it to handle variable-density data and identify noise. The HDBSCAN parameters used were $min_cluster_size = 1000$, $cluster_selection_epsilon = 5$, $alpha = 0.95$.

Characterization of the *bound* substates

For characterizing the *bound* substates, we performed further PCA using the radius of gyration of each chain individually and independently for each system —determined using the MDTraj *compute-rg* utility—. In this case, we kept the same dimensionality given that the input corresponded to only the radius of the gyration of both chains. As before, the two resultant principal components —PC1 and PC2— were used to generate new free energy surfaces using a temperature of 300 K.

Using the two principal components, we again performed HDBSCAN for clustering, using the following parameters depending on the specific system: *min_cluster_size* = 50—300 range, *cluster_selection_method* = "leaf", *alpha* = 0.5—1.5 range, *cluster_selection_epsilon* = 0.0—0.05 range.

Results

PBC copy interaction artifacts

An important condition in simulations with multiple chains is to confirm that these systems do not see either their copies or more than one single copy of their partner, to avoid simulation artefacts. In the following plots, the minimum distance determined by any of these combinations is shown, ensuring that during our REST2 simulations, chains did not see each other in a non-physical unexpected manner.

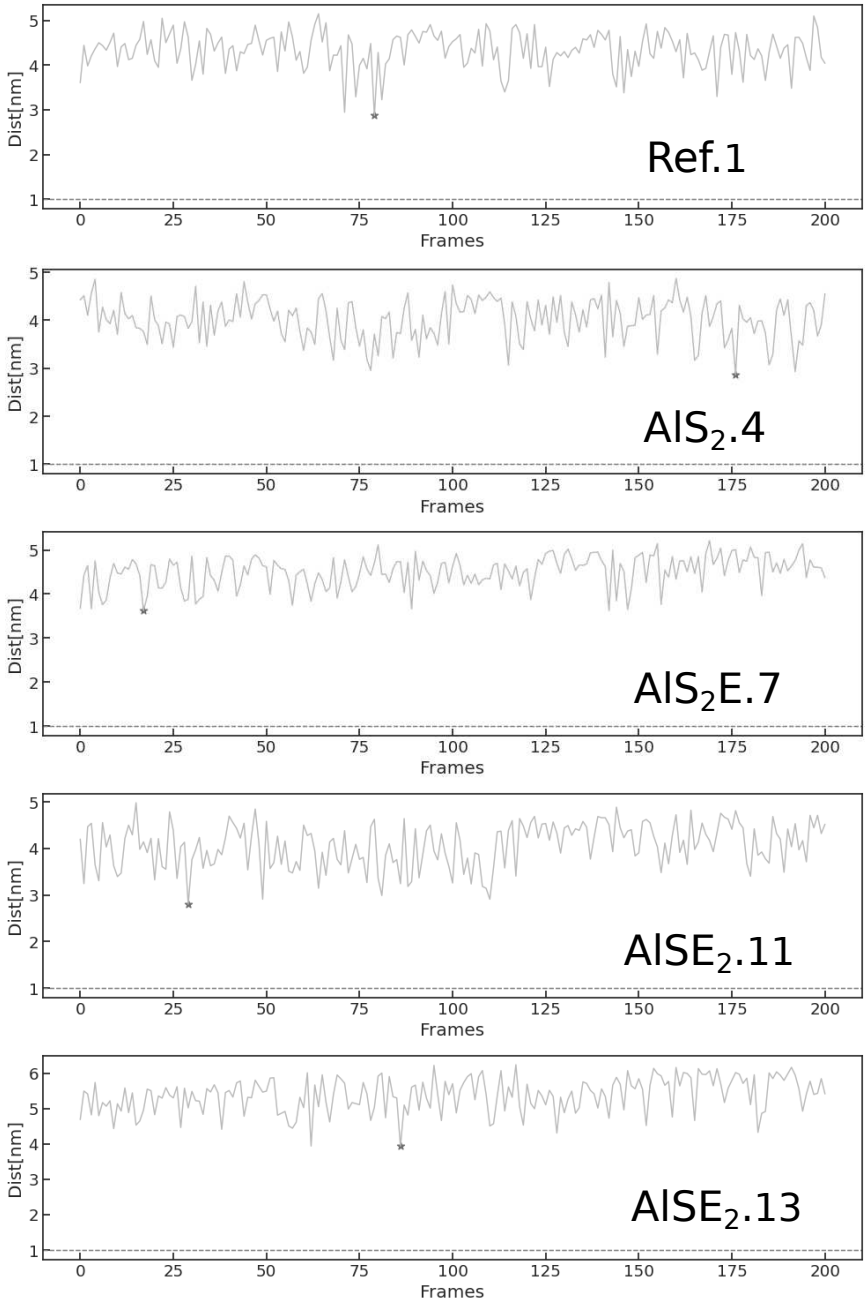


Figure C.2: Minimum distance analysis for selected systems. The blue star indicates the minimum distance measured, while the bottom red line is the cutoff. All structures have been confirmed to satisfy the minimum image distance condition.

Free energy surfaces and clustering

The resultant figures for the PCA free energy surface and HDBSCAN clustering of *bound* and *unbound* states are shown below:

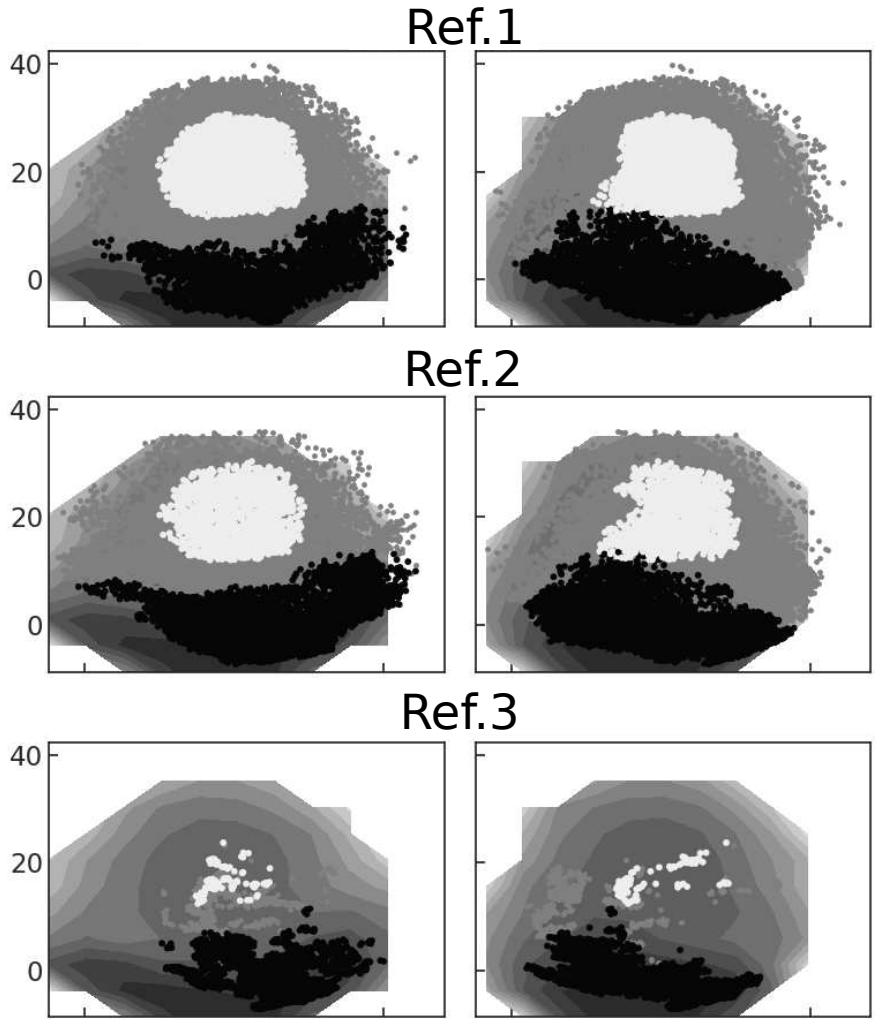


Figure C.3: PCA and HDBSCAN clustering results for reference systems, ordered from top to bottom by phosphorylation degree.

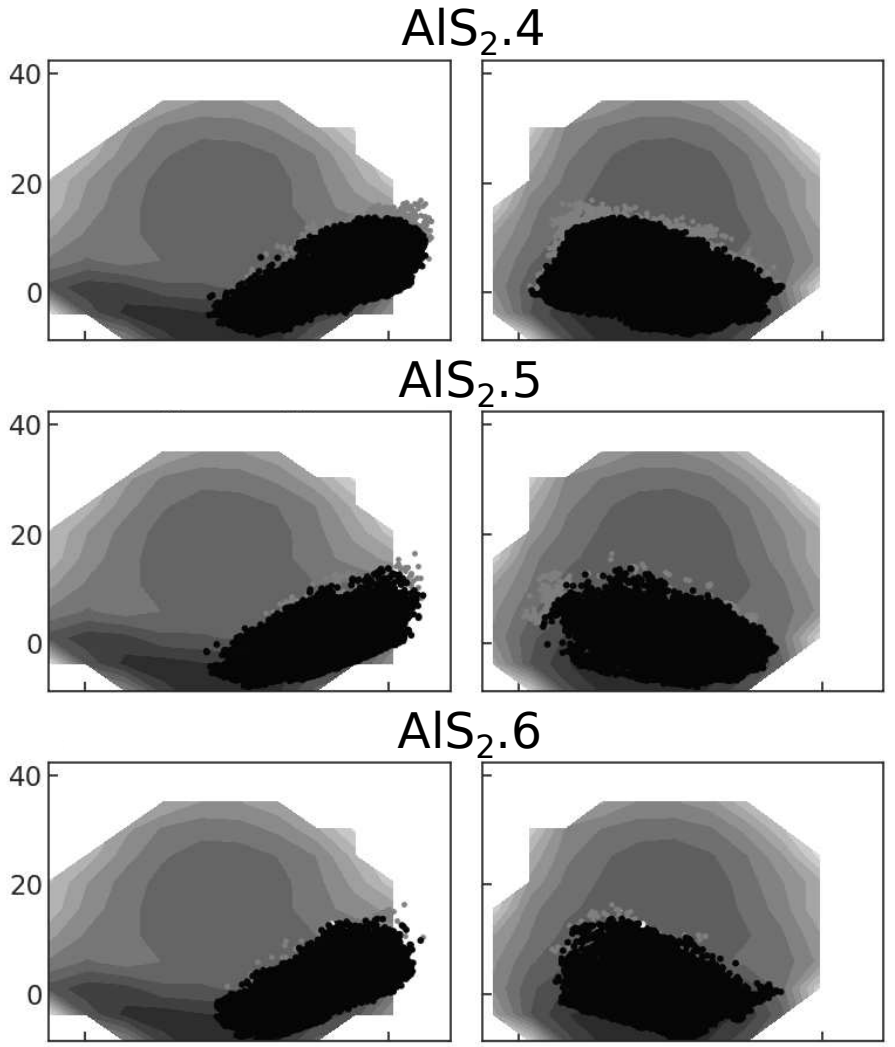


Figure C.4: PCA and HDBSCAN clustering results for AlS_2 systems, ordered from top to bottom by phosphorylation degree.

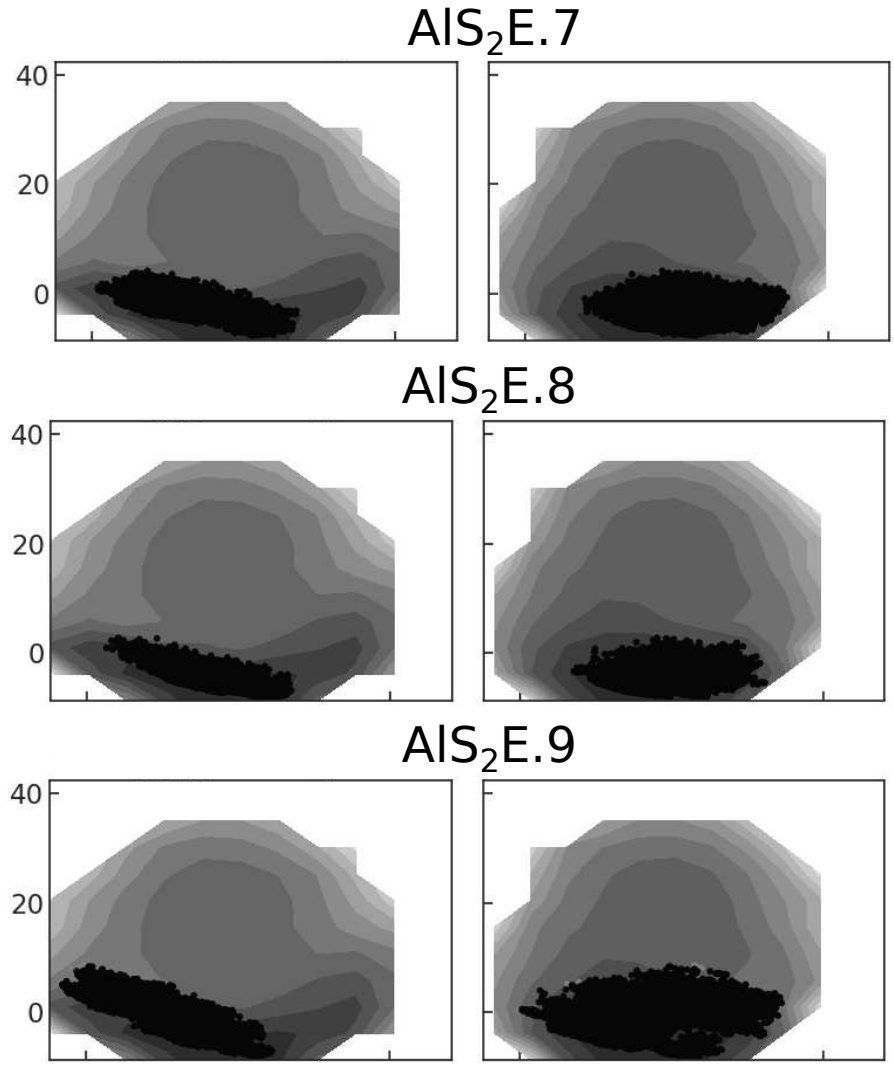


Figure C.5: PCA and HDBSCAN clustering results for AlS_2E systems, ordered from top to bottom by phosphorylation degree.

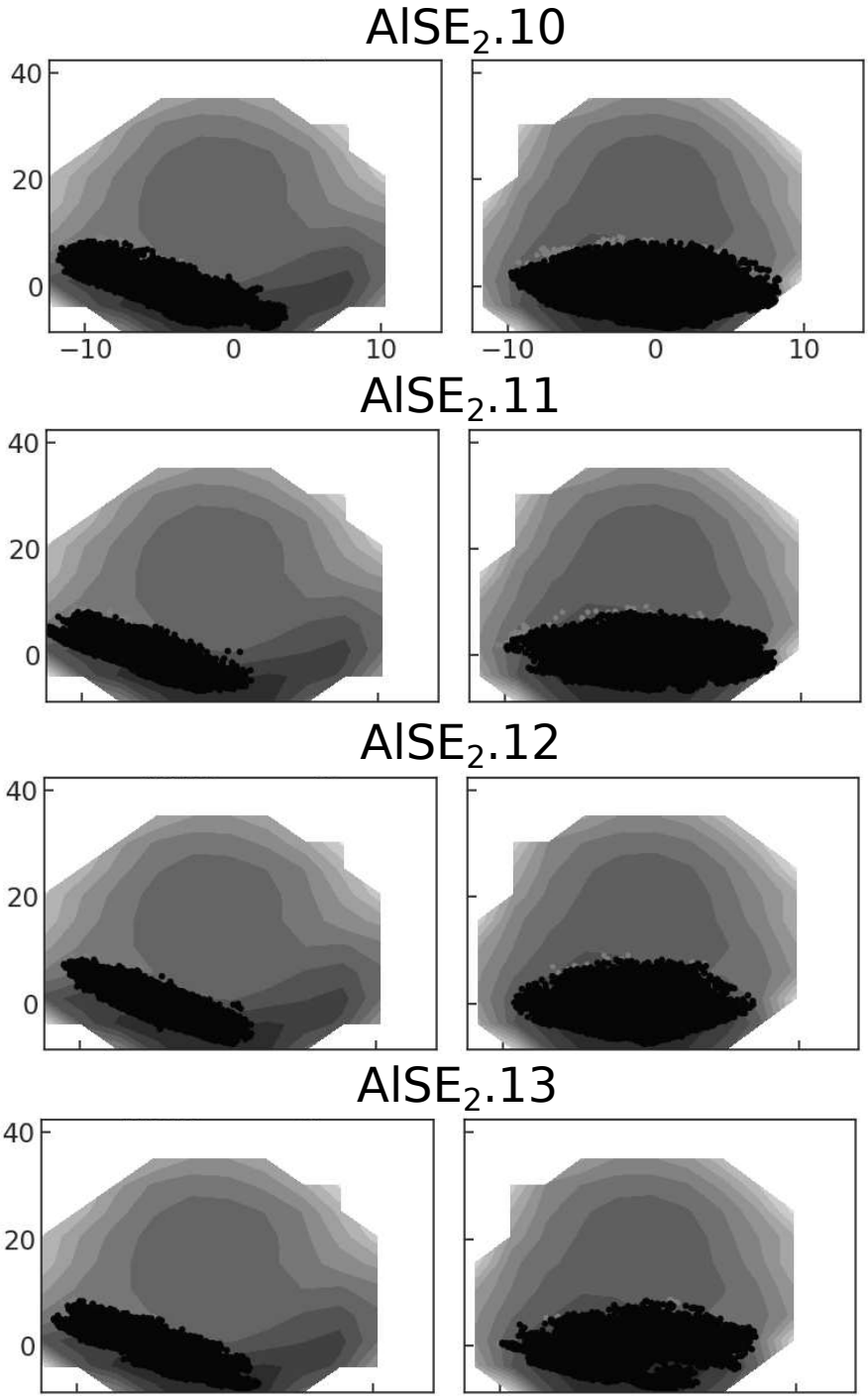


Figure C.6: PCA and HDBSCAN clustering results for $AlSE_2$ systems, ordered from top to bottom by phosphorylation degree.

Free energy surfaces and clustering of *bound* substates

In the following images, the results from the second PCA and HDBSCAN for characterizing the substates of the bound state are shown. For each system, a pie chart showing the occupancy of each cluster is shown. For each cluster, the intra-chain and inter-chain contact maps and the radius of gyration of each chain are displayed.

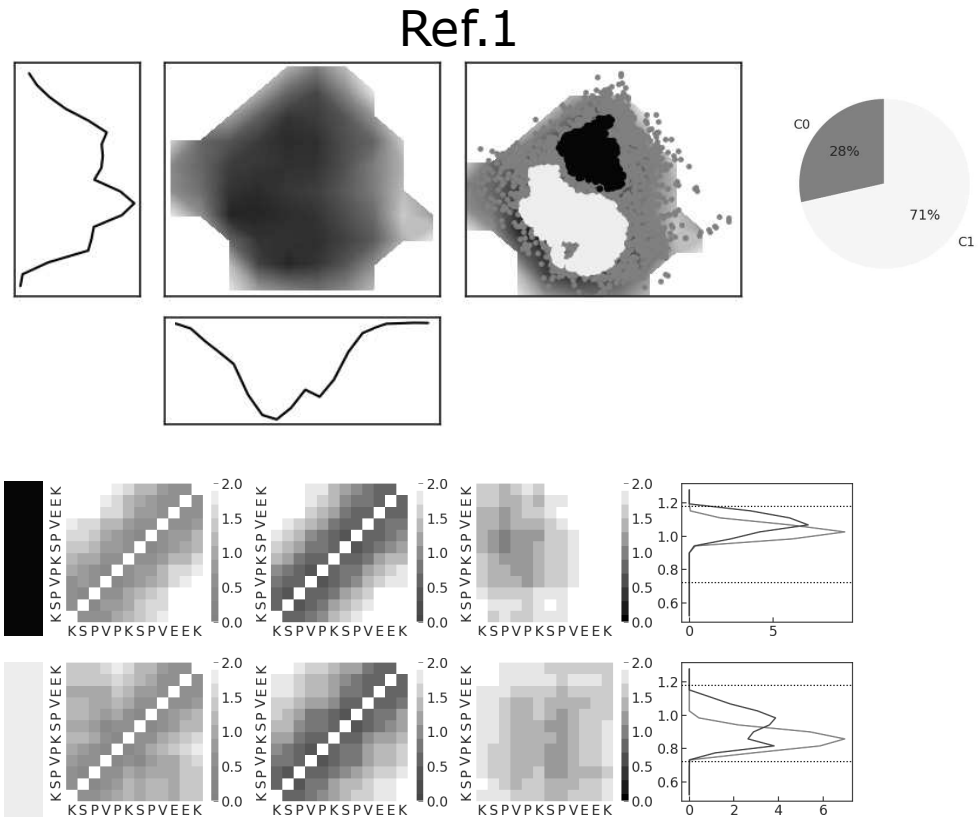


Figure C.7: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

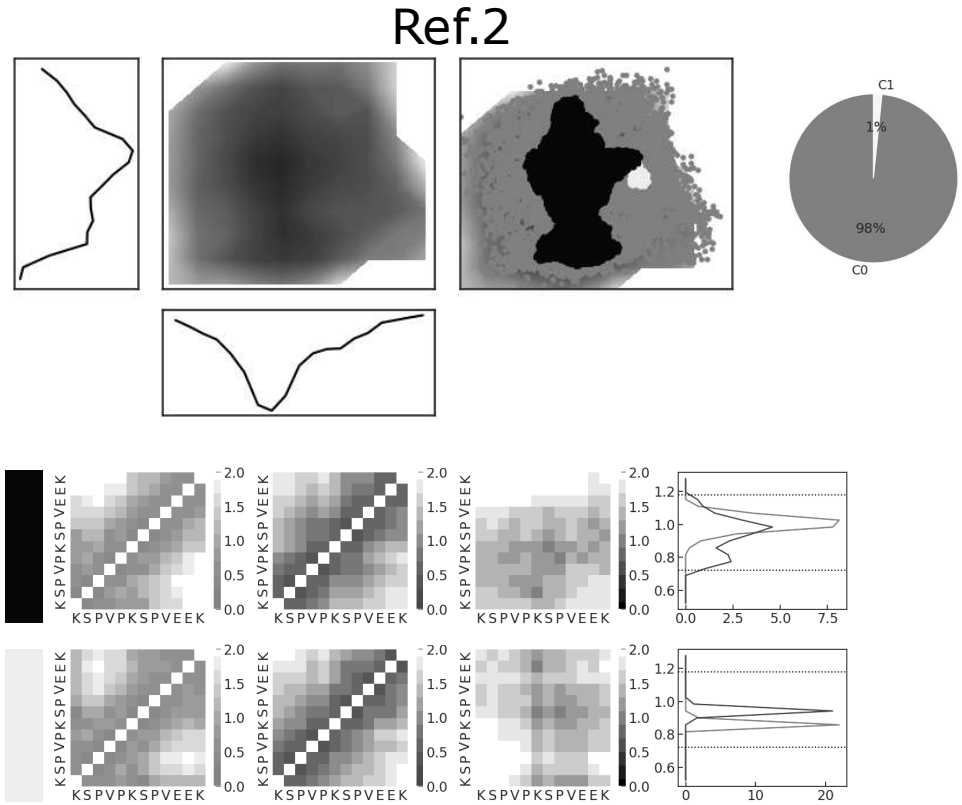


Figure C.8: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

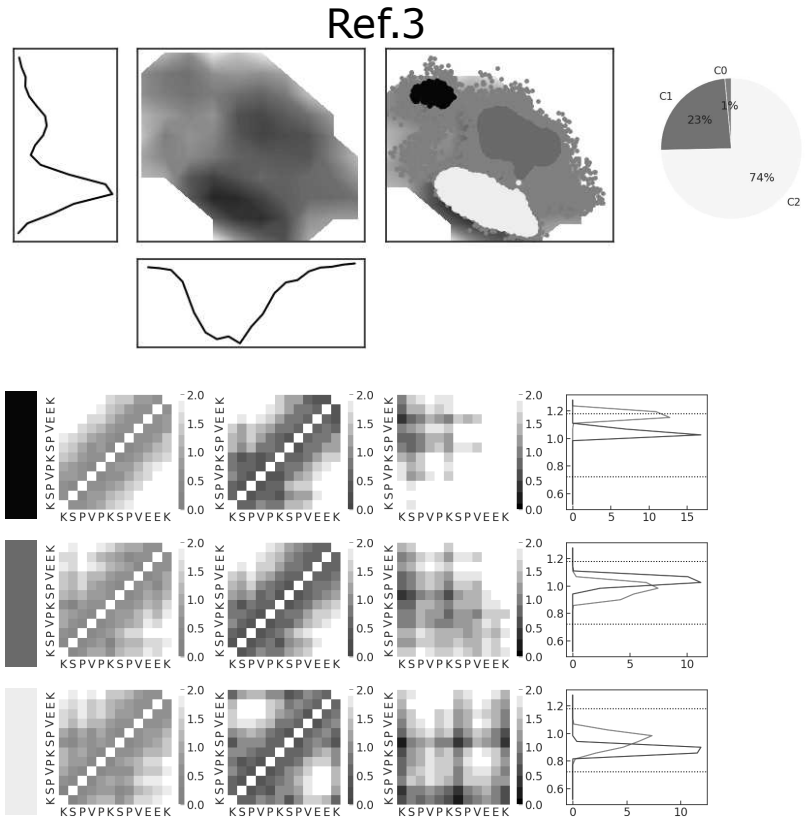


Figure C.9: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

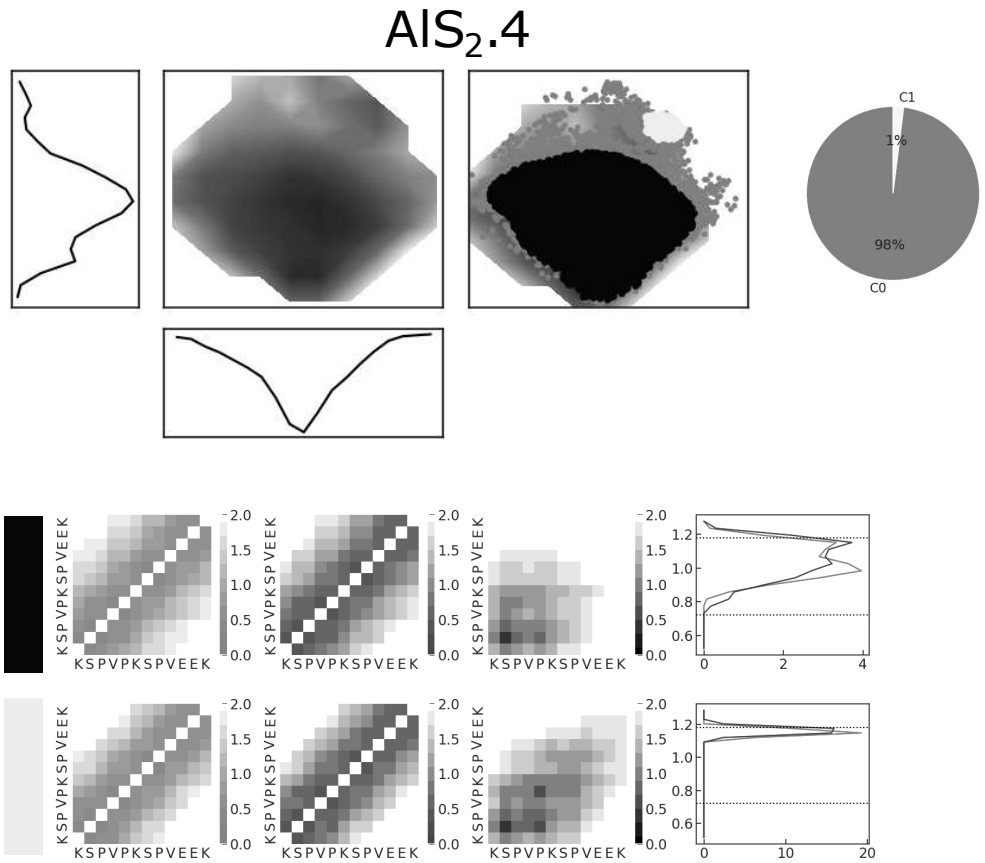


Figure C.10: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

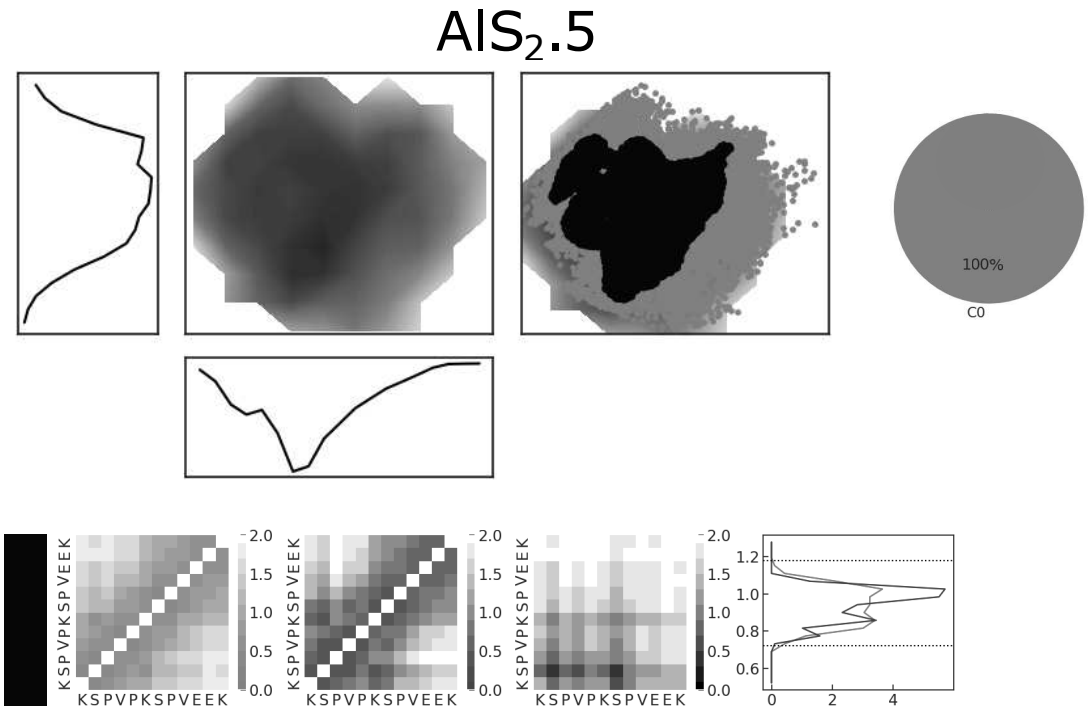


Figure C.11: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

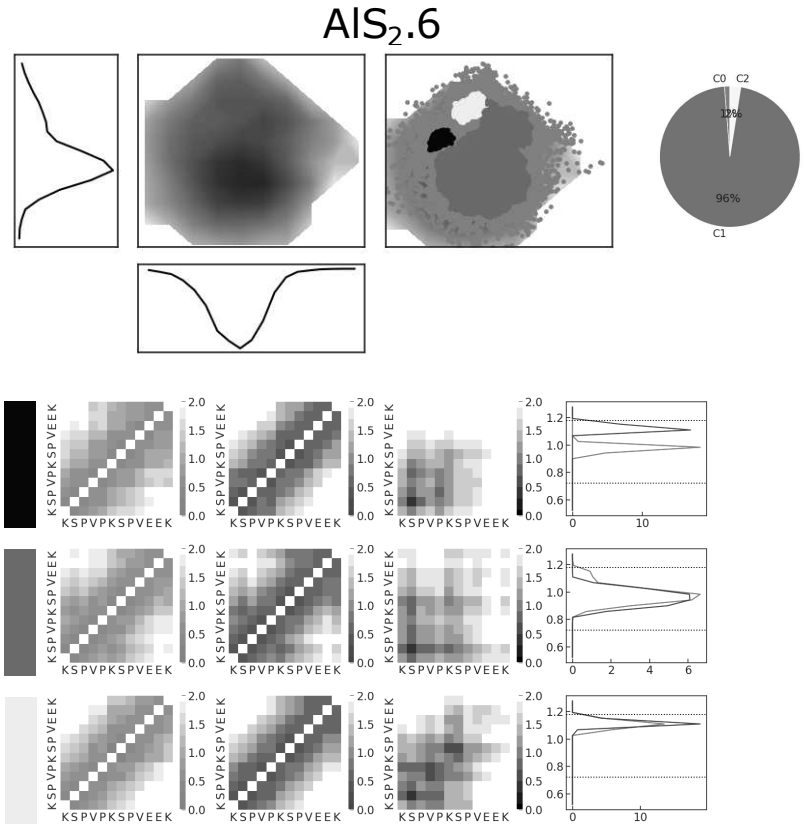


Figure C.12: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

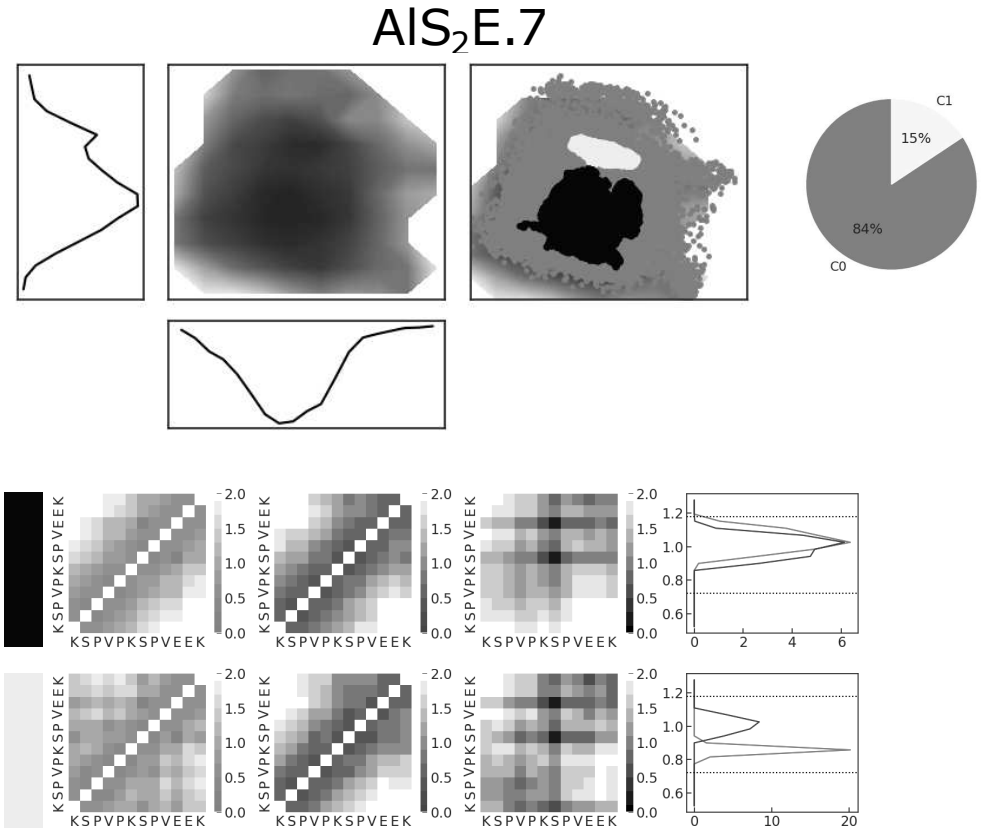


Figure C.13: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

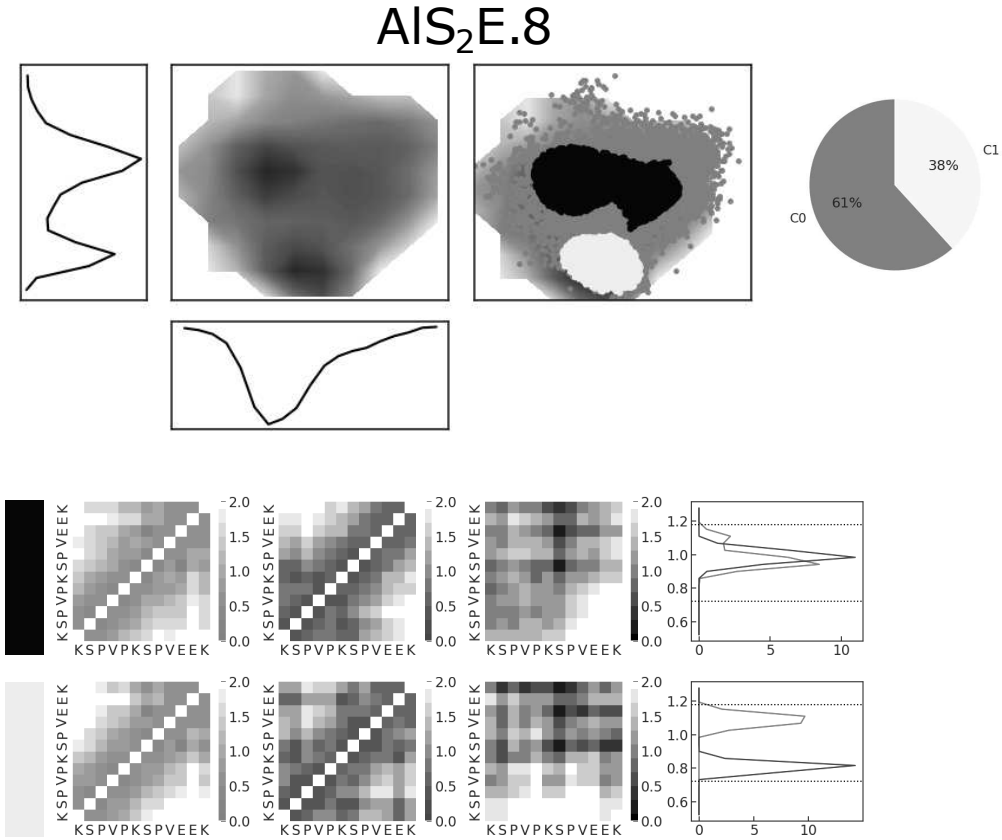


Figure C.14: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

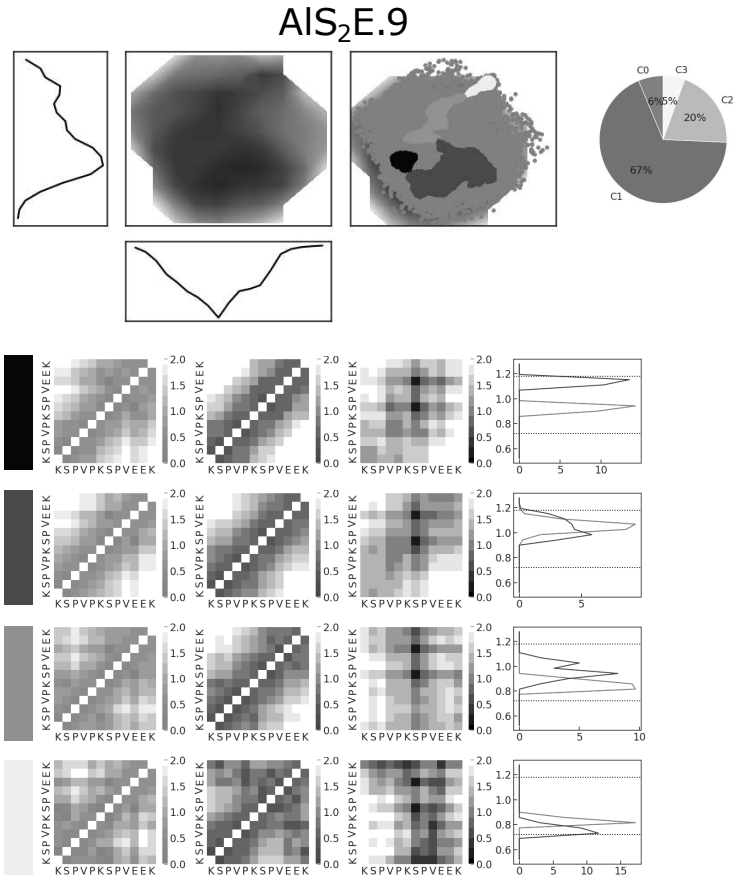


Figure C.15: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

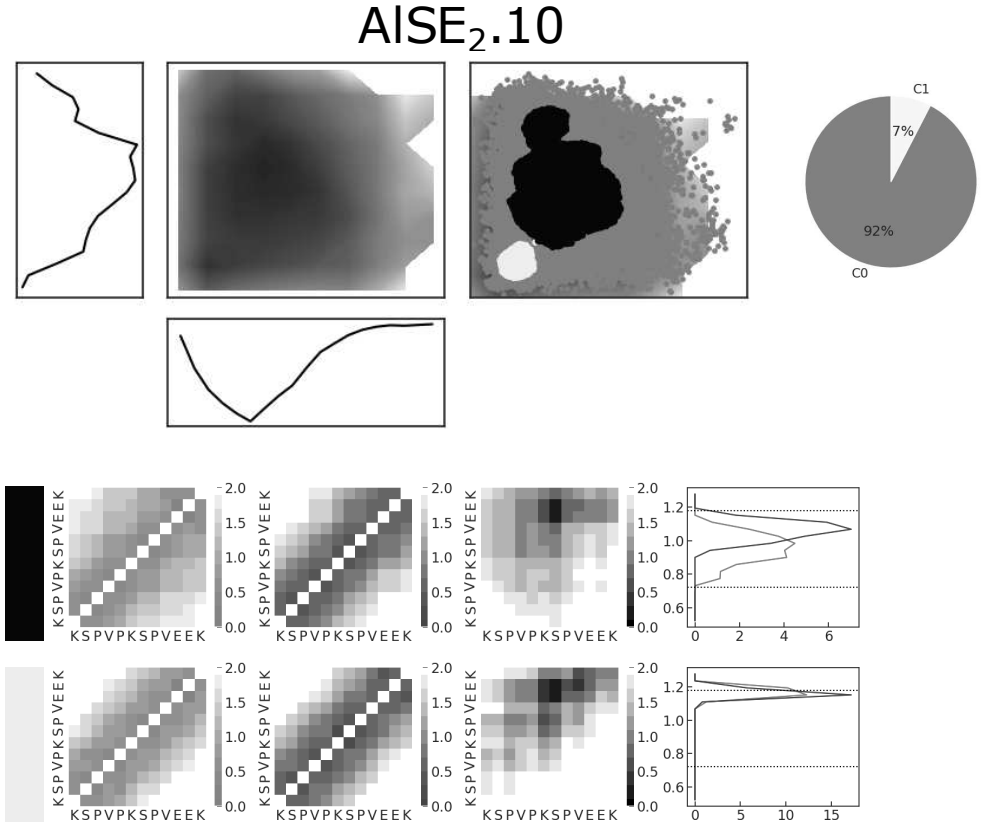


Figure C.16: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

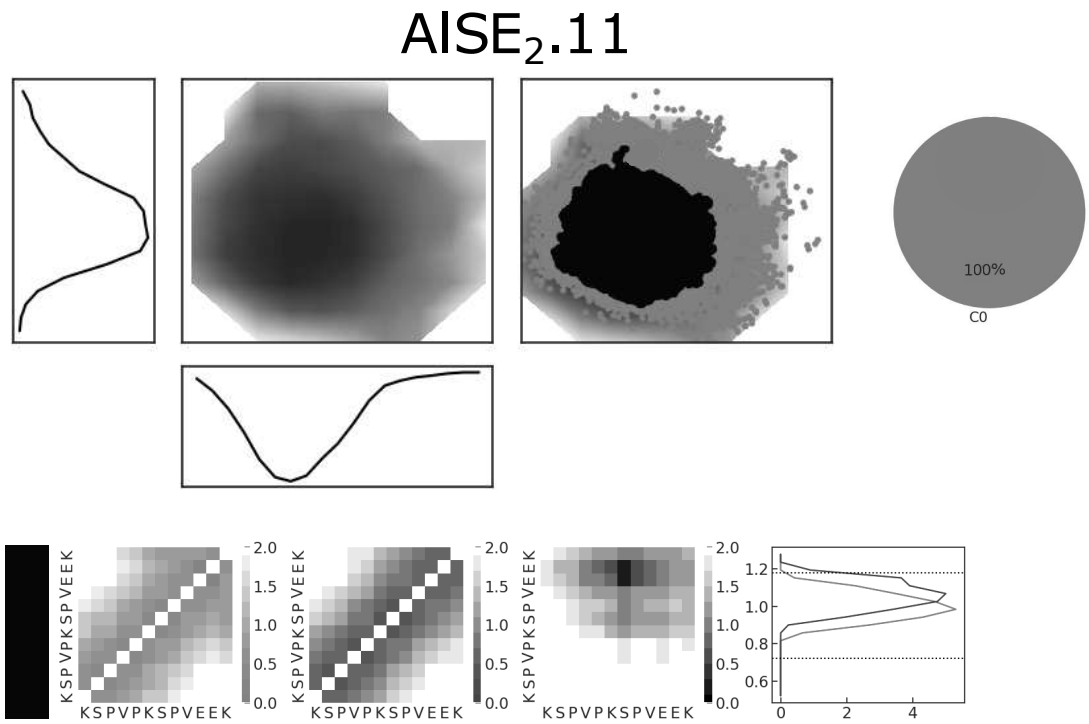


Figure C.17: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

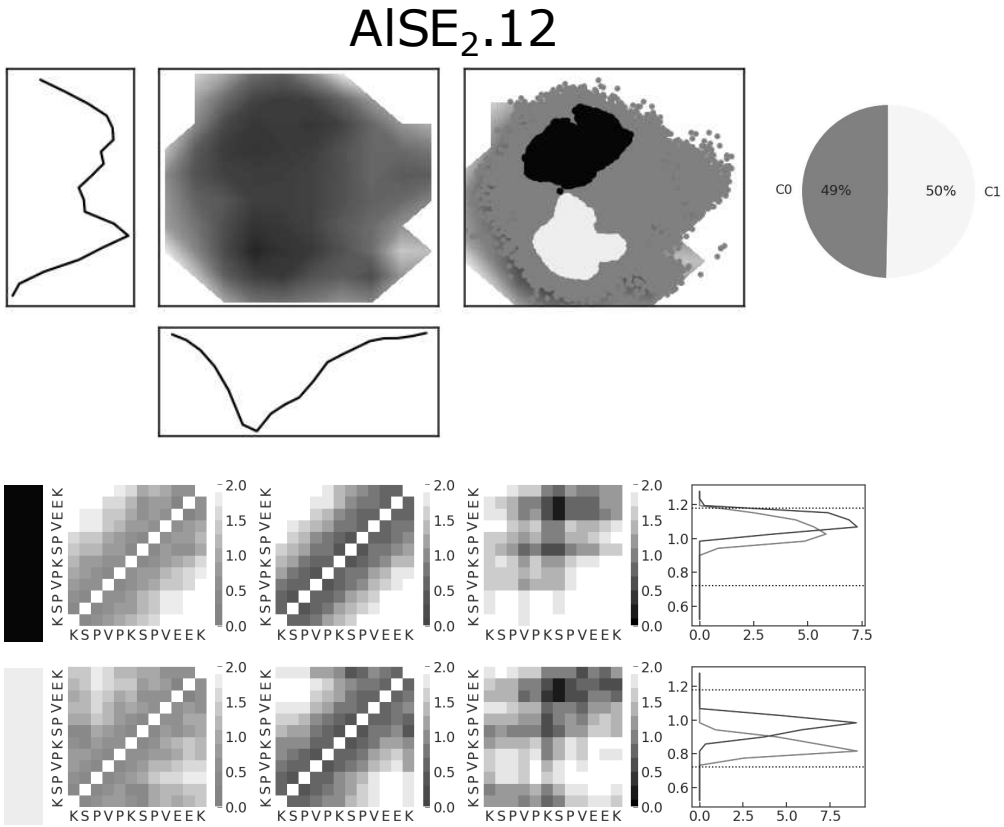


Figure C.18: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

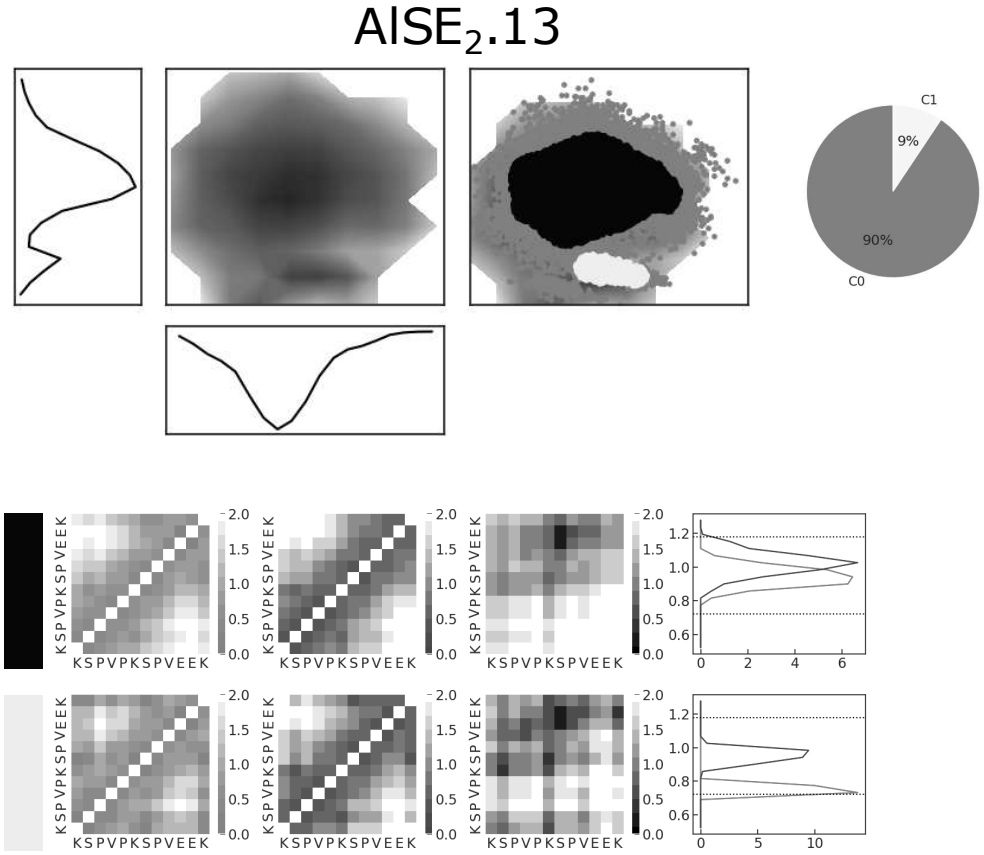


Figure C.19: PCA and HDBSCAN clustering, along with the occupancy of each cluster and the resultant intra-chain contact maps, inter-chain contact maps, and radius of gyration for each of the corresponding clusters.

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